

分子動力學模擬簡介

理論與實務應用

Min-Yeh Tsai

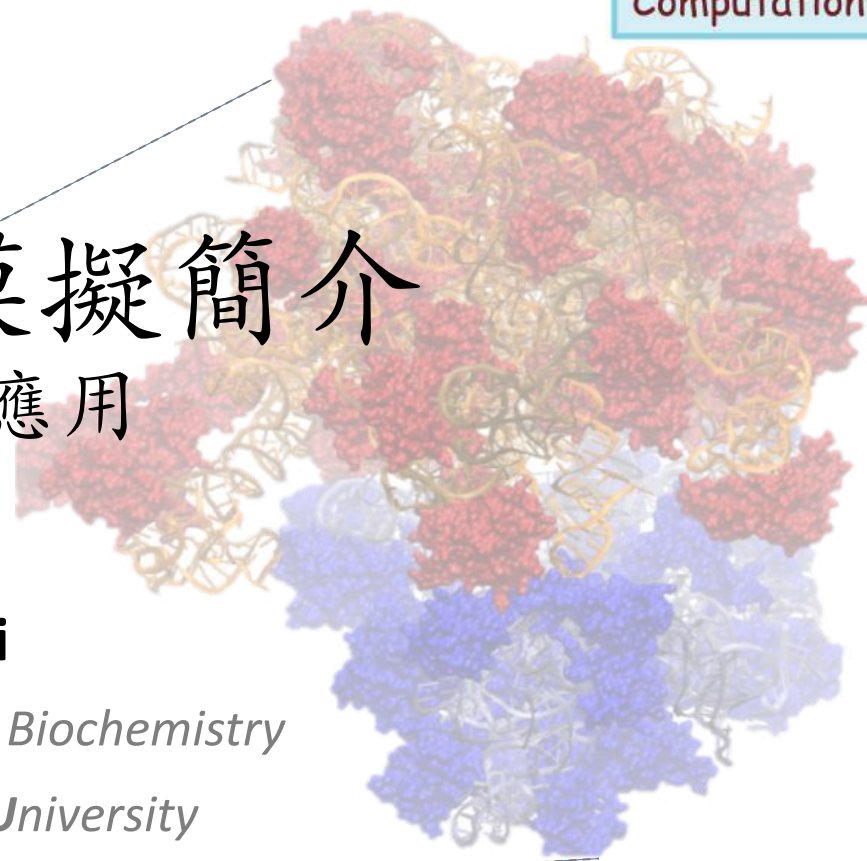
Department of Chemistry & Biochemistry

National Chung Cheng University

國立中正大學-化學暨生物化學系

Aug. 25-26, 2025

@臺大計中



請注意！

此投影片僅用於教學使用，其授權許可證屬於 **CC BY-NC**（非商業性使用）。

（投影片內容採中[**Chinese**]/英文[**English**]夾雜）

蔡旻燁
Min-Yeh Tsai

- Theoretical Phys. Chem.
- Computational Biophysics
- Structural Biology

Protein-protein & protein-DNA
interaction

Applied Chemistry
(Protein aggregation)

Physical Chemistry
(Protein folding)

Chemistry

BS

NTNU
(01-05)

師大



S. H. Lin

NTU (06-11)

台大

PhD

NCTU (11-14)

交大

Military
Duty

Rice (14-18)

萊斯大學



P. G. Wolynes

Postdoc

TKU (18-)

淡江大學

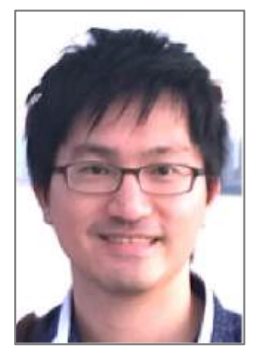
AP



AP

CCU (23-)

國立中正大學
化生系



助教 1



張育誠
Heery

助教 2



陳宏毅
Sam

這兩天 (十二個小時) 會學些什麼?

You will learn

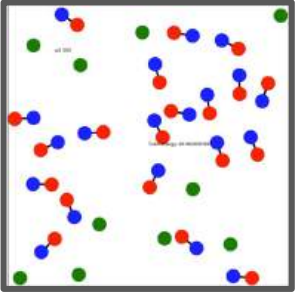
- 分子動力學模擬的 **ABC**
- 從頭快速建立一**簡單粒子模型**
- 使用軟體VMD**視覺化**複雜分子系統
- 使用 VMD **建立模型**以及 NAMD **進行動力學模擬**
- 模擬設定的輸入/輸出 (I/O)
- 快速建立用於**生物分子**、**奈米材料**的模擬系統
- 對模擬軌跡進行簡單**結構與能量分析**
- 了解MD模擬的整個流程，快速找到參考資料

You will **NOT** learn

- 進階的MD理論
- 分子力場開發
- 建立完全符合您研究上的模擬系統
(一天是不夠的...)
- 成為MD的專家
(**Rome wasn't built in a day...**)

讓各位對於建立模型與設定動態模擬的工具具有初步的認識
→ 激勵學習分子模擬的種子

Outline

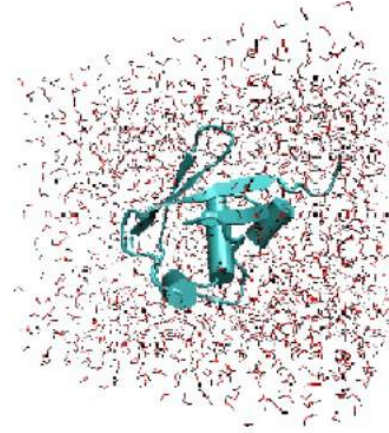


I. Simple MD

- Python
- JupyterLab

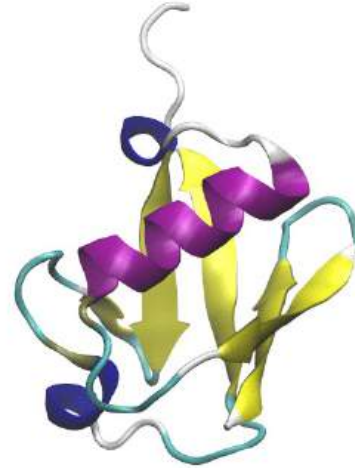


JupyterLab



II. Visualization

- VMD (Water)
- MolView
- Protein/DNA



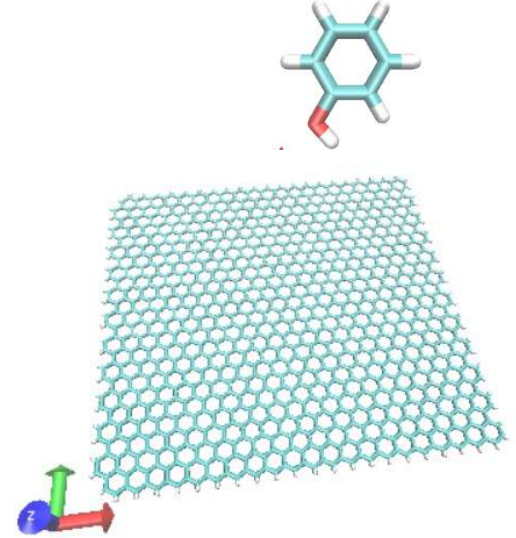
III. MD (Bio)

- NAMD/VMD
 - Protein
 - Solvation
 - Add ions



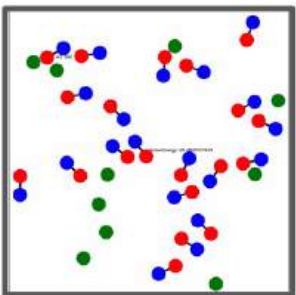
IV. MD (Nano)

- NAMD/VMD
 - Graphene
 - Small molecules



VMD
Visual Molecular Dynamics

NAMD
Scalable Molecular Dynamics



I. Simple MD

- Python
- JupyterLab



JupyterLab



II. Visualization

- MolView
- VMD



Part I

III. MD (Bio)

- NAMD/VMD



IV. MD (Nano)

- NAMD/VMD
 - Graphene
 - Small molecules

(分子動力學模擬ABC)

VMD
Visual Molecular Dynamics

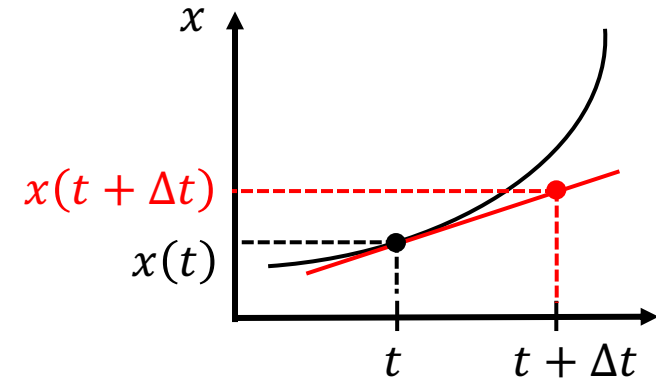
NAMD
Scalable Molecular Dynamics

預測運動軌跡好比預測未來發生的事

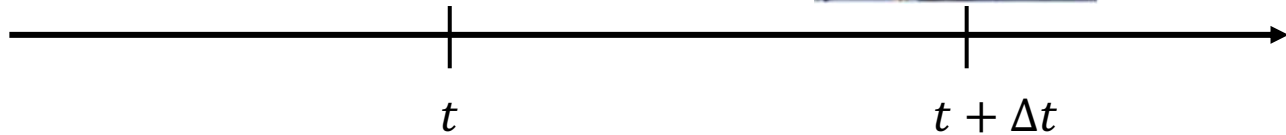
- 位置與時間關係 $x(t)$ 的一階線性關係

$$x(t + \Delta t) = x(t) + \dot{x}\Delta t \quad \dot{x} = \frac{dx}{dt} = v \text{ (velocity)}$$

粒子現在位於 $x(t)$ ，經過 Δt 之後到達 $x(t + \Delta t)$



- 線性思維



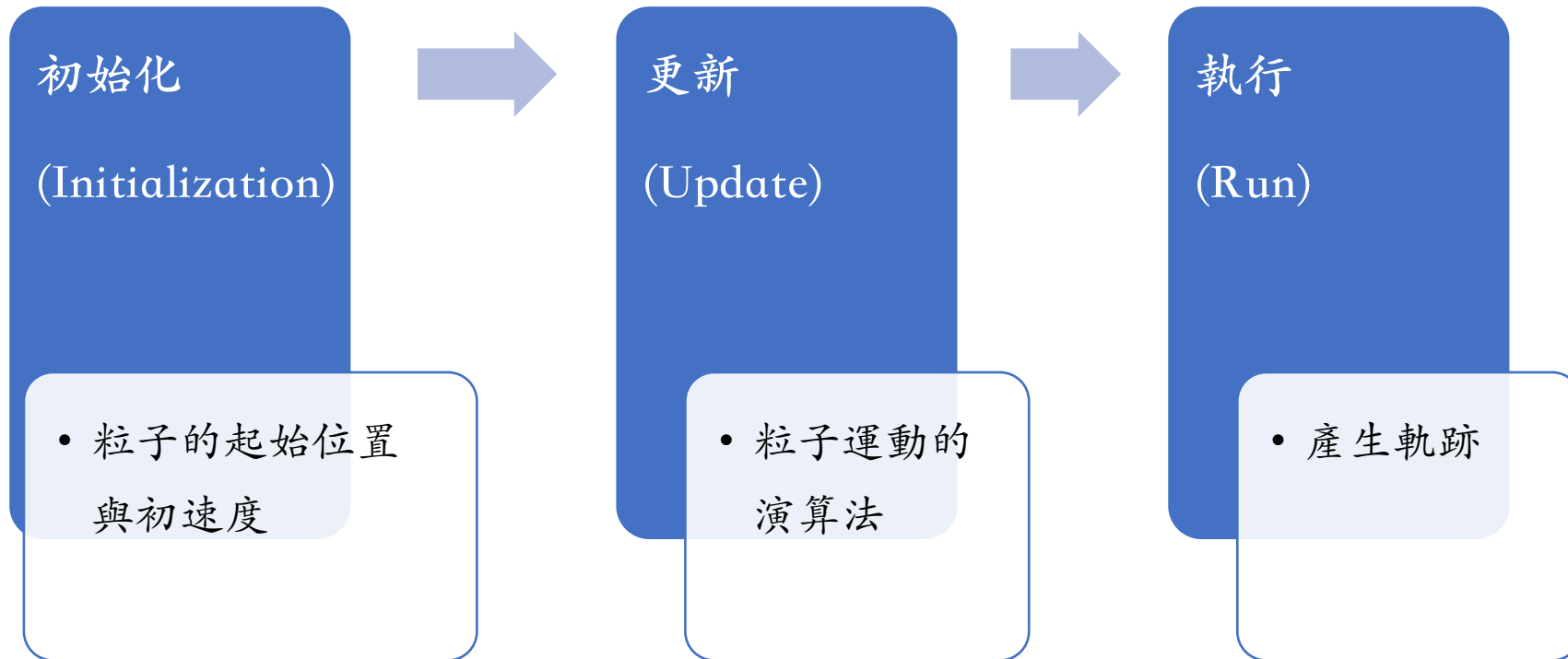
今日(t)不讀書

明日($t + \Delta t$)考試就GG囉

$$\text{考試表現} = \text{目前能力} + \text{用功程度} \times \text{時間}$$

$$\begin{array}{ccccccc} \uparrow & \uparrow & \uparrow & \uparrow \\ x(t + \Delta t) & x(t) & \dot{x} & \Delta t \end{array}$$

讓粒子動起來的三部曲



視覺化 (Visualization) & 分析 (Analysis)

1. 從Dropbox下載 113.2 MD workshop.zip

2. 解壓縮檔案至 113.2 MD workshop\

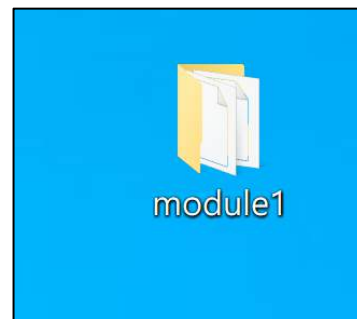
3. 進入113.2 MD workshop\ 資料夾，將 **module1** 移至桌面 (Desktop)

4. 開啟 Anaconda → (wait...) → 在 **base (root)** 環境下開啟 JupyterLab → 點 **Launch**

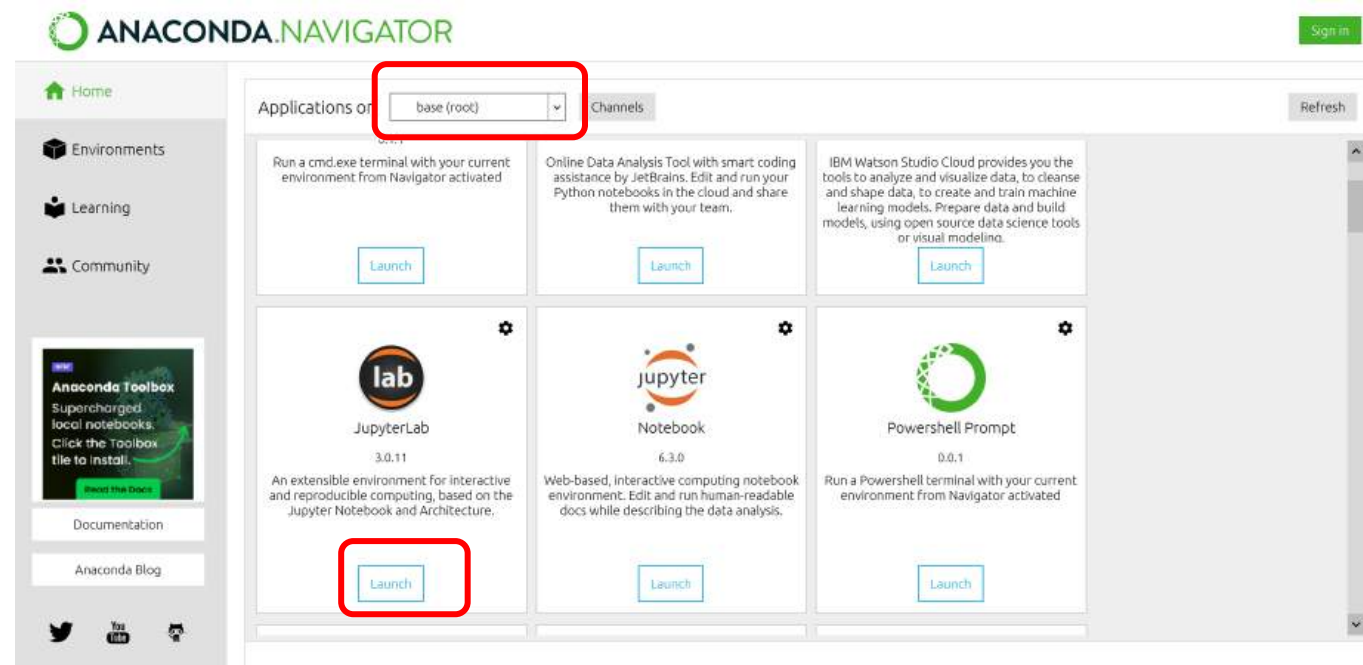
5. JupyterLab將開啟於瀏覽器新分頁

將左側之目錄切換至『桌面』

→ 進入 **module1**。



JupyterLab



將左側之目錄切換至『桌面』→ 進入 **module1**

→ 開啟 **00_pythonIntro_ntu.ipynb**

The screenshot displays a Jupyter Notebook environment. On the left, the file explorer shows the directory structure: Desktop / module1. The file **00_pythonIntro_ntu.ipynb** is selected. The main area shows the notebook content, which includes a title 'Python basics', a description of the notebook's purpose, creation and update information, a list of references, and a section titled 'The python world (初探python的世界)' with three code cells.

00_pythonIntro_ntu.ipynb

Python basics

This Notebook is used to introduce python basics, including **Basic Math, Data Type, Function, Array, and Plotting** 此份Notebook 介紹基本的python指令與使用，包含【基本運算】、【資料型態】、【函式】、【陣列】與【繪圖】。

Created by Min-Yeh Tsai (MYTLab)

Nov 23, 2022 updated (MYT)

參考資料

- J.R. Johansson, Intro to Python Programming
- J.R. Johansson, Numpy

此文件僅供教學使用

All rights reserved.

The python world (初探python的世界)

```
[ ]: print("hello world!") #印出字串 "hello world!"
```

```
[ ]: "test" #字串以 "" 包圍表示
```

```
[ ]: seq_1="hello " #宣告一變數 seq_1 指派成為一字串 "hello"
```

Simple 0 3 Python 3 | Idle Saving completed Mode: Command Ln 1, Col 1 pythonIntro_ntu.ipynb

將左側之目錄切換至『桌面』→ 進入 **module1**

→ 開啟 **01_MDfromScratch.ipynb**

01_MDfromScratch.ipynb

The screenshot displays the JupyterLab environment. On the left, the file browser shows the directory structure: `/ ... / 桌面 / module1 /`. The file `01_MDfromScratch.ipynb` is selected and highlighted with a red box. A red arrow points to this file with the text "雙擊開啟" (Double-click to open). The main area shows the notebook content for `01_MDfromScratch.ipynb`. The title is "讓粒子動起來!" (Let particles move!). The content includes a list of steps: 1. 初始化 (準備粒子的起始位置・初速度), 2. 更新 (粒子運動的演算法), and 3. 執行 (反覆迭代產生軌跡). A reference to [PolymerTheory](#) is provided. The notebook is in "Basic version" mode. The first cell, labeled "1. 初始化", contains the following code:

```
[1]: #import numerical library (匯入函式庫)
import numpy as np

[14]: #parameters
n = 10 # number of particles (粒子的數量)
D = 3 # dimensions (維度)
dt = 0.01 # time interval (時間的間隔)
t_steps = 100 # time steps (時間步數)
#arrays of variables
r = 100*np.random.rand(n,D) # initialize random positions for n particles in D dimensions
v = 100*(np.random.rand(n,D)-0.5) # initialize random velocities for n particles in D dimensions

[8]: r
[8]: array([[57.41157716,  6.31160163, 78.19677118],
```

The bottom status bar shows "Simple" mode, "0" lines of code, and "2" cells. The bottom right corner indicates "Launcher".

將左側之目錄切換至『桌面』→ 進入 **module1**

→ 開啟 **02_MDfromScratch-v1.ipynb**

02_MDfromScratch.ipynb



Launcher × MDfromScratch.ipynb MDfromScratch-v1.ipynb Python 3

Markdown

"看見"粒子動起來! 📺

1. 初始化 (準備粒子的起始位置、初速度+盒子大小+週期邊界條件)
2. 更新 (粒子運動的演算法+週期邊界條件)
3. 執行 (反覆迭代產生軌跡+3D繪圖)

參考資料 [PolymerTheory](#)

Visualization version

1. 初始化

```
[4]: #import numerical library (匯入函式庫)
import numpy as np

[18]: #parameters
n = 5 # number of particles (粒子的數量)
D = 3 # dimensions (維度)

#---- new addition ----#
LL = 100 # box size (盒子大小)
BC = 1 # PBC switch; on/off=1/0
#-----#

dt = 0.1 # time interval (時間的間隔)
t_steps = 100 # time steps (時間步數)
```

Mode: Command Ln 1, Col 1 MDfromScratch-v1.ipynb

預測的品質與採取的策略有關

- **Verlet algorithm**
(韋爾萊算法)

$$x(t + \Delta t) = 2x(t) - x(t - \Delta t) + \ddot{x}\Delta t^2$$

表現 目前 實力 過去 實力 有效努力 x 時間²

實際 表現 是根據 過去 與 目前 的差異，透過一定時間內 有效的努力 所達成

了解歷史，掌握當下，加上持續的努力 → 成功

[補充] 運用數值方法解運動方程式

Position and dynamic properties can be approximated as Taylor series expansions

Position → $\mathbf{r}(t + \delta t) = \mathbf{r}(t) + \delta t \mathbf{v}(t) + \frac{1}{2} \delta t^2 \mathbf{a}(t) + \frac{1}{6} \delta t^3 \mathbf{b}(t) + \frac{1}{24} \delta t^4 \mathbf{c}(t) + \dots$

Velocity → $\mathbf{v}(t + \delta t) = \mathbf{v}(t) + \delta t \mathbf{a}(t) + \frac{1}{2} \delta t^2 \mathbf{b}(t) + \frac{1}{6} \delta t^3 \mathbf{c}(t) + \dots$

Acceleration → $\mathbf{a}(t + \delta t) = \mathbf{a}(t) + \delta t \mathbf{b}(t) + \frac{1}{2} \delta t^2 \mathbf{c}(t) + \dots$

$\mathbf{b}(t + \delta t) = \mathbf{b}(t) + \delta t \mathbf{c}(t) + \dots$

Truncated

Verlet algorithm

$$\mathbf{r}(t + \delta t) = \mathbf{r}(t) + \delta t \mathbf{v}(t) + \frac{1}{2} \delta t^2 \mathbf{a}(t) + \dots$$

$$\mathbf{r}(t - \delta t) = \mathbf{r}(t) - \delta t \mathbf{v}(t) + \frac{1}{2} \delta t^2 \mathbf{a}(t) - \dots$$



Summation

**Truncated at $O(\delta t^4)$
→ Fourth-order method**

$$\mathbf{r}(t + \delta t) = 2\mathbf{r}(t) - \mathbf{r}(t - \delta t) + \delta t^2 \mathbf{a}(t)$$

Tutorial: 簡諧運動模擬 (Harmonic Oscillator)

File Edit View Run Kernel Tabs Settings Help

+

↑

↺

↓

桌面

/ module1 /

Name	Modified
00_pythonIntro_ntu.ipynb	11 months ago
01_MDfromScratch.ipynb	22 minutes ago
02_MDfromScratch-v1.ipynb	20 minutes ago
03_Harmonic_oscillator_MD.ipynb	24 minutes ago

Launcher

03_Harmonic_oscillator_MiX

+

✂

📄

📄

▶

⏮

⏪

⏩

⏭

Markdown

Share

Notebook

Python [conda env:base]

Simple Molecular Dynamics Simulation

Use harmonic oscillator as an example

by Min-Yeh Tsai (2024/4/24)

```
[ ]: # Import
import math
import numpy as np
```

I. Anytical solution

The Hamiltonian $H(q, p)$ for a harmonic oscillator is $H = \text{kinetic energy} + \text{potential energy} = \frac{p^2}{2m} + \frac{1}{2}kq^2$,

where m is the mass, k is the force constant, and that E is the total energy.

Since the equations of motion for a harmonic oscillator, $q(t)$, follows a periodic motion, we can define its angular frequency (ω) and the period ($T = \frac{1}{\nu}$):

$$\omega = \sqrt{\frac{k}{m}} = 2\pi\nu \text{ and that } \nu = \frac{\omega}{2\pi}$$

The parameters described above can be implemented into python through the step called *declaration* shown in the code block below.

```
[ ]: # Parameters
mass = 2 # m
force_const = 9 # k
energy = 1.5 # E
delta_t = 0.3 # dt
t_min = 0
t_max = 10
```

Simple

0

2

Python [conda env:base] * | Idle

Mode: Command

Ln 1, Col 1

03_Harmonic_oscillator_MD.ipynb

0

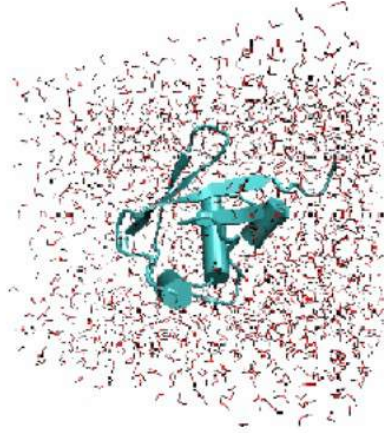


I. Simple MD

- Python
- JupyterLab



JupyterLab



II. Visualization

- VMD (Water)
- MolView
- Protein/DNA



Visual Molecular Dynamics



Part II

(分子結構 & 視覚化)

III.

- NAMD/VMD
- Protein
- Water



IV. MD (Nano)

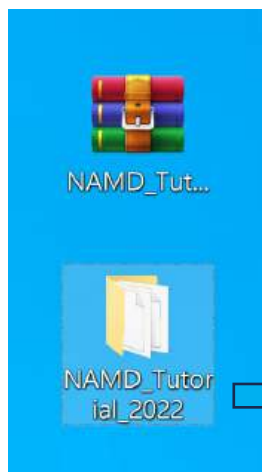
- NAMD/VMD
- Graphene
- Small molecules



方法一：

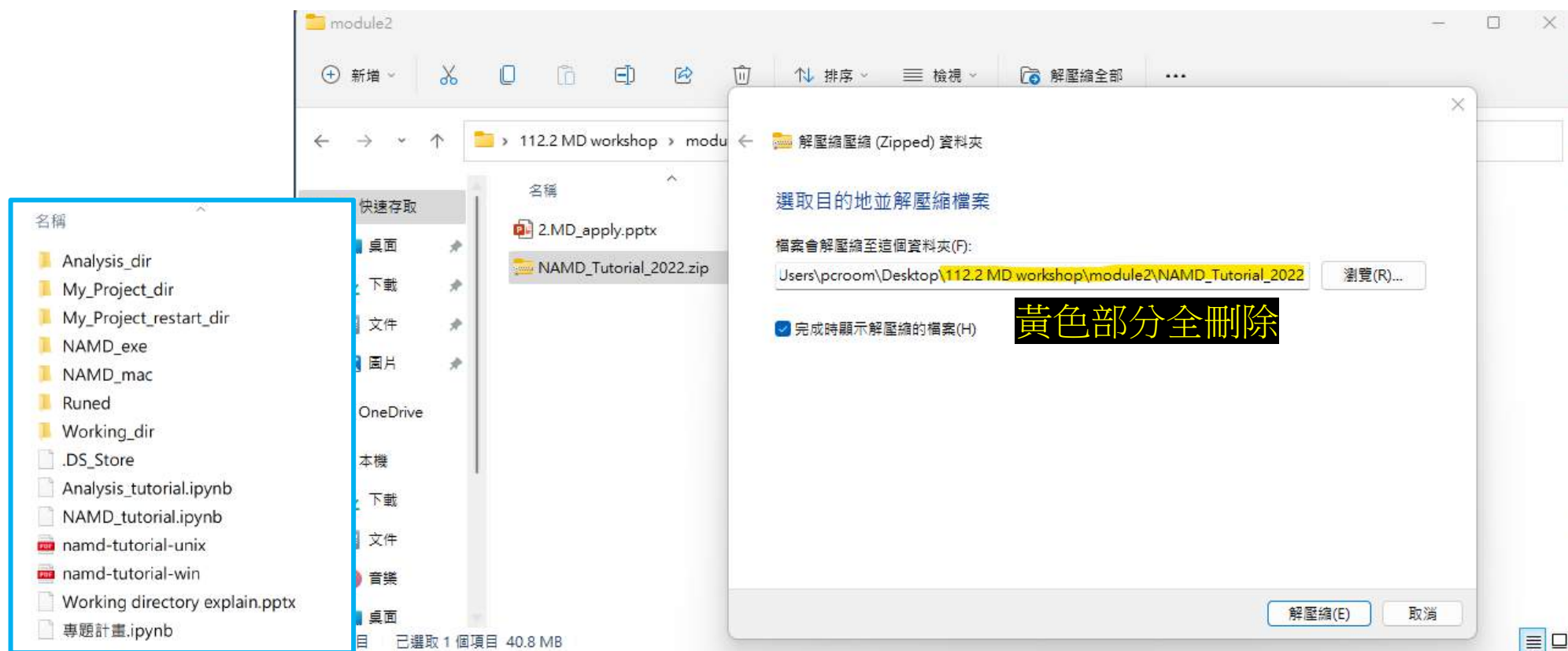
1. 從 113.2 MD workshop/module3 中，將 **NAMD_Tutorial_2025.zip** 資料夾拖曳至桌面
2. 按滑鼠右鍵 → 解壓縮檔案... → 目的地路徑 \.....\.....\Desktop\ (此處全刪除)
3. 解壓縮

桌面



方法二：

或是直接從 **NAMD_Tutorial_2025.zip** 上按右鍵 → 解壓縮檔案...

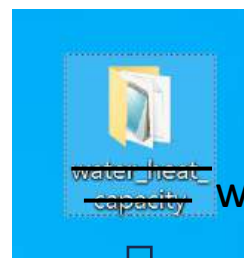


Tutorial: 水盒模擬 (Water Box)

從 113.2 MD workshop/module2 中，將 **water** 資料夾拖曳至桌面

桌面

開啟 JupyterLab



water



名稱

- ✓ .ipynb_checkpoints
- ✓ simulation_output_100000
- ✓ Get_all_atoms_velvector_mass_trajecto...
- ✓ Get_atom_index_select
- ✓ par_all27_prot_lipid
- ✓ water_simulation
- ✓ water_tutorial.ipynb

File Edit View Run Kernel Tabs Settings Help

Filter files by name

/ Desktop / water /

Name	Last Modified
simulation_output_100000	5 minutes ago
Get_all_atoms_velvector_mass_tr...	6 days ago
Get_atom_index_select.tcl	6 days ago
par_all27_prot_lipid.inp	6 days ago
water_simulation.conf	6 days ago
water_tutorial.ipynb	a minute ago

Launcher x water_tutorial.ipynb x

Python 3

Water Tutorial

2023/08 completed by Yoyo@MYT Lab

2025/08 updated by Heery@MYT Lab

MYT Lab website: <https://mytsai.cc>

參考資料：

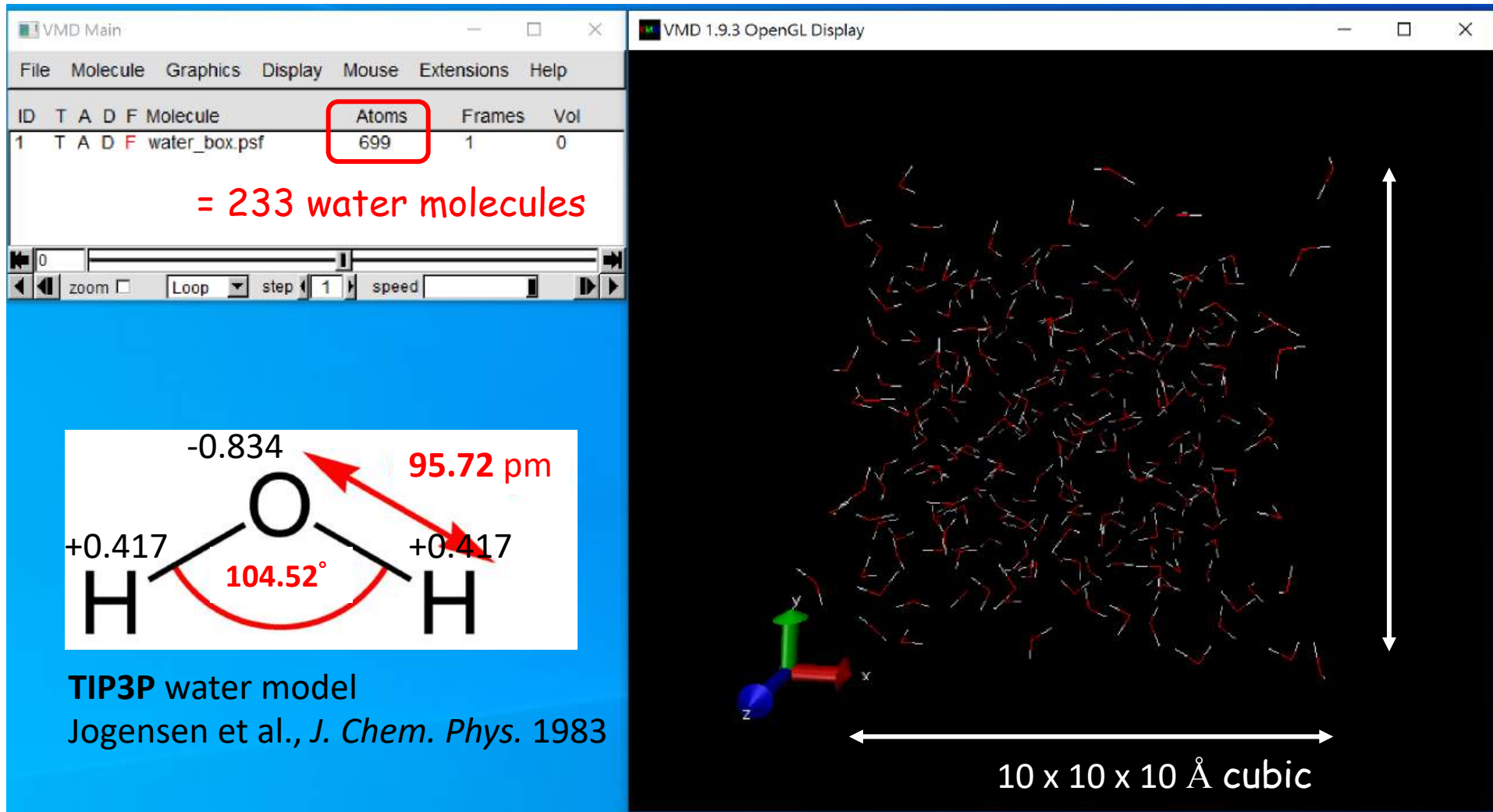
- NAMD tutorial ver.2017
- Levitt et al., J. Phys. Chem. B 1997

此教程僅供教學用途

All rights reserved.

Part1. 透過實際創建水盒並運行分子動力學模擬，計算出與參考文獻實驗值相近的水的熱容量值

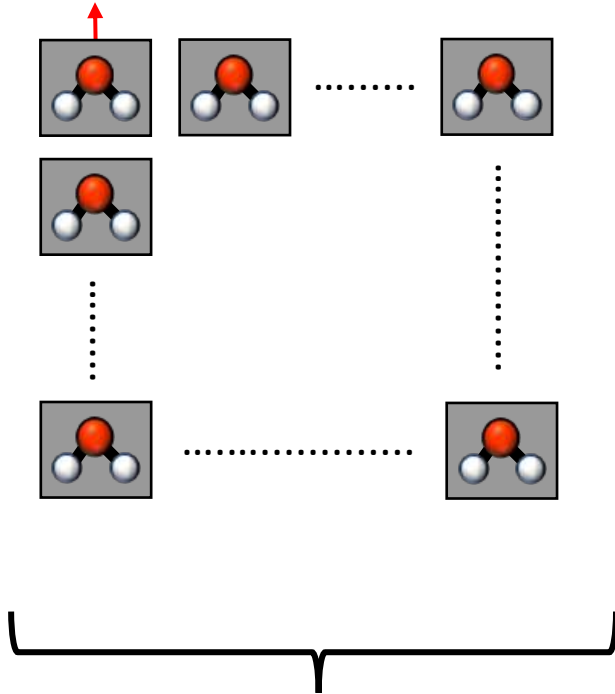
Tutorial: 水盒模擬 (Water Box)



從水盒分子能量變化計算水的熱容量/比熱

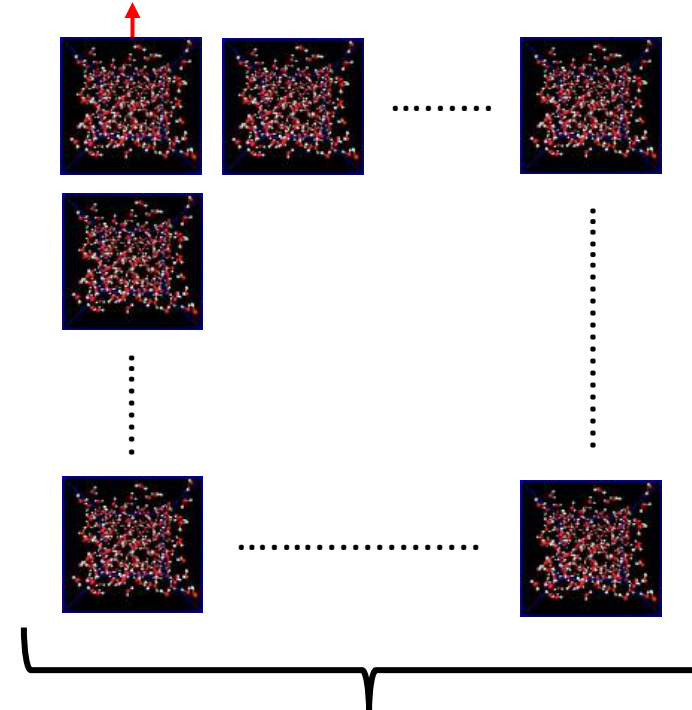
Tutorial: 水盒模擬 (Water Box)

1 個水分子



1 mole = 6.022×10^{23}

1 個 233 個水分子的系統



1 mole = 6.022×10^{23}

Ex. 請注意，此計算出的數值是整個水盒(233 個水分子)的總熱容量，為了得到「單一水分子的熱容量」，必須用「總熱容量 / 233」，這樣結果才能代表「單一水分子」的特性。

Tutorial: 水盒模擬 (Water Box)

請試著思考為什麼實際模擬計算出的單一水分子熱容量結果不等於 $17 \text{ (cal/mol}\cdot\text{K)}$?

對於模擬，計算結果會與實驗結果有誤差，且由於每次模擬的獨立性，每次模擬的熱容量計算誤差會有所不同。

可能造成模擬誤差的原因有：

- 水模型本來就不完美: 模擬裡的水是用「簡化模型」表達的，不是真正的水，所以本來就會有誤差。
- 模擬的水太少: 由於系統粒子數有限，能量波動的統計不充分，導致熱容量計算存在有限尺寸效應與統計誤差，無法完全反映宏觀系統的真實行為。
- 模擬時間不夠長: 熱容量是靠能量的波動計算的，但模擬時間不夠長，這些波動還未收斂，就會算不準。

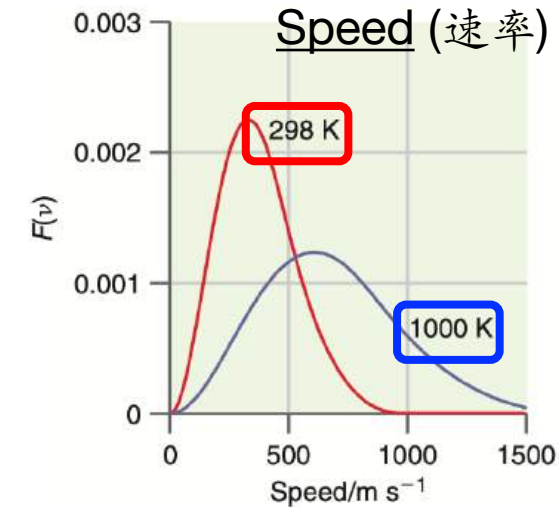
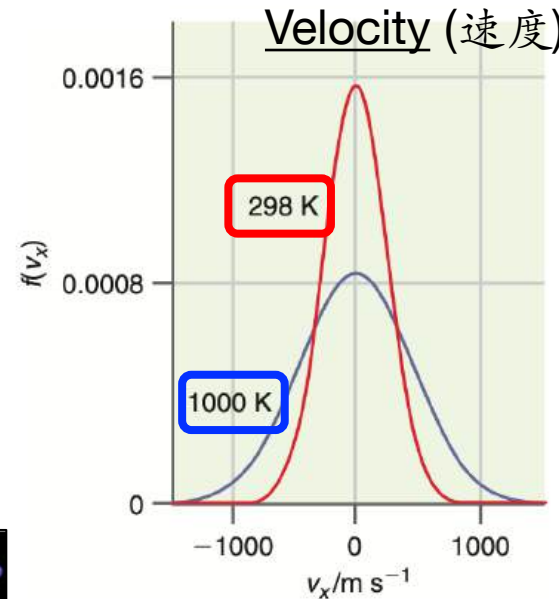
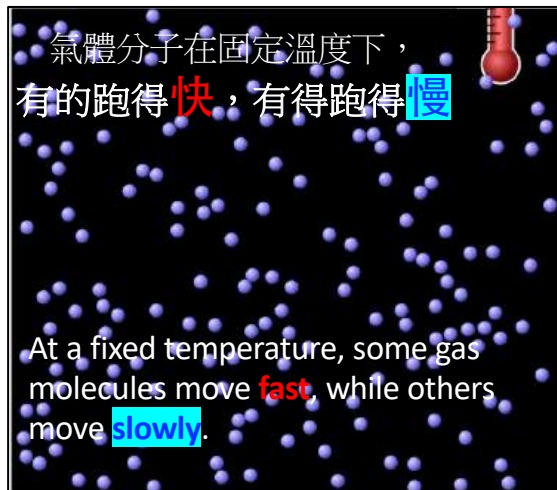
因此若想計算出更準確的熱容量值，可以試著增加模擬步數使模擬系統穩定，並選取足夠收斂的能量部分再進行熱容量計算!

[補充] 統計熱力學

Distribution function (分布函數)



Ludwig E. Boltzmann
(1844-1906) from wiki

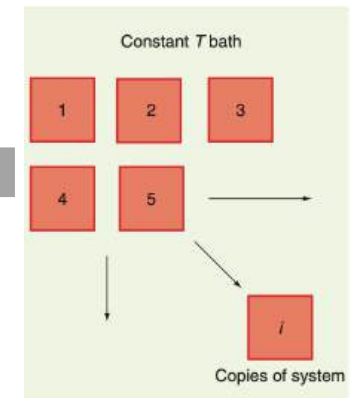


$$F(v) dv = 4\pi \left(\frac{M}{2\pi RT} \right)^{3/2} v^2 e^{-Mv^2/2RT} dv$$

$$f(v_j) = \left(\frac{M}{2\pi RT} \right)^{1/2} e^{-Mv_j^2/2RT}$$

巨觀 (macro) $\rightarrow$ $Q = \sum_n e^{-\beta E_n}$ $\leftarrow$ 微觀 (micro)

Canonical partition function
(正則配分函數)



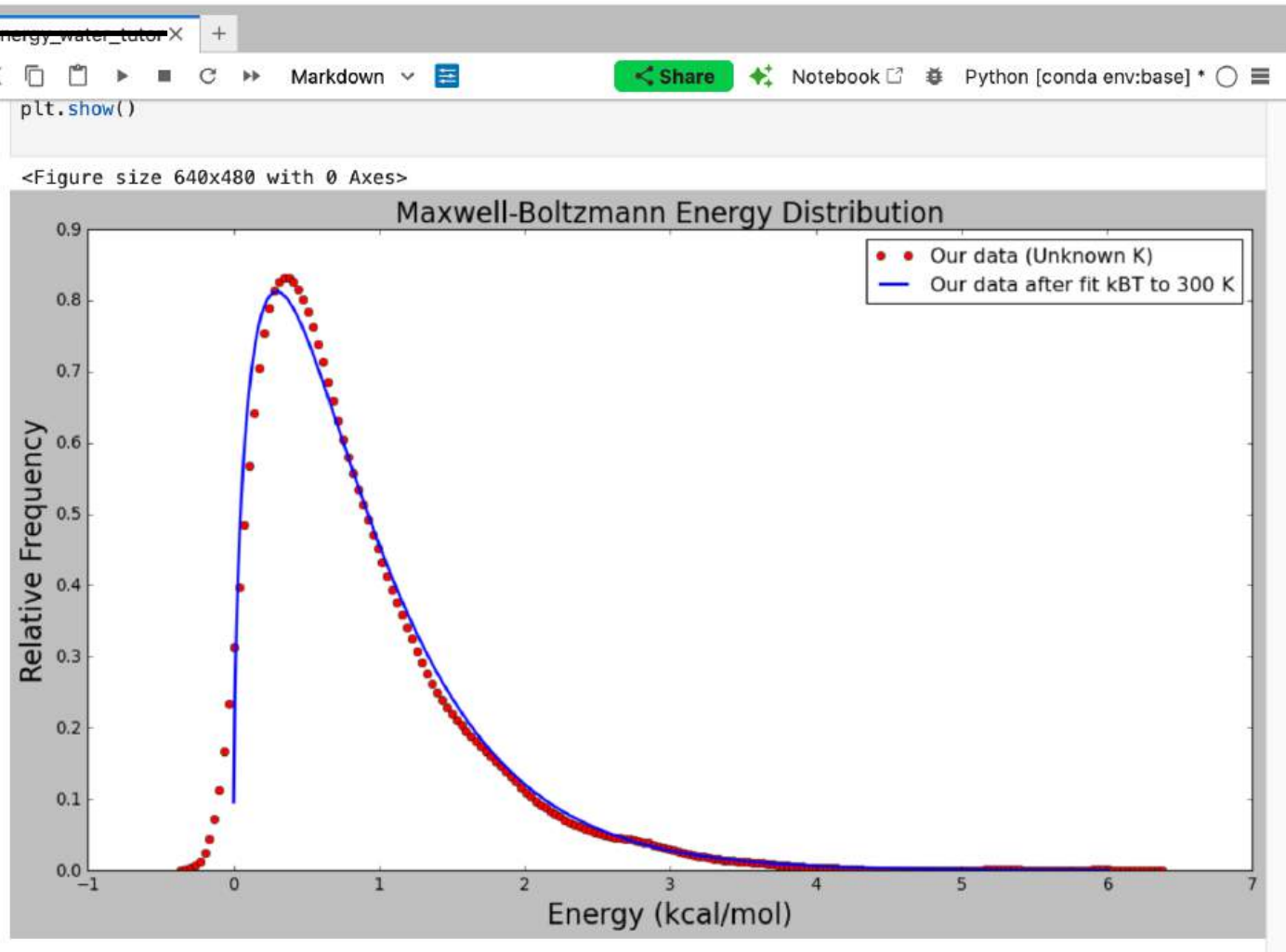
Tutorial: 數據分析—提取分子動能與能量分布

water_tutorial.ipynb

File Edit View Run Kernel Tabs Settings Help

+/ Desktop / water /

Name	Last Modified
simulation_output_100000	5 minutes ago
Get_all_atoms_velvector_mass_tr...	6 days ago
Get_atom_index_select.tcl	6 days ago
par_all27_prot_lipid.inp	6 days ago
water_simulation.conf	6 days ago
water_tutorial.ipynb	a minute ago



模擬結果動能計算流程

VMD 內操作 (擷取原始模擬數據)

Get_all_atoms_velvector_mass_trajectory.tcl

計算系統個別原子速度向量和質量，產生：

→ my_atoms_velocity_vector.txt

→ my_atoms_mass.txt

Get_atom_index_select.tcl

計算選取範圍原子的 index，產生：

→ my_atom_indexs_select.txt



JupyterLab 內操作 (原始模擬數據整理)

B.2.(b1) 讀取 my_atoms_velocity_vector.txt

整理檔案內原子的速度向量

→ 以 my_velocity_vector_all_F 列表暫存

B.2.(b2) 讀取 my_atoms_mass.txt

整理檔案內原子的質量

→ 以 my_mass_all_F 列表暫存

B.2.(b3) 讀取 my_atom_indexs_select.txt

整理檔案內選取原子的 index

→ 以 my_select_atom_index_all_F 列表暫存



JupyterLab 內操作

(參照選取原子 index，選出速度向量和質量)

B.2.(c) 參照 my_select_atom_index_all_F

選取出列表 my_velocity_vector_all_F

對應的速度向量

→ 以 select_velocity_vectors 列表暫存

選取出列表 my_mass_all_F 對應的質量

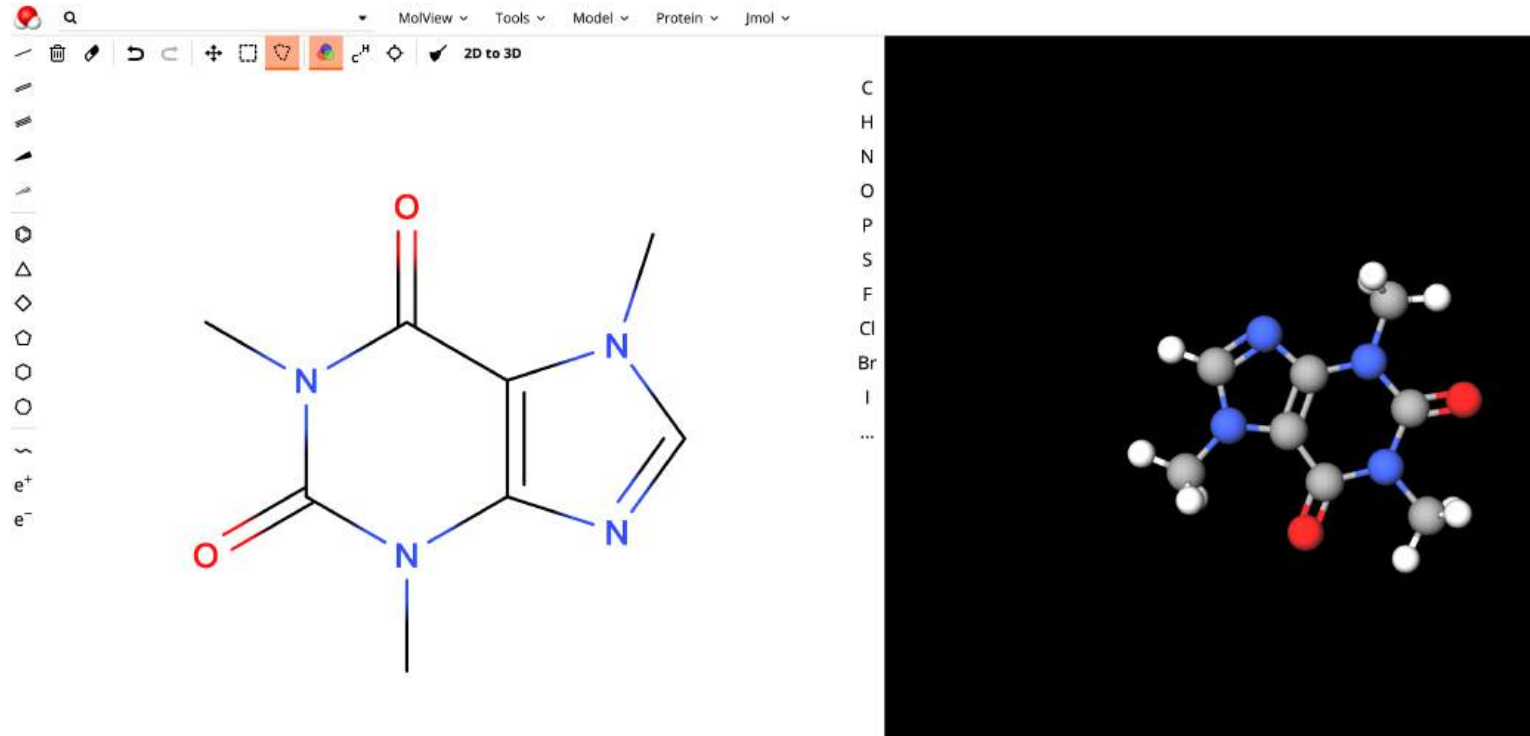
→ 以 select_mass 列表暫存

JupyterLab 內操作



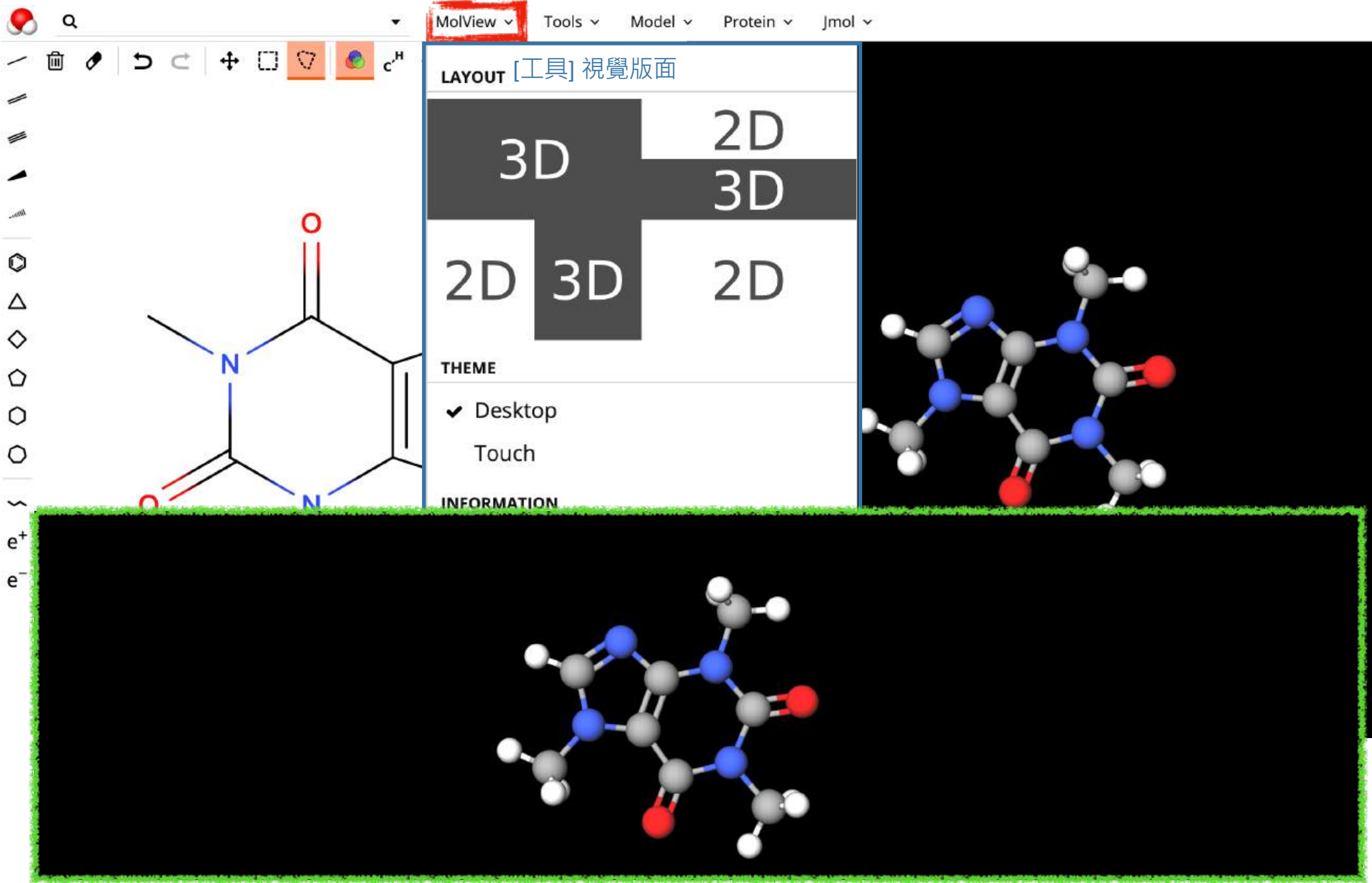
B.2.(d1~d4) 使用列表 select_velocity_vectors、select_mass 中速度以及原子質量資訊計算動能

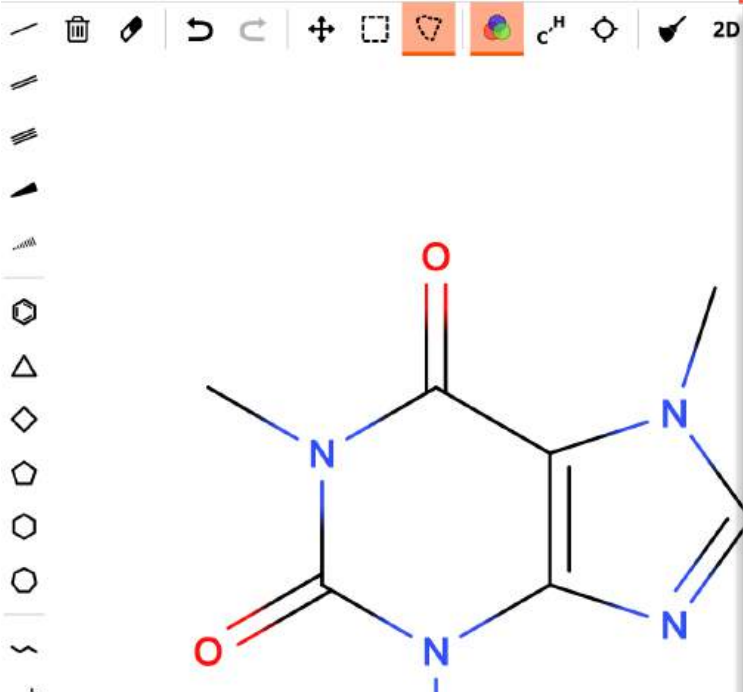
MolView



<https://molview.org/>

- Video tutorial:
<https://www.youtube.com/watch?v=ZesFyuFj30s>
<https://www.youtube.com/watch?v=af6PAAfVpRc>





[模型] 結構輸出、資料

LINK

</> Embed

EXPORT

Structural formula image

3D model image

MOL file

CHEMICAL DATA

Information card

Spectroscopy

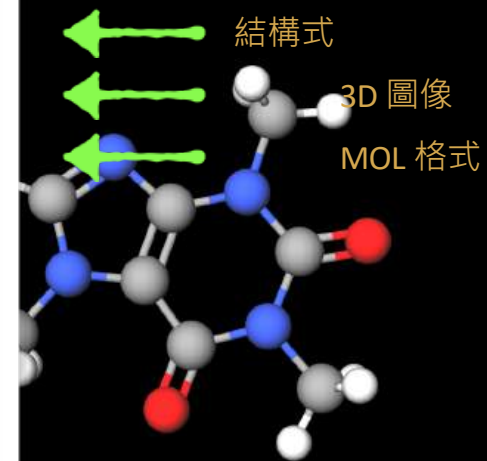
3D model source

ADVANCED SEARCH

arity

structure

rstructure



```

2519
-0EChem-09212006583D
24 25 0 0 0 0 0 0999 V2000
0.4700 2.5688 0.0006 O 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
-3.1271 -0.4436 -0.0003 O 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
-0.9686 -1.3125 0.0000 N 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
2.2182 0.1412 -0.0003 N 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
-1.3477 1.0797 -0.0001 N 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
1.4119 -1.9372 0.0002 N 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0.8579 0.2592 -0.0008 C 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0.3897 -1.0264 -0.0004 C 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0.0307 1.4220 -0.0006 C 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
-1.9061 -0.2495 -0.0004 C 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
2.5032 -1.1998 0.0003 C 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
-1.4276 -2.6960 0.0008 C 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
3.1926 1.2061 0.0003 C 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
-2.2969 2.1881 0.0007 C 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
3.5163 -1.5787 0.0008 H 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
-1.0451 -3.1973 -0.8937 H 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
-2.5186 -2.7596 0.0011 H 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
-1.0447 -3.1963 0.8957 H 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
4.1992 0.7801 0.0002 H 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
3.0468 1.8092 -0.8992 H 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
3.0466 1.8083 0.9004 H 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
-1.8087 3.1651 -0.0003 H 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
-2.9322 2.1027 0.8881 H 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
-2.9346 2.1021 -0.8849 H 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
    
```


MolView Tools Model Protein Jmol

Systematic name
1,3,7-trimethylpurine-2,6-dione

Canonical SMILES
CN1C=NC2=C1C(=O)N(C(=O)N2C)C

Isomeric SMILES
CN1C=NC2=C1C(=O)N(C(=O)N2C)C

InChIKey
RYYVLZVUVIJVGH-UHFFFAOYSA-N

InChI
InChI=1S/C8H10N4O2/c1-10-4-9-6-5(10)7(13)12(3)8(14)11(6)2/h4H,1-3H3

CAS Number
58-08-2

PubChem Compound ID [2519](#)

[模型] 結構輸出、資料

LINK

</> Embed

EXPORT

Structural formula image

3D model image

MOL file

CHEMICAL DATA

Information card

Spectroscopy

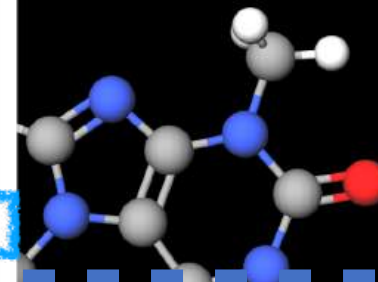
3D model source

ADVANCED SEARCH

Similarity

Substructure

Superstructure



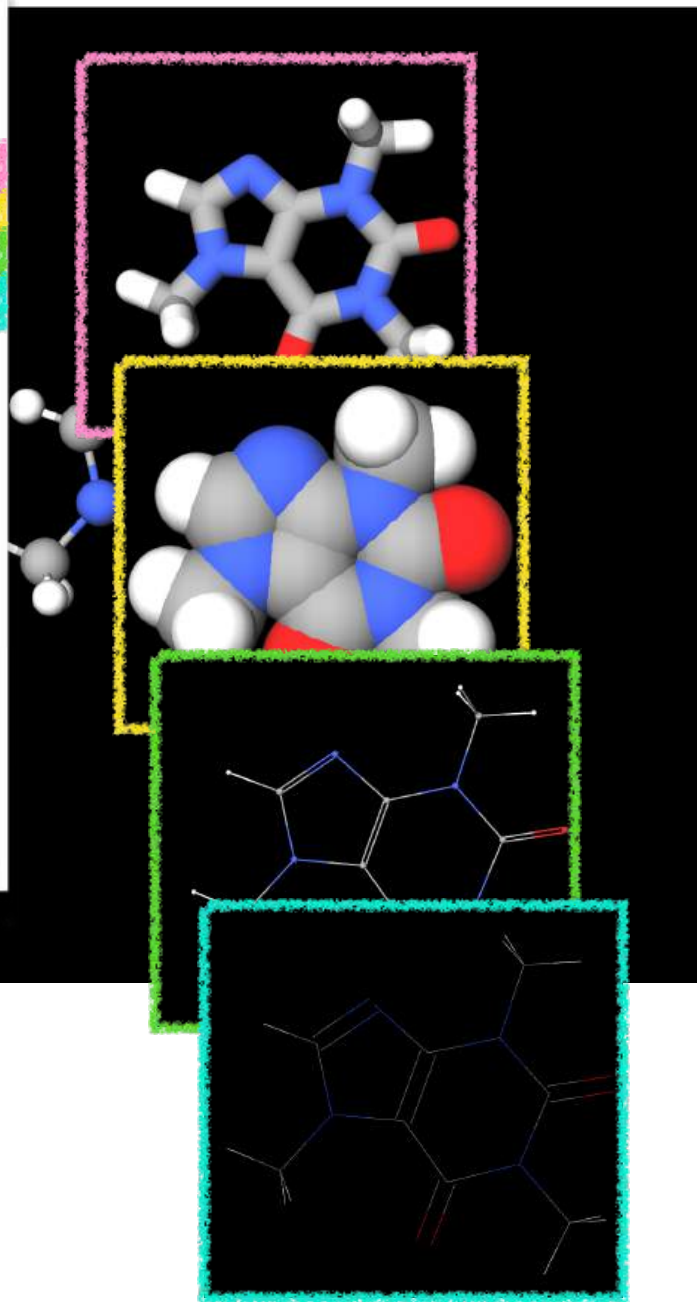
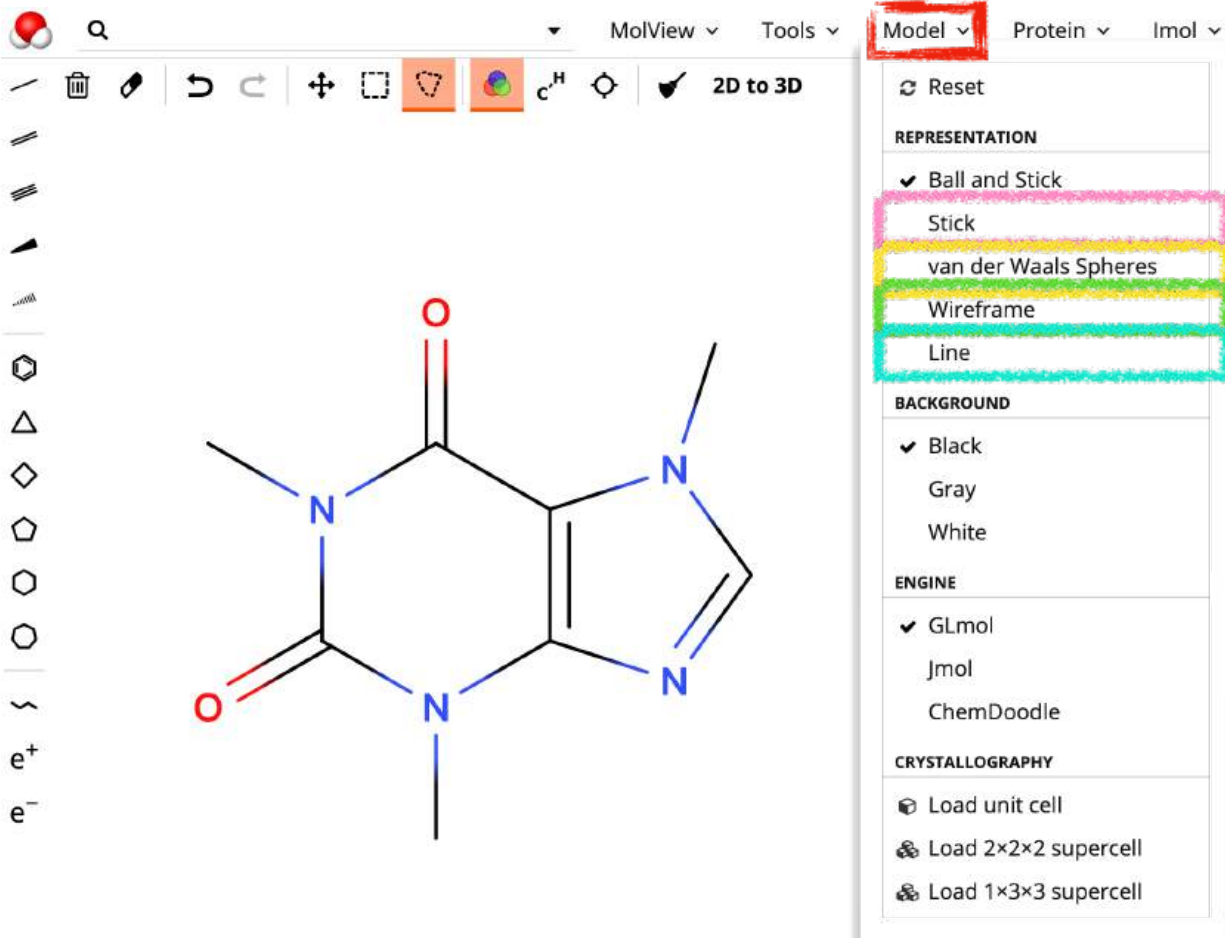
Caffeine

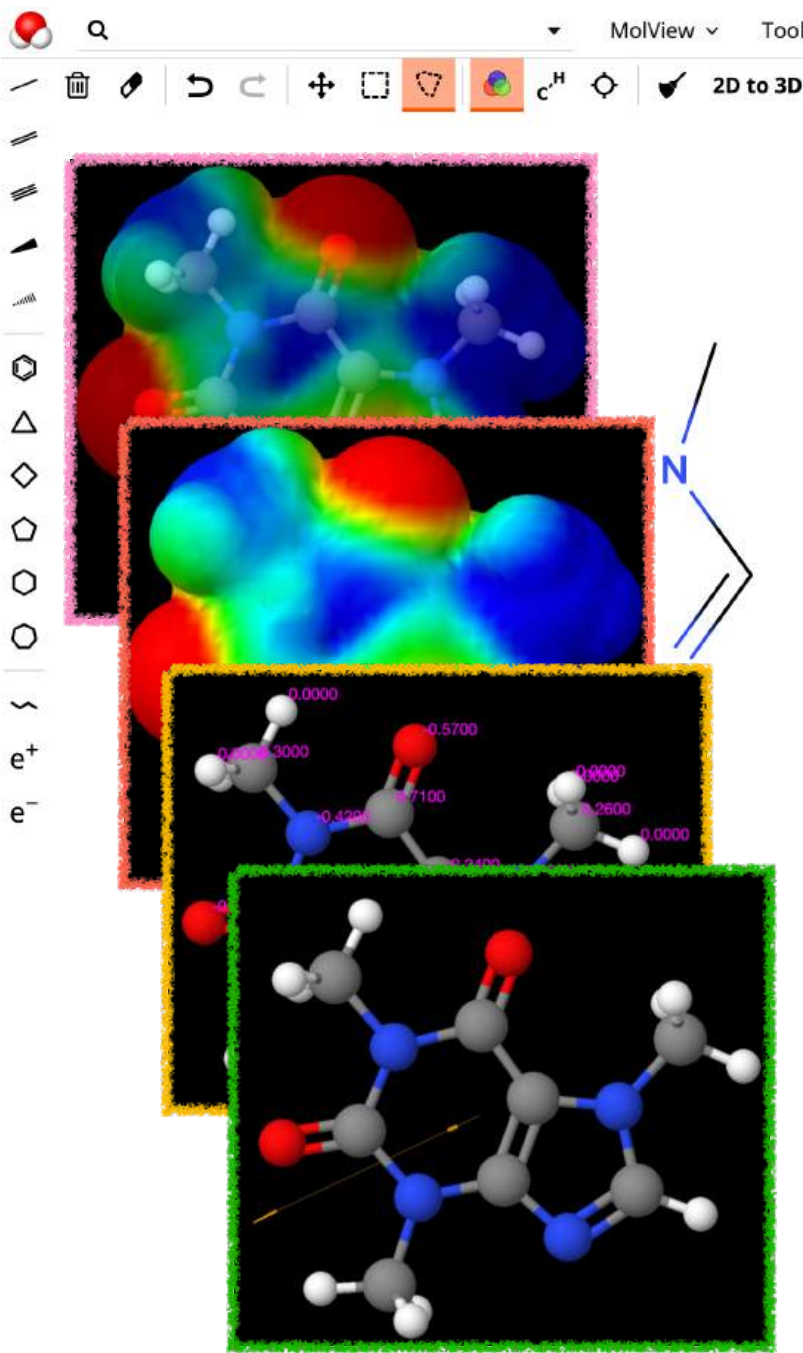
Caffeine is a trimethylxanthine in which the three methyl groups are located at positions 1, 3, and 7. A purine alkaloid that occurs naturally in tea and coffee. It has a role as a central nervous system stimulant, an EC 3.1.4.* (phosphoric diester hydrolase) inhibitor, an adenosine receptor antagonist, an EC 2.7.11.1 (non-specific serine/threonine protein kinase) inhibitor, a ryanodine receptor agonist, a fungal metabolite, an adenosine A2A receptor antagonist, a psychotropic drug, a diuretic, a food additive, an adjuvant, a plant metabolite, an environmental contaminant, a xenobiotic, a human blood serum metabolite, a mouse metabolite and a mutagen. It is a purine alkaloid and a trimethylxanthine.

Formula	C ₈ H ₁₀ N ₄ O ₂
Molecular weight	194.19 u
Hydrogen bond donors	0
Hydrogen bond acceptors	3

Percent composition

C	12.0107 u × 8	49.480 %
H	1.00794 u × 10	5.1905 %
N	14.0067 u × 4	28.851 %
O	15.9994 u × 2	16.478 %





C
H
N
O
P
S
F
Cl
Br
I
...

✓ High Quality

Clean

CALCULATIONS

MEP surface lucent

MEP surface opaque

Charge

Bond dipoles

Overall dipole

Energy minimization

MEASUREMENT

Distance

Angle

Torsion

結構些微調整至能量較低的結構
(移除原子間之立體衝突 steric clash)



Q



MolView



Tools



Model



Protein



Jmol



2D to 3D























































































































































































































































































































































MolView ▾

Tools ▾

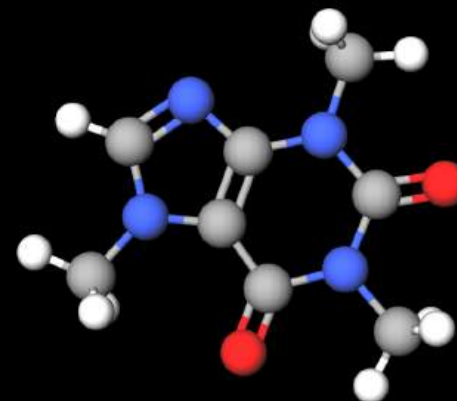
Model ▾

Protein ▾

Jmol ▾

Catechol	Compound
Catechin	Compound
Catecholborane	Compound
Catechuic acid	Compound
Catechinic acid	Compound
Catechol violet	Compound
Catechol diacetate	Compound
Catechol diethyl ether	Compound
Catechol dimethyl ether	Compound
Catechol Dimethylether-d6	Compound
Cahnite	Mineral
Calcite	Mineral
Caoxite	Mineral
Cavoite	Mineral
Catalase	Enzymes
Arcanite	Mineral
Cafetite	Mineral
Caminite	Mineral
Canasite	Mineral
Cattiite	Mineral
Cejkaite	Mineral
Cytochrome c	Energy Production
Collagen	Infrastructure

C
H
N
O
P
S
F
Cl
Br
I
...





Q Catechin



MolView



Tools



Model



Protein



Jmol

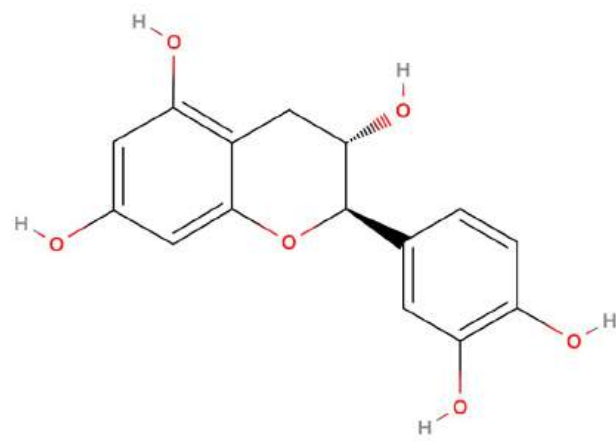


2D to 3D

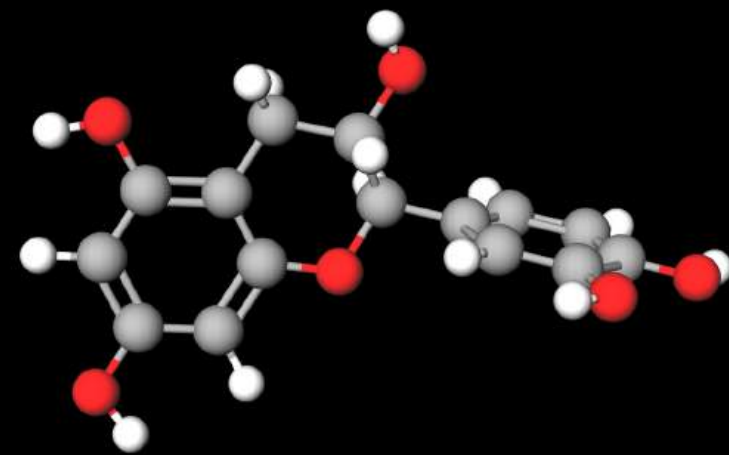


e⁺

e⁻



C
H
N
O
P
S
F
Cl
Br
I
...



Your Turn!

找尋日常生活會用到的小分子藥物



MolView

Tools

Model

Protein

Jmol



2D to 3D



e⁺

e⁻

- 搜尋PubChem compounds
- 名稱/分類/介紹
- 結構式 & 3D影像下載
- MOL格式
- 量測分子大小長度（近似即可）截圖或下載
- 將分子用surface表示後截圖或下載

PubChem Classification Browser

Browse PubChem data using a classification of interest, or search for PubChem records annotated with DNA repair). [More...](#)

Select classification

MeSH

Search selected classification by

Keyword

Enter desired search term

Classification description (from MeSH)

MeSH (Medical Subject Headings) is the NLM controlled vocabulary thesaurus used for indexing articles for PubMed.

Data type counts to display

None

Compound

Substance

PubMed

Display zero count nodes?

Yes

No

Browse MeSH Tree

MeSH Tree

130,583

Analytical, Diagnostic and Therapeutic Techniques and Equipment Category

Anatomy Category

Anthropology, Education, Sociology and Social Phenomena Category

Chemicals and Drugs Category

Amino Acids, Peptides, and Proteins

Biological Factors

Biomedical and Dental Materials

Carbohydrates

Chemical Actions and Uses

Complex Mixtures

Enzymes and Coenzymes

Heterocyclic Compounds

Hormones, Hormone Substitutes, and Hormone Antagonists

Inorganic Chemicals

Lipids

Macromolecular Substances

Nucleic Acids, Nucleotides, and Nucleosides

Organic Chemicals

Pharmaceutical Preparations

Polycyclic Compounds

Chemical structure of a phospholipid bilayer, showing the hydrophilic head groups (phosphate and choline) and the hydrophobic tails (fatty acid chains).

- Macrolide antifungal antibiotic complex produced by *Streptomyces noursei*, *S. aureus*, and other *Streptomyces* species. The biologically active components of the complex are nystatin A1, A2, and A3.

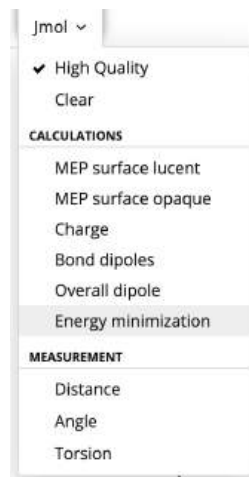
```

C47H75N017
APtclcactv10122200512D 0 0.00000 0.00000
140142 0 0 1 0 0 0 0 0999 V2000
12.3100 0.6569 0.0000 O 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
12.3100 1.6569 0.0000 C 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
12.1495 2.2558 0.0000 H 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
11.3100 1.6569 0.0000 C 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
10.7274 1.4449 0.0000 H 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
11.4177 1.0463 0.0000 H 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
10.8100 2.5230 0.0000 C 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
11.4300 2.5230 0.0000 H 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
9.8100 2.5230 0.0000 C 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
9.5000 1.9860 0.0000 H 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
9.3100 3.3890 0.0000 C 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
9.4177 3.9996 0.0000 H 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
8.3100 3.3890 0.0000 C 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
8.0000 2.8520 0.0000 H 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
7.8100 4.2550 0.0000 C 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
8.1200 4.7919 0.0000 H 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
6.8100 4.2550 0.0000 C 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
6.5000 4.7919 0.0000 H 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
6.3100 3.3890 0.0000 C 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
6.6200 2.8520 0.0000 H 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
5.3100 3.3890 0.0000 C 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
5.0000 3.9259 0.0000 H 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

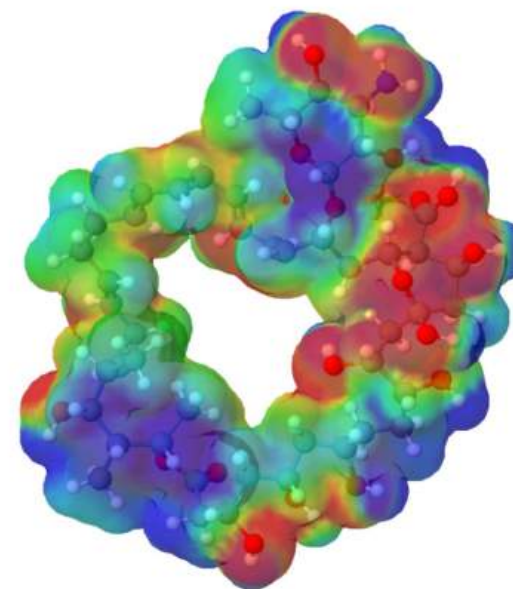
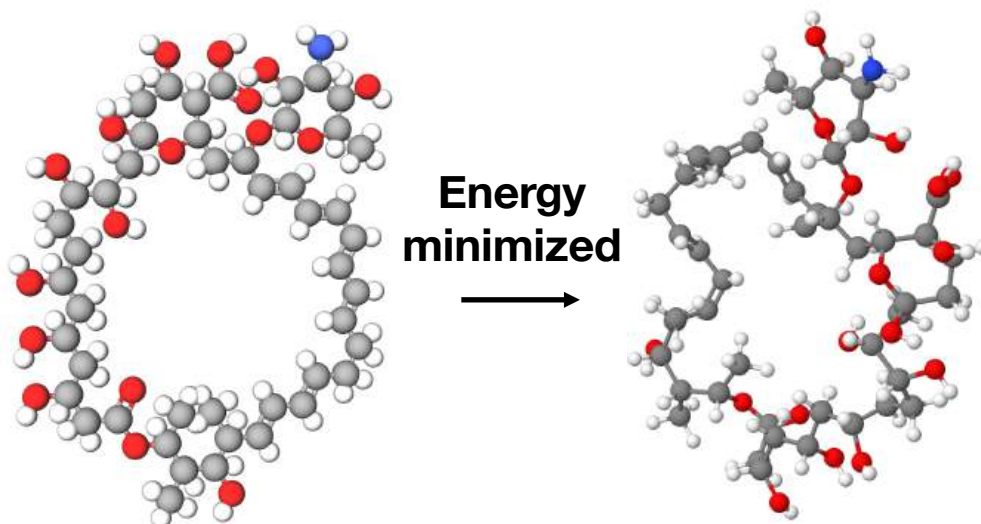
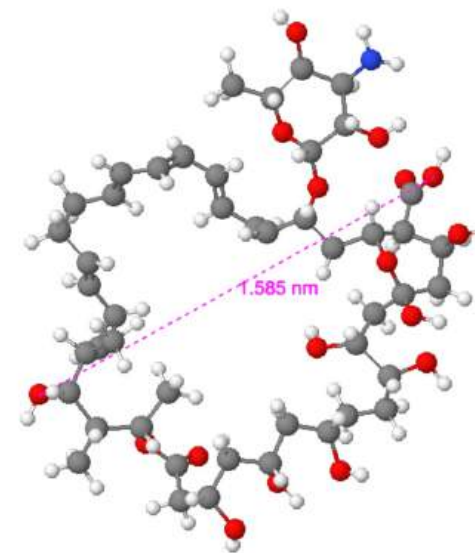
```

- 結構式 & 3D影像下載
- MOL格式
- 量測分子大小長度（近似即可）截圖或下載
- 將分子用**surface**表示後截圖或下載

Nystatine as an example



注意分子結構會動!

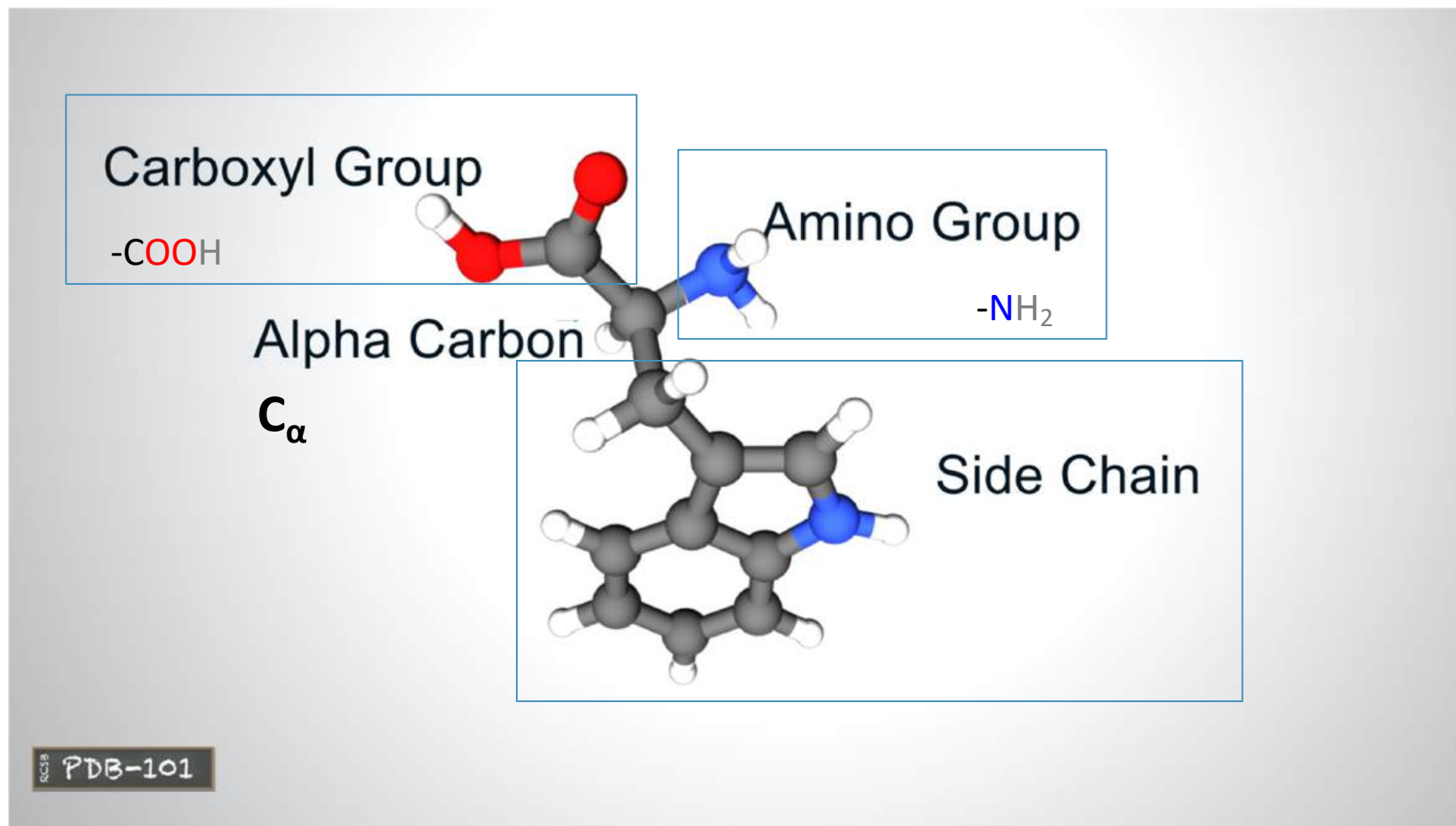


生活常見藥物

- Acetaminophen 乙醯胺酚 (普拿疼有效成分)
- Amphetamine 安非他命
- Methamphetamine 冰毒 --> 絕命毒師主要製毒
- Remdesivir 瑞德西韋 --> 老藥新用來治療Covid-19
- Biopterin 生物喋呤 ([link](#)) —> 缺乏會導致苯丙酮尿症([罕見疾病](#))
- Cephalosporin 抗生素([link](#)) —> <https://www.nejs.app/2018/10/cephalosporins.html>
- Aspirin 退燒
- Artemisinin 青蒿素 (屠呦呦發現, 獲2015諾貝爾生理醫學獎)
- Barbitol 巴比妥 (安眠藥 [link](#)。)
- Bilirubin 膽紅素 [link](#) (血紅素的主要代謝產物; 膽汁的主要色素; 肝功能的重要指標; [Can 'toxic' bilirubin treat a variety of illnesses?](#))
- Vancomycin (萬古黴素) --> 治療細菌感染
- Clindamycin (克林黴素) --> 治療細菌感染(痘痘、挫瘡)
- Tamsulosin (坦索羅辛) --> 受體拮抗劑 (調整泌尿到機能、使尿路平滑肌鬆弛, 幫助排出腎結石、治療前列腺肥大症)
- Flopropione --> 調整胰膽道、泌尿道、機能。緩解消化管平滑肌、胰膽道及尿路系統平滑肌收縮之痙攣作用

The Building Blocks of Proteins

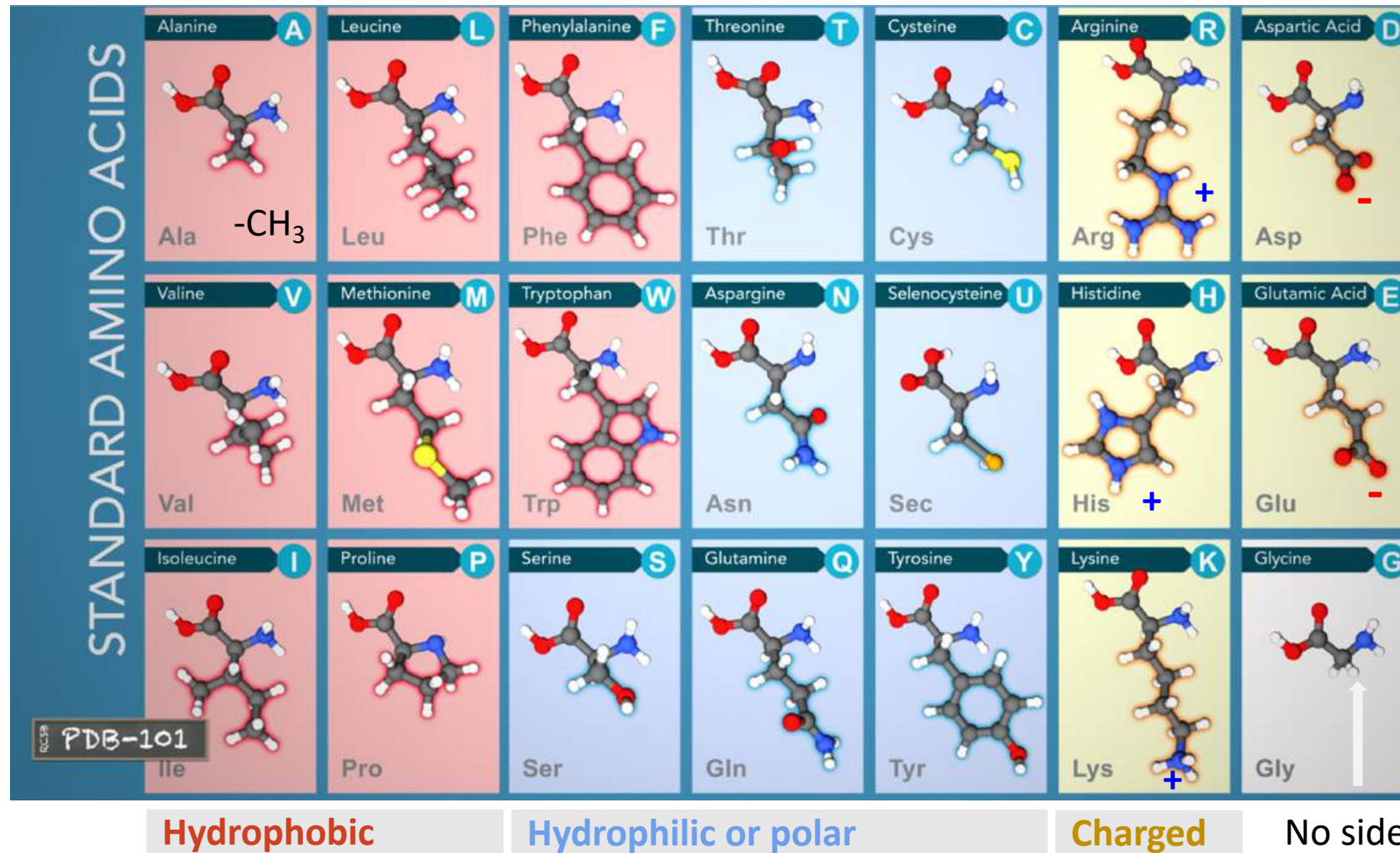
pdb101.rcsb.org



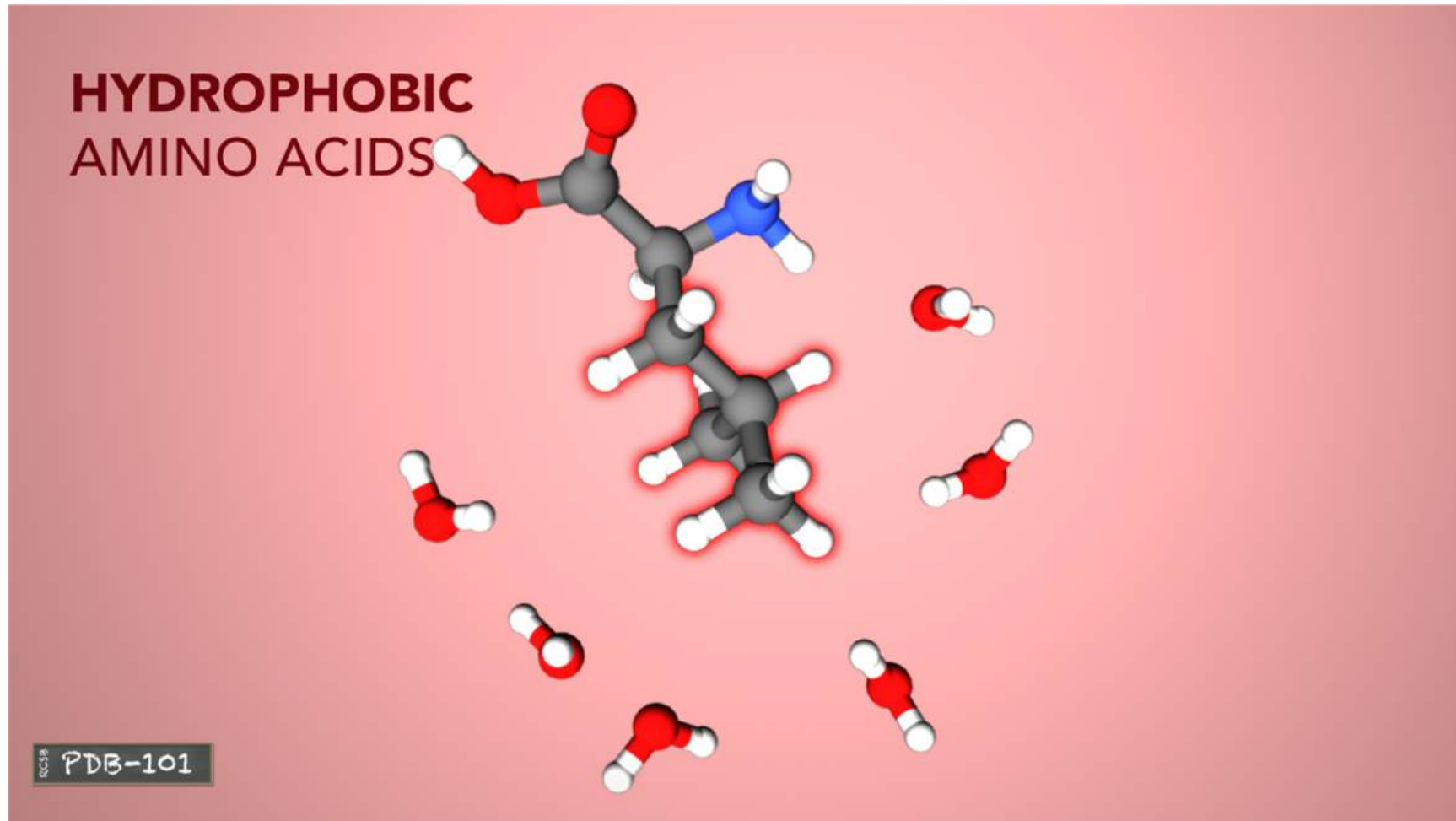
The atoms of each amino acid form an **amino group**, a **carboxyl group**, and a **side chain** attached to a central carbon atom.

The Building Blocks of Proteins

pdb101.rcsb.org



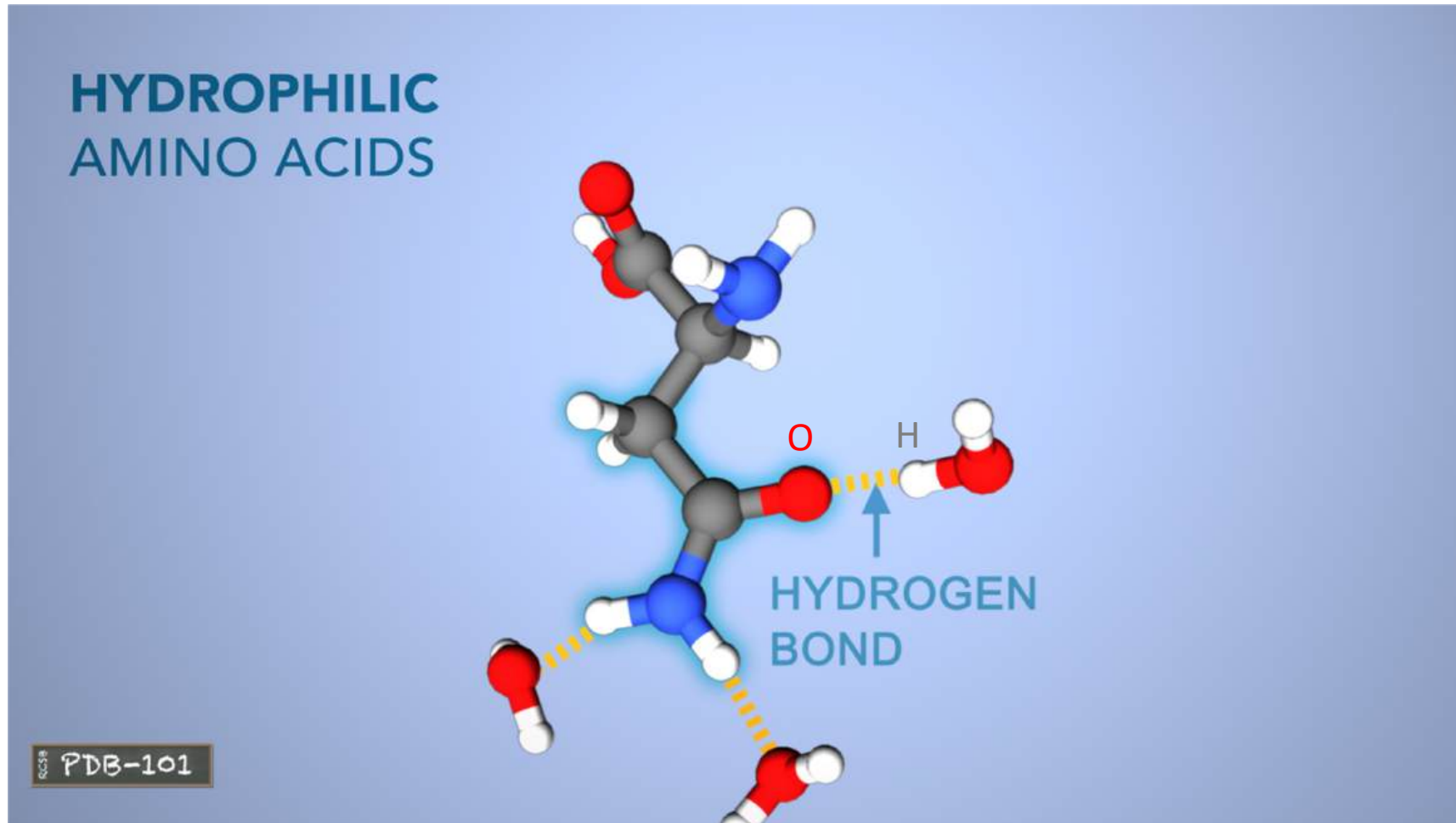
The side chain is the only part that varies from amino acid to amino acid and determines its chemical properties.



Hydrophobic amino acids have carbon-rich side chains, which don't interact well with water.

The Building Blocks of Proteins

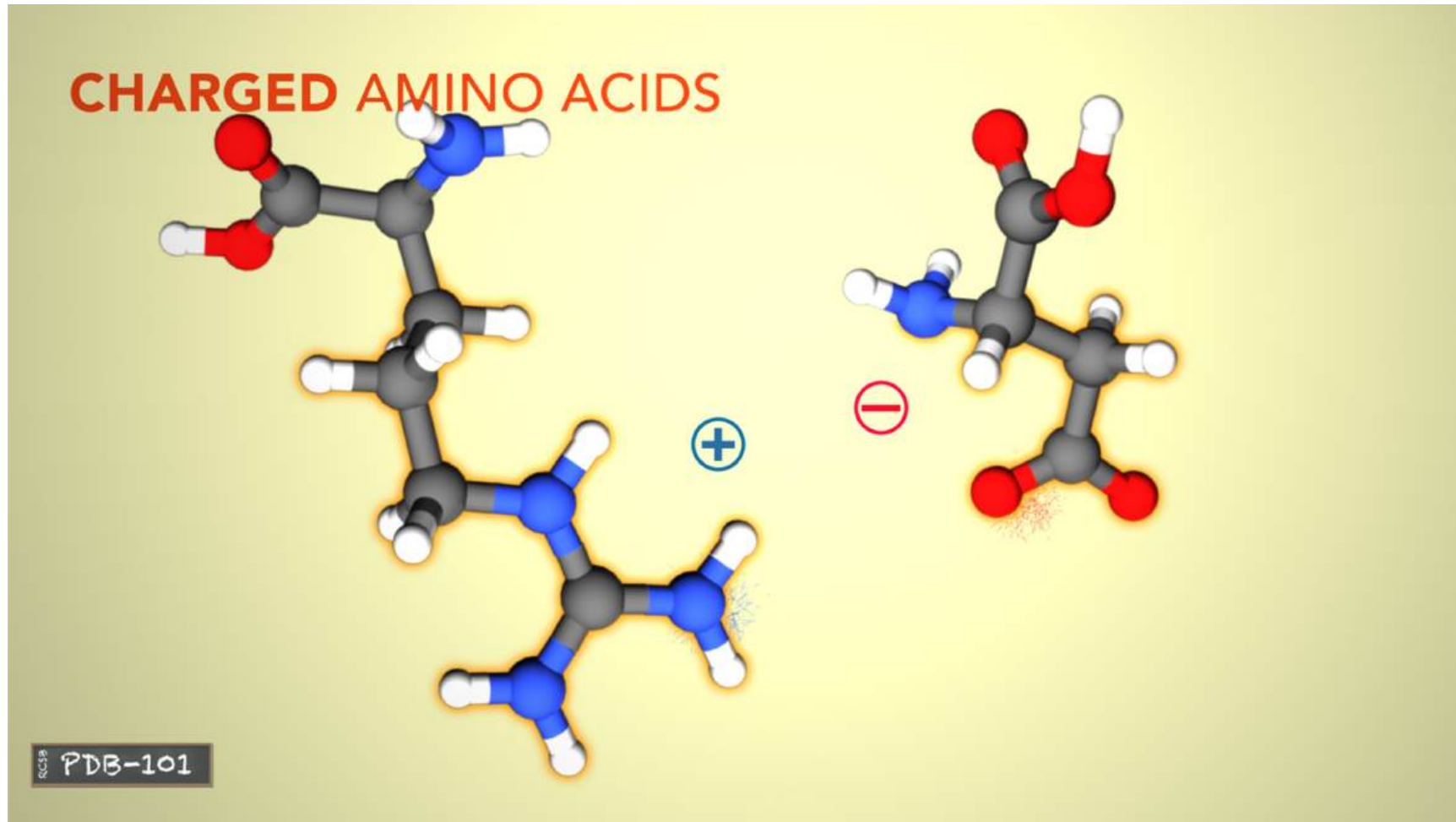
pdb101.rcsb.org



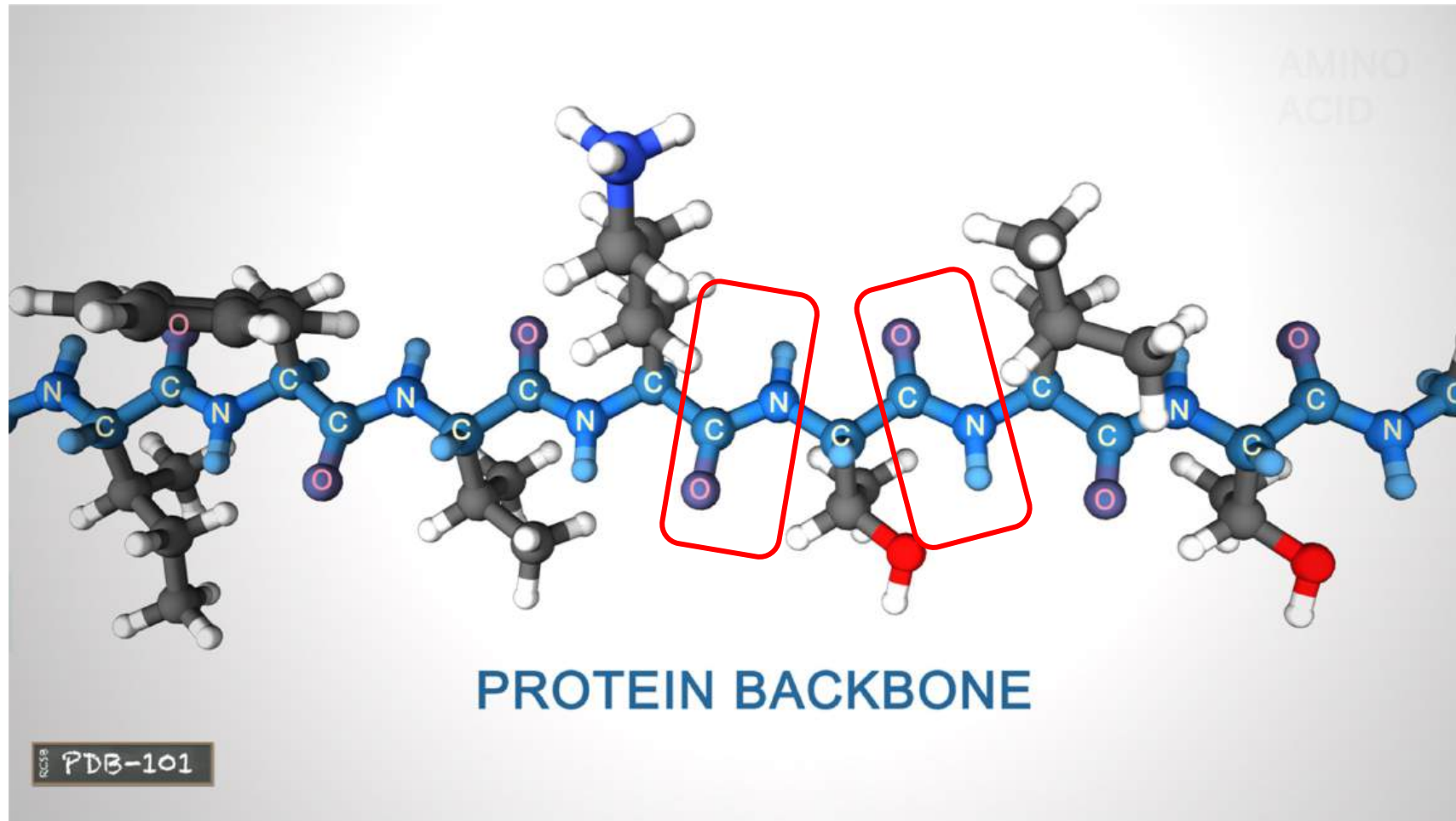
Hydrophilic or **polar** amino acids interact well with water.

The Building Blocks of Proteins

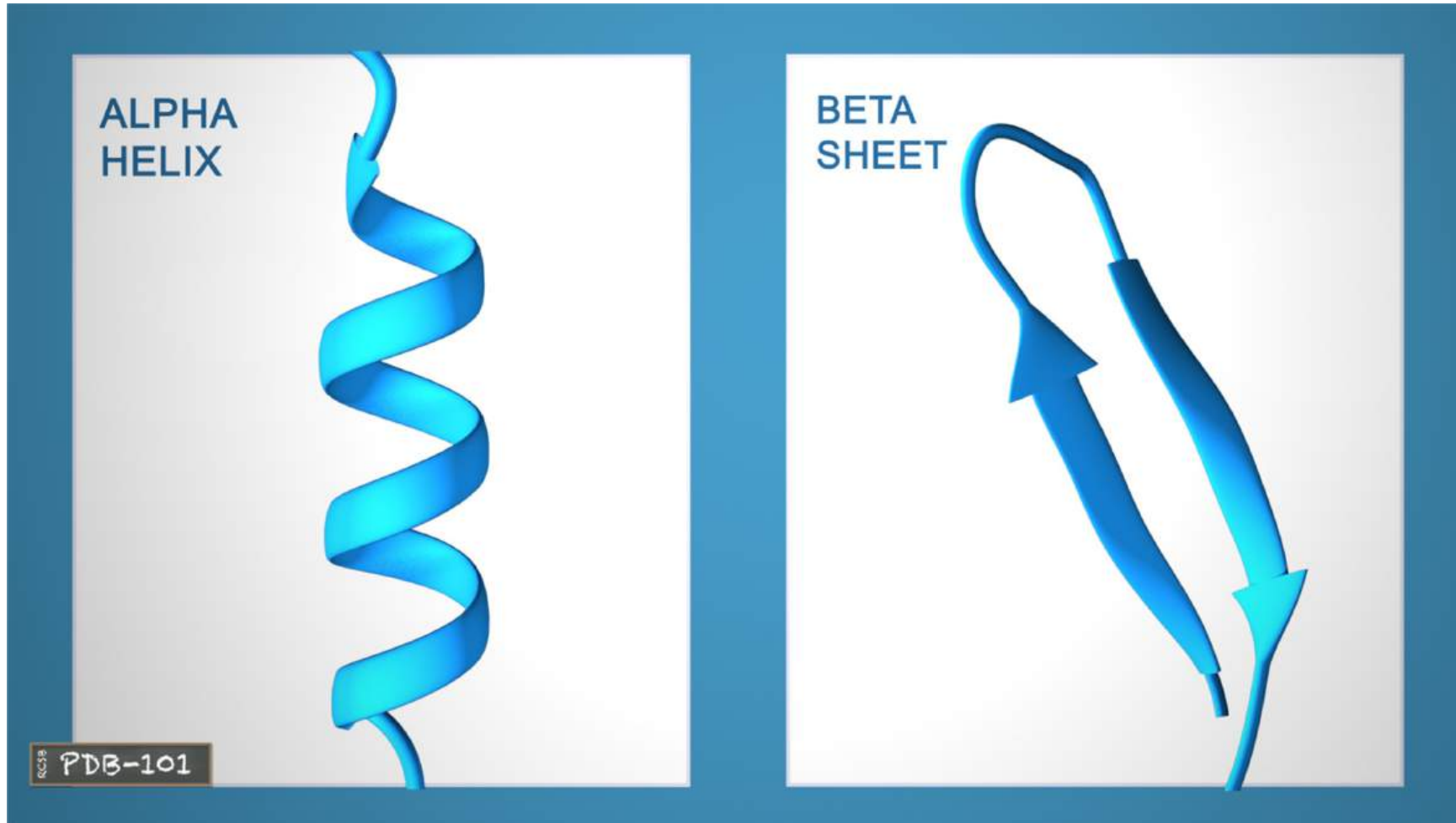
pdb101.rcsb.org



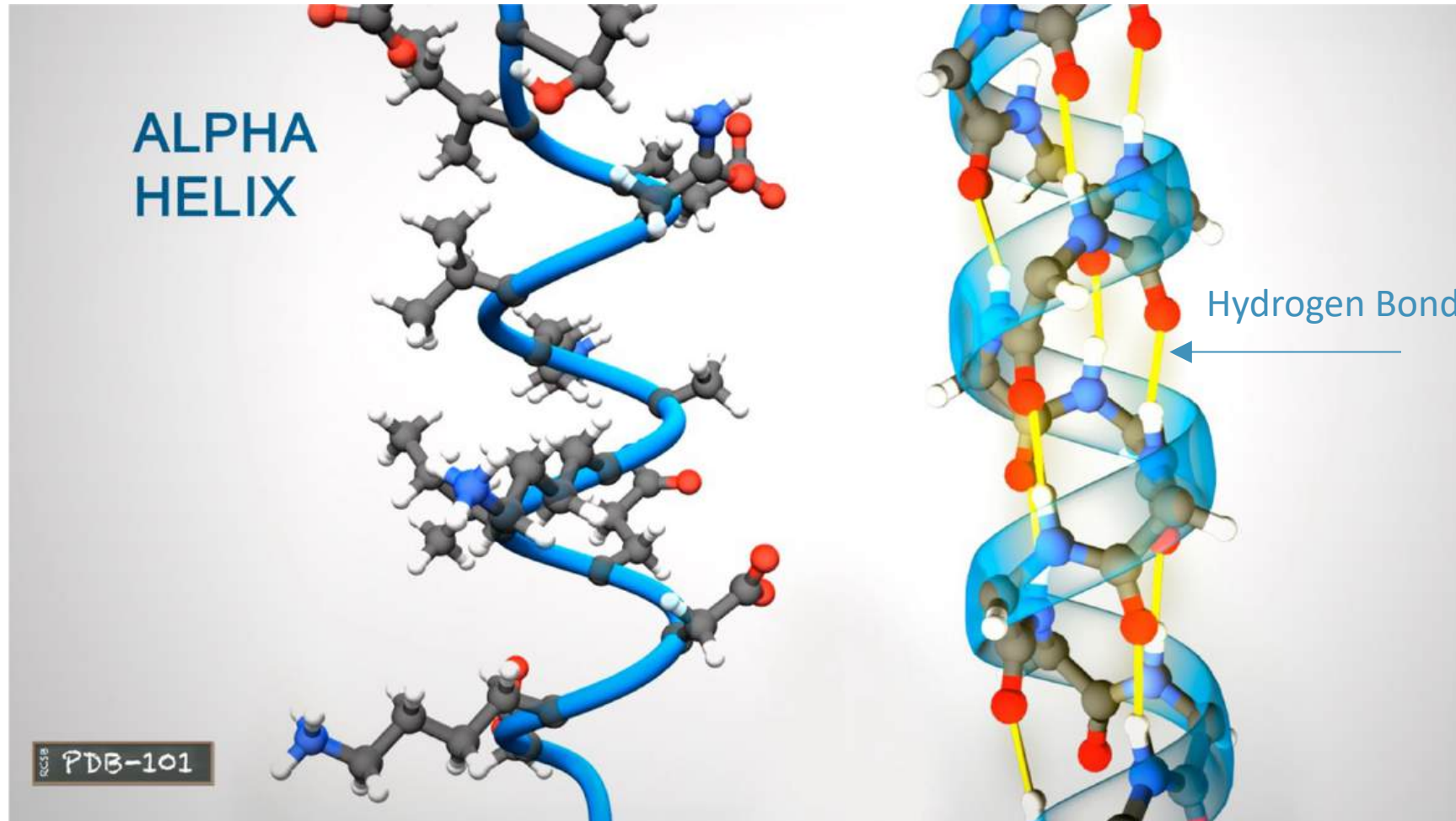
Charged amino acids interact with oppositely charged amino acids or other molecules.



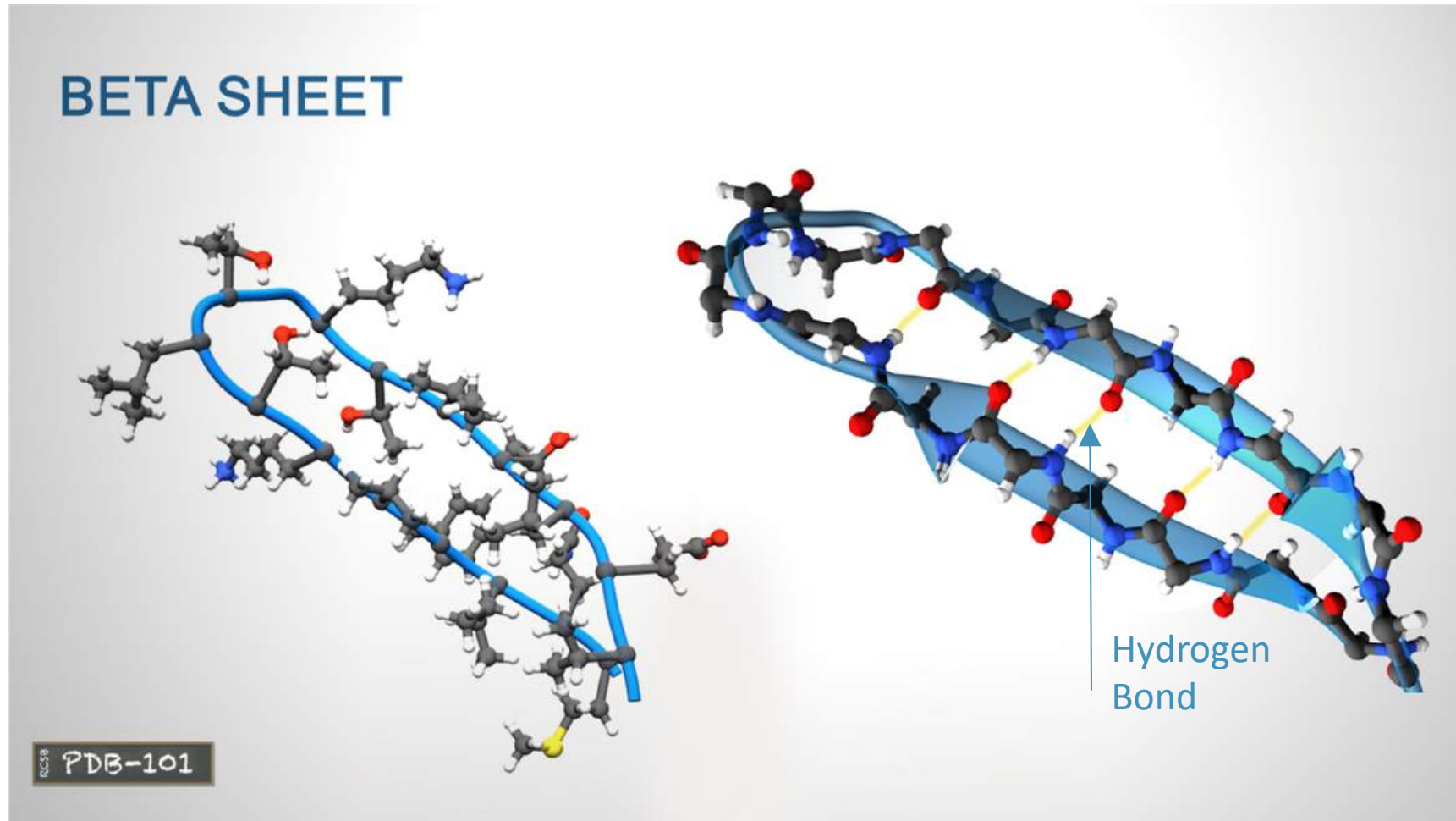
The linked series of carbon, nitrogen, and oxygen atoms make up the **protein backbone**.



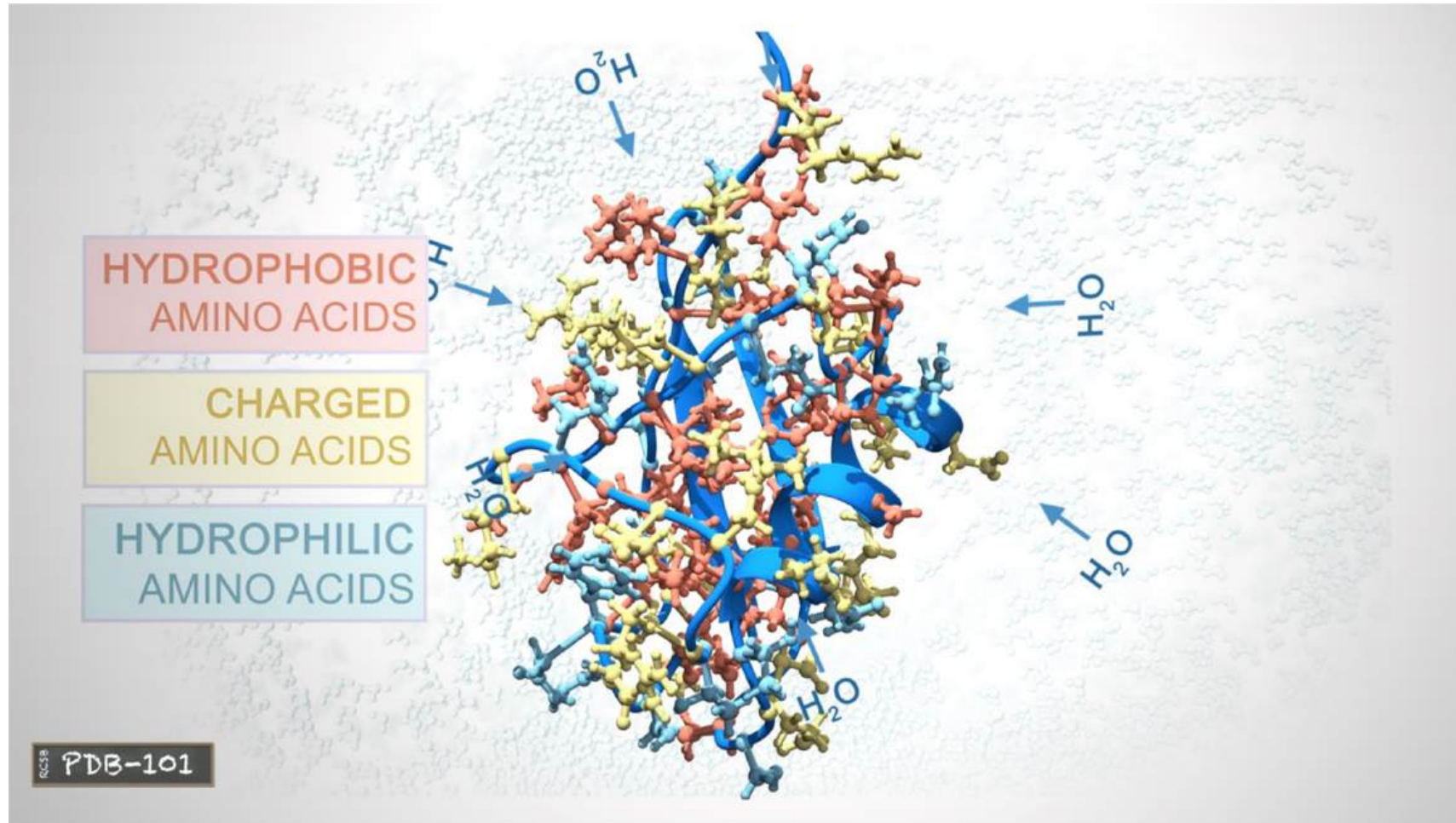
The protein chains often fold into two types of secondary structures: **alpha helices**, or **beta sheets**.



An alpha helix is a **right-handed coil** stabilized by **hydrogen bonds** between the amine and carboxyl groups of nearby amino acids



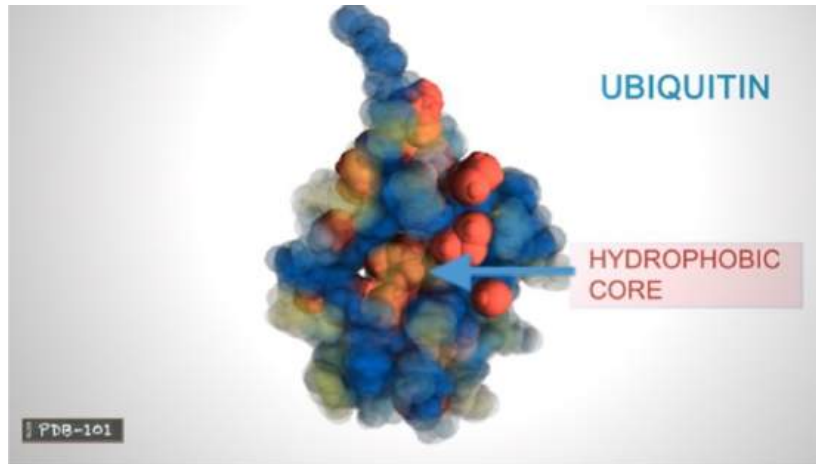
Beta-sheets are formed when hydrogen bonds stabilize **two or more adjacent strands**.



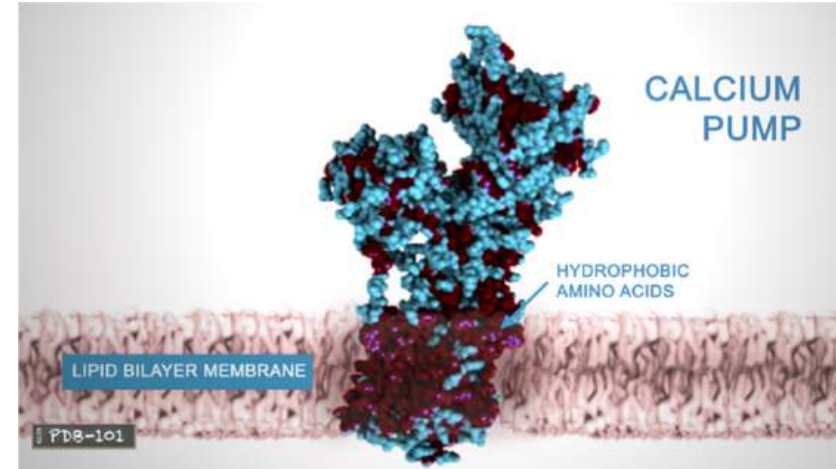
The tertiary structure of a protein is the **three-dimensional shape of the protein chain**. This shape is determined by the characteristics of the amino acids making up the chain.

Protein Structure: Tertiary Structure

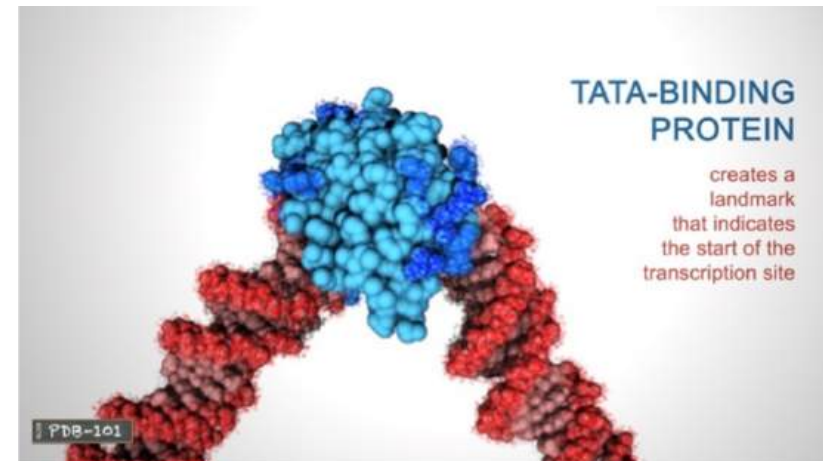
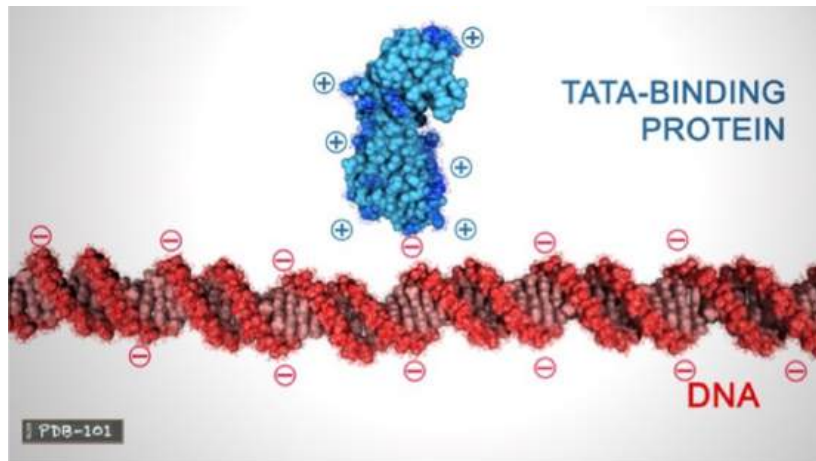
pdb101.rcsb.org



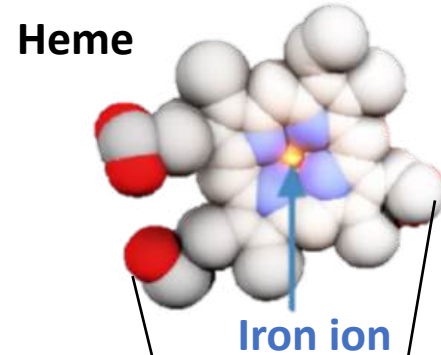
Many proteins form globular shapes with hydrophobic side chains sheltered inside, away from the surrounding water.



Membrane-bound proteins have hydrophobic residues clustered together on the outside, so that they can interact with the lipids in the membrane.

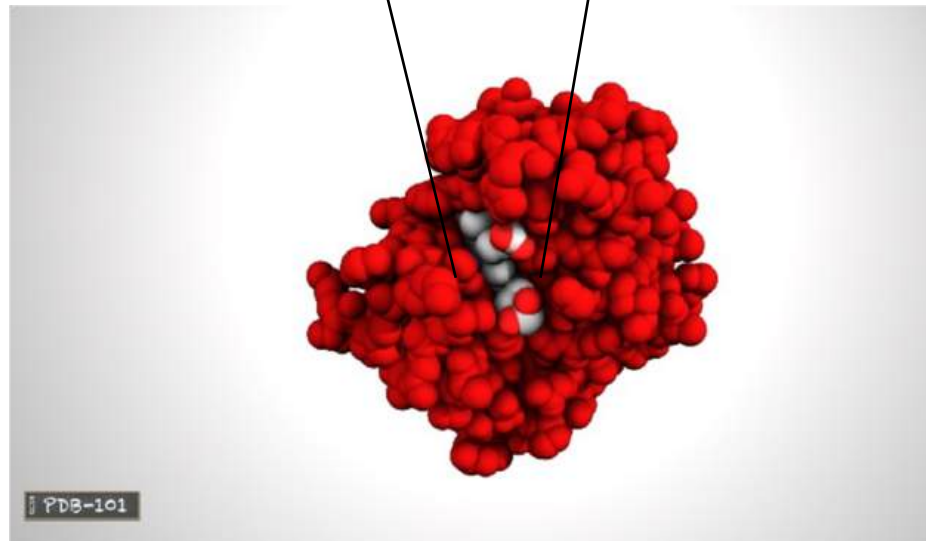


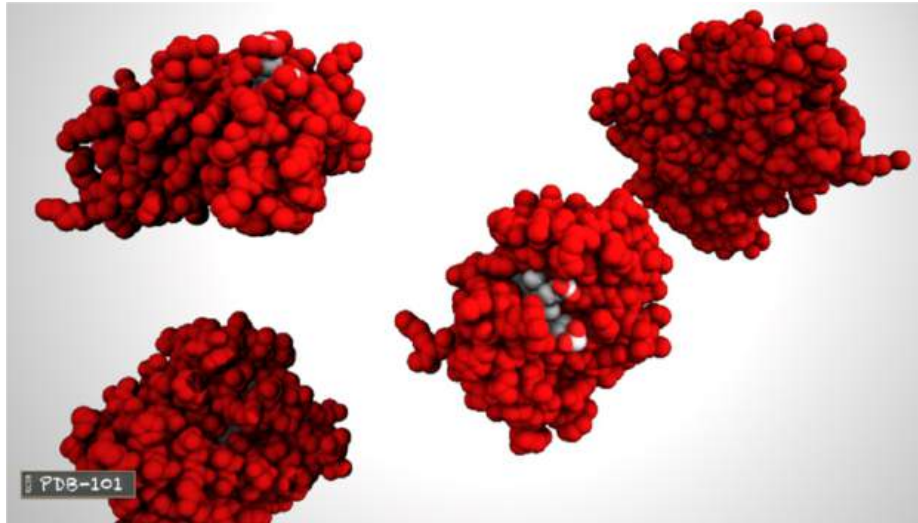
Charged amino acids allow proteins to interact with molecules that have complementary charges.



The functions of many proteins rely on their three-dimensional shapes.

For example, **hemoglobin** forms a pocket to hold **heme**, a small molecule with an **iron** atom in the center that binds oxygen.





Two or more polypeptide chains can come together to form one functional molecule with several subunits.

For example, the four subunits of hemoglobin cooperate so that the complex can pick up more oxygen in the lungs and release it in the body.



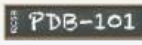
Protein Data Bank (PDB)



138464 Biological
Macromolecular Structures
Enabling Breakthroughs in
Research and Education

Go

[Advanced Search](#) | [Browse by Annotations](#)





Welcome

Deposit

Search

Visualize

Analyze

Download

Learn

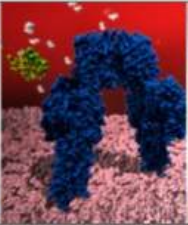
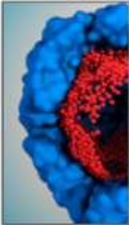
A Structural View of Biology

This resource is powered by the Protein Data Bank archive—information about the 3D shapes of proteins, nucleic acids, and complex assemblies that helps students and researchers understand all aspects of biomedicine and agriculture, from protein synthesis to health and disease.

As a member of the wwPDB, the RCSB PDB curates and annotates PDB data.

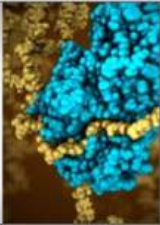
The RCSB PDB builds upon the data by creating tools and resources for research and education in molecular biology, structural biology, computational biology, and beyond.

New Video: What is a Protein?

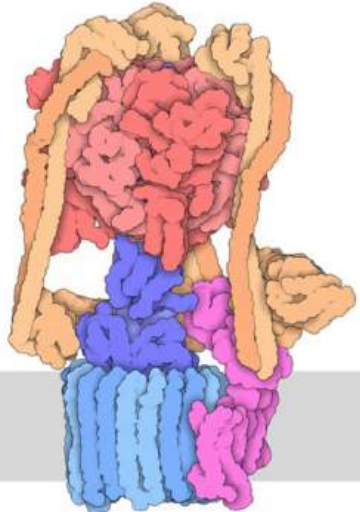




VIDEO
WHAT IS A
PROTEIN?



March Molecule of the Month



Vacuolar ATPase

Protein Data Bank (PDB)



138878 Biological Macromolecular Structures
Enabling Breakthroughs in Research and Education



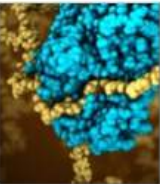
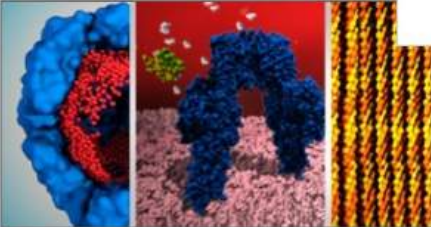
[Welcome](#)
[Deposit](#)
[Search](#)
[Visualize](#)
[Analyze](#)
[Download](#)
[Learn](#)

A Structural View of Bi

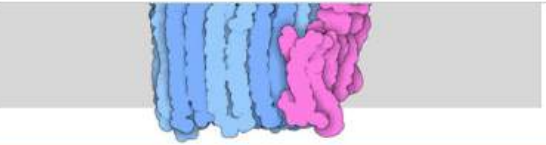
This resource is powered by the Protein Data Bank, which provides the 3D shapes of proteins, nucleic acids, and other macromolecules. This helps students and researchers understand all aspects of life, from protein synthesis to health and disease.

As a member of the wwPDB, the RCSB PDB builds upon the data by providing research and education in molecular biology, and beyond.

New Video: What is a Protein?



VIDEO
WHAT IS A PROTEIN?



Vacuolar ATPase

Go

UniProt Molecule Name

- Hemoglobin beta chain (330)
- Hemoglobin alpha chain (326)
- Hemoglobin subunit beta (325)
- Dimeric hemoglobin (64)
- Hemoglobin (19)
- Neural hemoglobin (17)

[More - Find all](#)

Structural Domains

- Hemoglobin... (245)
- Alpha-hemoglobin... (8)
- Hemoglobin... (237)
- Hemoglobin [SCOP] (1)
- Hemoglobin I [SCOP] (20)
- Protozoan... hemoglobin... (15)

[More](#)

Membrane Proteins

- Hbp (hemoglobin... (2)

Sequence Cluster Name

- GLOBIN LI637

[Find all](#)

Ontology Terms

- hemoglobin... (33)
- hemoglobin... (427)
- hemoglobin... (265)
- AT : HEMOGLOBIN... (2)
- D12.776.124.400.405: Hemog... [... (36)
- hemoglobin... (4)

[More](#)

Protein Data Bank (PDB)

Search Parameter:

Text Search for: hemoglobin

[Refine Search](#)

[Save Search to MyPDB](#)

Refinements

ORGANISM

[Homo sapiens \(270\)](#)
[Scapharca inaequivalvis \(66\)](#)
[Amphitrite ornata \(26\)](#)
[Equus caballus \(18\)](#)
[Lupinus luteus \(17\)](#)
[Cerebratulus lacteus \(17\)](#)
[Physeter catodon \(14\)](#)
[Other \(282\)](#)

UNIPROT MOLECULE NAME

[Hemoglobin subunit alpha \(326\)](#)
[Hemoglobin subunit beta \(325\)](#)
[Globin-1 \(64\)](#)
[Dehaloperoxidase A \(24\)](#)
[Leghemoglobin-2 \(17\)](#)
[Neural hemoglobin \(17\)](#)
[Hemoglobin subunit alpha-A \(16\)](#)
[Refine Query](#)

Currently showing 1 - 25 of 710

Page: 1 of 29

[← Previous](#) [Next →](#)

Displaying 25 Results

View:

Detailed

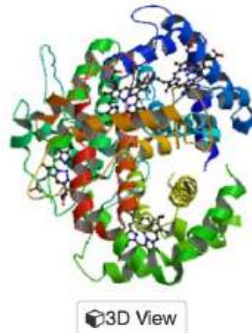
Reports:

Select a Report

Sort:

↓ Match score: Higher to Lower

[Download Files](#)



1FN3

[Download File](#)

[View File](#)



CRYSTAL STRUCTURE OF NICKEL RECONSTITUTED HEMOGLOBIN-A CASE FOR PERMANENT, T-STATE HEMOGLOBIN

[Venkatesh Rao, S.](#), [Deepthi, S.](#), [Pattabhi, V.](#), [Manoharan, P.T.](#)

PubMed ID is not available.

Released: 10/7/2003

Method: X-ray Diffraction

Resolution: 2.48 Å

Residue Count: 574

Macromolecule:

HEMOGLOBIN ALPHA CHAIN (protein)

HEMOGLOBIN BETA CHAIN (protein)

Unique Ligands: HNI

Search term match score: 268.69

Matched fields in 1FN3.cif:

•
•
•

Protein Data Bank (PDB)

[Structure Summary](#) [3D View](#) [Annotations](#) [Sequence](#) [Sequence Similarity](#) [Structure Similarity](#) [Experiment](#)

Biological Assembly 1 ?

1FN3

CRYSTAL STRUCTURE OF NICKEL RECONSTITUTED HEMOGLOBIN-A CASE FOR PERMANENT, T-STATE HEMOGLOBIN

DOI: [10.2210/pdb1FN3/pdb](https://doi.org/10.2210/pdb1FN3/pdb)

Classification: [OXYGEN STORAGE/TRANSPORT](#)

Organism(s): [Homo sapiens](#)

Deposited: 2000-08-20 Released: 2003-10-07

Deposition Author(s): [Venkatesh Rao, S.](#), [Deepthi, S.](#), [Pattabhi, V.](#), [Manoharan, P.T.](#)

Experimental Data Snapshot

Method: X-RAY DIFFRACTION
Resolution: 2.48 Å
R-Value Free: 0.324
R-Value Work: 0.242

wwPDB Validation

3D Report

Full Report

Metric	Percentile Ranks	Value
Clashscore		66
Ramachandran outliers		5.5%
Sidechain outliers		35.7%
RSRZ outliers		15.3%

Worse Better
■ Percentile relative to all X-ray structures
□ Percentile relative to X-ray structures of similar resolution

Literature

Download Primary Citation

Crystal Structure of Nickel Reconstituted Hemoglobin - A Case for Permanent, T-State Hemoglobin

[Venkatesh Rao, S.](#), [Deepthi, S.](#), [Pattabhi, V.](#), [Manoharan, P.T.](#)

(2003) CURR.SCI. **84**: 179-187

3D View: **Structure** | Electron Density | Ligand Interaction

Standalone Viewers
[Protein Workshop](#) | [Ligand Explorer](#)

Protein Data Bank (PDB)

Structure Summary **3D View** Annotations Sequence Sequence Similarity Structure Similarity Experiment

1FN3

CRYSTAL STRUCTURE OF NICKEL RECONSTITUTED HEMOGLOBIN-A CASE FOR PERMANENT, T-STATE

Note: Use your mouse to drag, rotate, and zoom in and out of the structure. Mouse-over to identify atoms and bonds. [Mouse controls documentation](#).

Sequence format

Display Files Download Files

- FASTA Sequence
- PDB Format**
PDB Format (gz)
- PDBx/mmCIF Format
PDBx/mmCIF Format (gz)
- PDBML/XML Format (gz)
- Biological Assembly 1
- Structure Factors (CIF)
Structure Factors (CIF - gz)
- 2fo-fc Map (DSN6)
fo-fc Map (DSN6)

☐ Water ☒ Ions
☒ Hydrogens ☐ Clashes

Default Structure View

HEADER OXYGEN STORAGE/TRANSPORT 20-AUG-00 1FN3
 TITLE CRYSTAL STRUCTURE OF NICKEL RECONSTITUTED HEMOGLOBIN-A CASE FOR
 TITLE 2 PERMANENT, T-STATE HEMOGLOBIN
 .
 .
 .
 SOURCE MOL_ID: 1;
 SOURCE 2 ORGANISM_SCIENTIFIC: HOMO SAPIENS;
 SOURCE 3 ORGANISM_COMMON: HUMAN;
 SOURCE 4 ORGANISM_TAXID: 9606;
 .
 .
 KEYWDS PERMANENT T-STATE, METAL ION COORDINATION, SUBUNIT INEQUIVALENCE,
 KEYWDS 2 SPECTROSCOPY, CRYSTALLOGRAPHY., OXYGEN STORAGE-TRANSPORT COMPLEX
 EXPDTA X-RAY DIFFRACTION
 AUTHOR S.VENKATESHRAO,S.DEEPTHI,V.PATTABHI,P.T.MANOHARAN
 .
 .
 REMARK 2 RESOLUTION. 2.48 ANGSTROMS
 .
 .

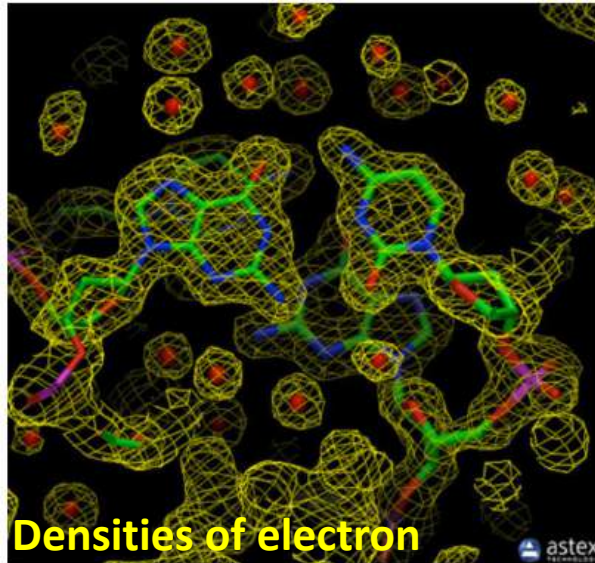
ATOM	1	N	VAL	A	1	42.219	19.998	3.819	1.00	28.12	N
ATOM	2	CA	VAL	A	1	43.152	19.932	2.662	1.00	30.16	C
ATOM	3	C	VAL	A	1	42.742	20.917	1.574	1.00	26.03	C
ATOM	4	O	VAL	A	1	41.698	20.768	0.947	1.00	29.47	O
ATOM	5	CB	VAL	A	1	43.197	18.504	2.062	1.00	33.08	C
ATOM	6	CG1	VAL	A	1	41.808	18.088	1.572	1.00	36.44	C
ATOM	7	CG2	VAL	A	1	44.230	18.448	0.938	1.00	34.80	C
ATOM	8	N	LEU	A	2	43.570	21.931	1.360	1.00	23.04	N
ATOM	9	CA	LEU	A	2	43.289	22.947	0.349	1.00	17.29	C
ATOM	10	C	LEU	A	2	44.042	22.680	-0.952	1.00	12.46	C

Atomic coordinates

Methods for Determining Atomic Structures

X-ray Crystallography

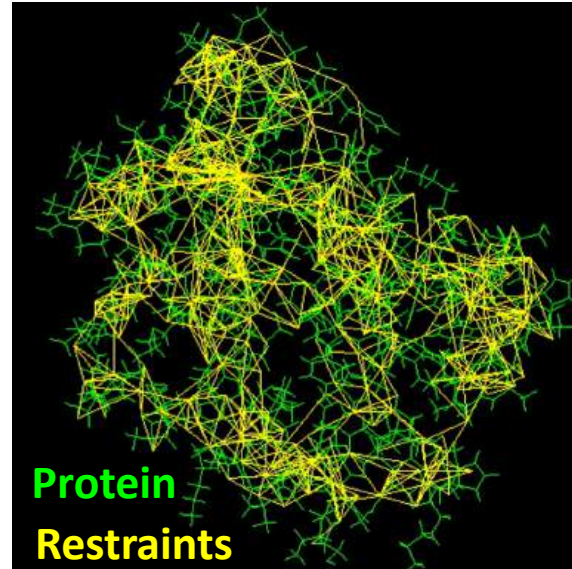
diffraction pattern



- 需要純化與結晶
- 只能解析重原子
- PDB 資料最多!
- 誤差 1-2埃

NMR Spectroscopy

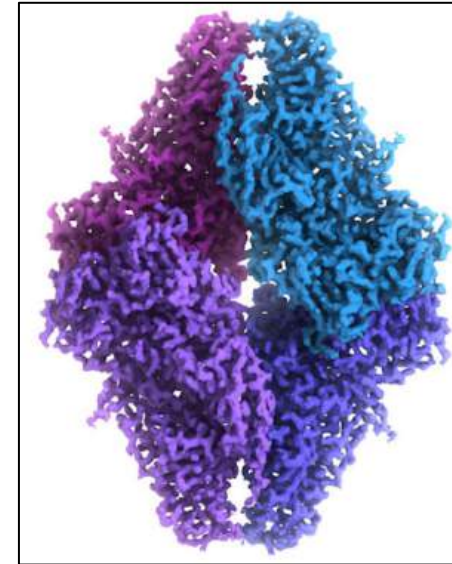
distance between atoms



- 溶液或固態 (無需結晶)
- 多種結構(系綜)
- 誤差2-4埃

cryo-Electron Microscopy

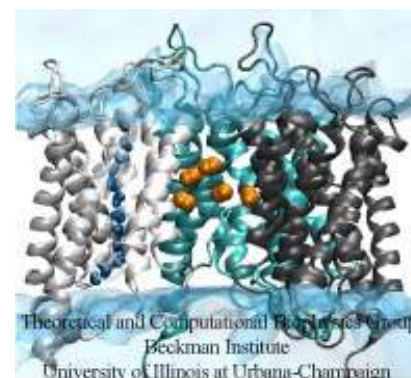
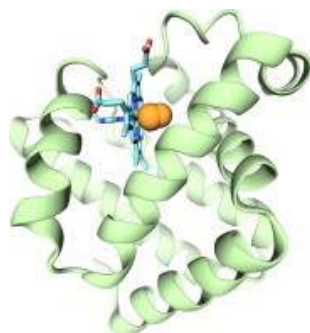
image of the overall shape



- 專解巨型大分子
- 解析度稍低
- 多種方法混搭使用能完善解析度
- 誤差2-9埃

<http://pdb101.rcsb.org>

VMD is designed for **visualization**, **analysis**, and modeling of biological systems such as proteins, nucleic acids, lipid bilayer assemblies, etc. VMD can read standard Protein Data Bank (**PDB**) files and display the contained structure.



- Support all major computer platforms (ex. MacOS X, Windows, Linux)
- Many molecular rendering and coloring methods

新世代 **VMD 2** 正推出中 … (先觀望…)

A New Era in Molecular Visualization

After nearly 30 years of evolution, VMD 2.0 is here with a brand-new user interface, making molecular visualization more intuitive and powerful than ever. With enhanced GPU acceleration, streamlined workflows, and advanced analysis tools, VMD 2.0 is designed to push the boundaries of molecular modeling and simulation.



Visual Molecular Dynamics

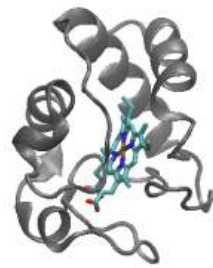
Quick Guide

Download VMD 2

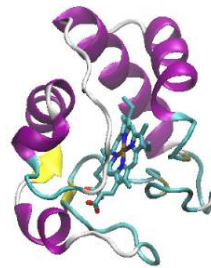
<https://www.ks.uiuc.edu/Research/vmd/vmd2intro/index.html>

What Can We Do with VMD?

- Visualize the structures that have been solved and reported

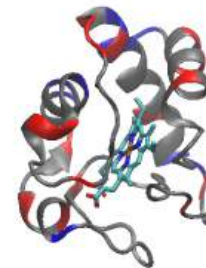


Cartoon
diagram



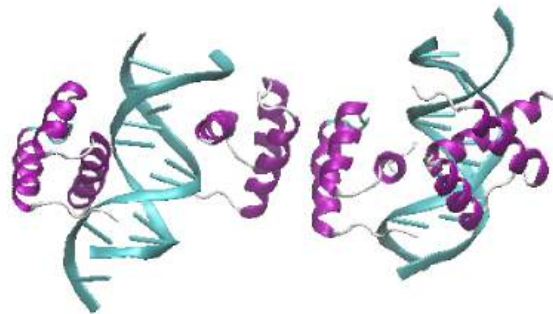
Secondary structure

PDB: 1hrc



Positively charged
Negatively charged

蛋白質 + 小分子



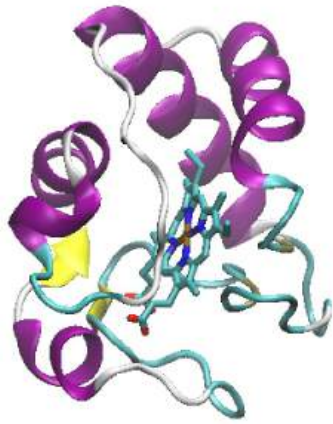
Multiple protein assembly

PDB: 4qtr

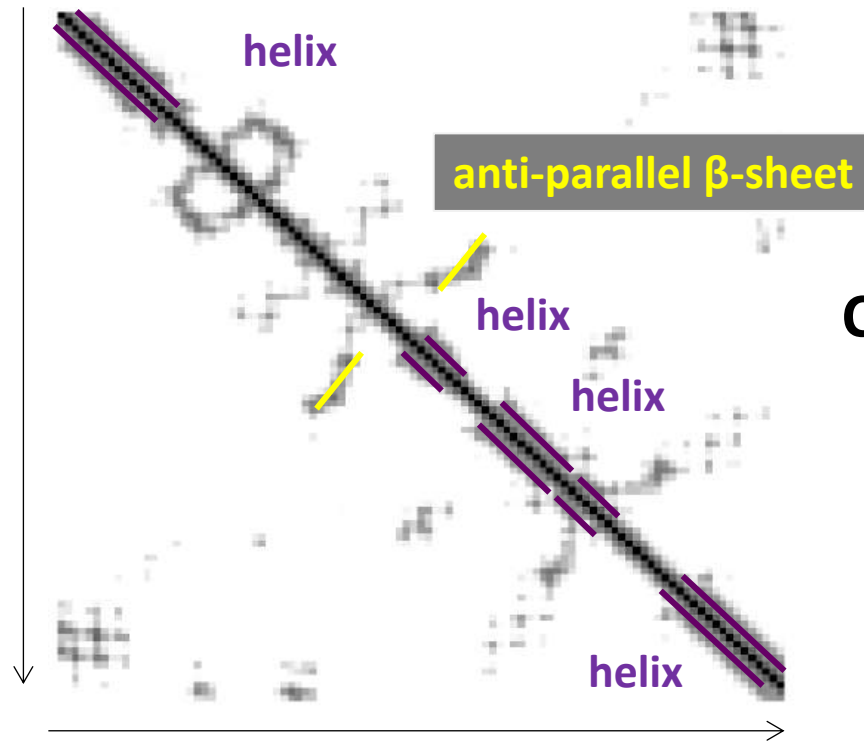
蛋白質 + 核酸
(複雜組裝)

What Can We Do with VMD?

- Structural analysis (Contact map)



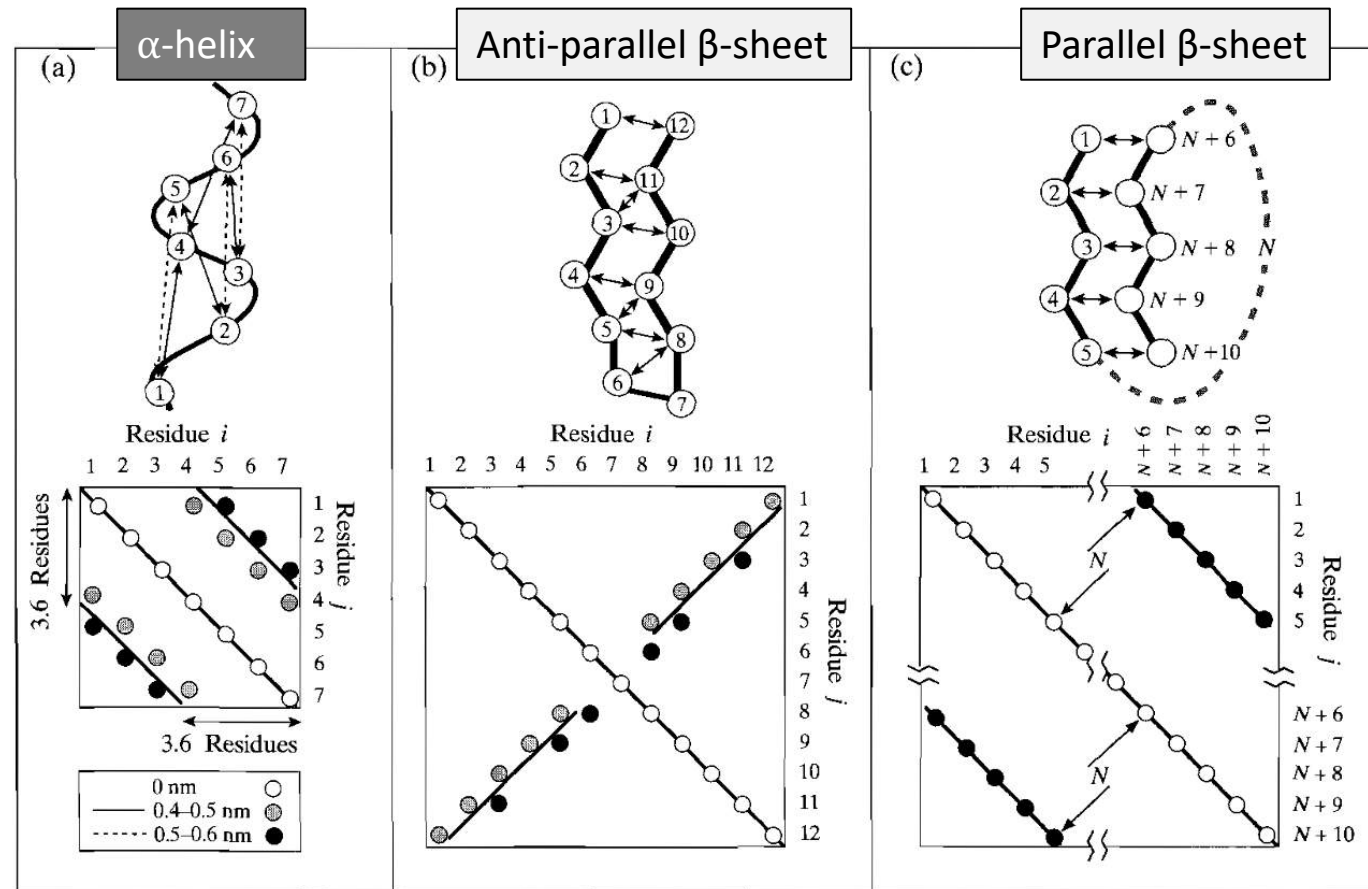
PDB: 1hrc



Contact map

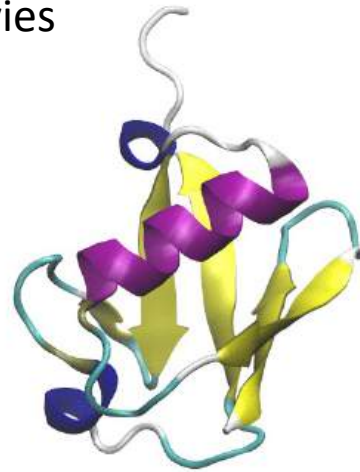
What Can We Do with VMD?

- Structural analysis (Contact plots for α -helix and β -sheets)

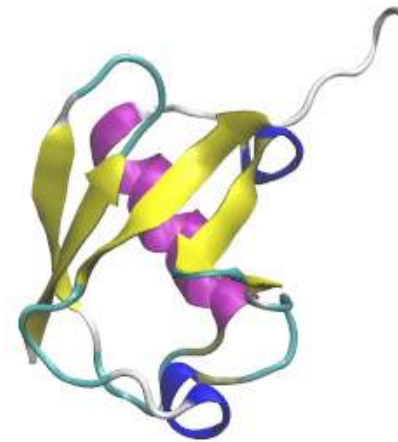


What Can We Do with VMD?

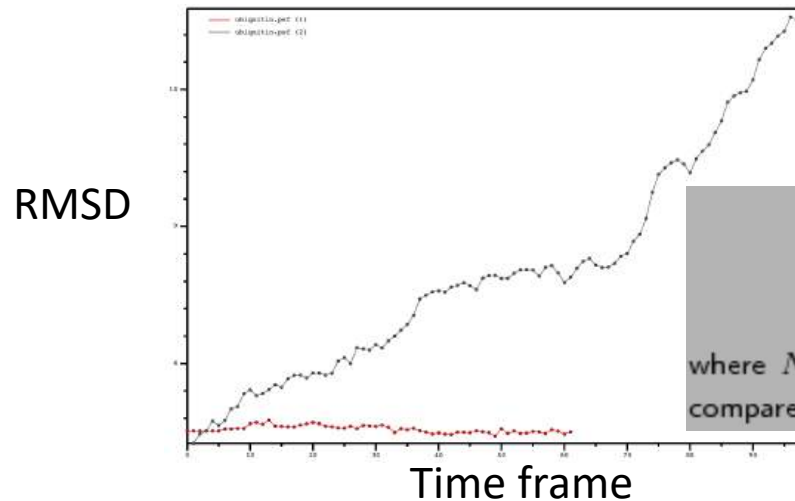
- Make movies



PDB: 1ubq



- Trajectory analysis



Root Mean Square Deviation

$$RMSD = \sqrt{\frac{\sum_{i=1}^{N_{atoms}} (r_i(t_1) - r_i(t_2))^2}{N_{atoms}}} \quad (1)$$

where N_{atoms} is the number of atoms whose positions are being compared, and $r_i(t)$ is the position of atom i at time t .

Now You Can Try to Launch VMD

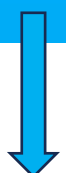
桌面
(Desktop)



vmd

進入113.2 MD workshop\module2 資料夾，將 **vmd** 移至桌面 (Desktop)

See *VMD-Installation-Guide.pdf* if you want to install it on your computer



名稱	修改日期
output	2024/8/25
1ubq	2021/8/16
4qtr	2021/8/16
par_all36_prot.prm	2021/3/12
VMD-Installation-Guide	2018/3/15

1. 主視窗

3. 圖形表示視窗

Selected molecule

0: ribosome.psf

Create Rep Delete Rep

Style	Color	Selection
NewCartoon	ColorID 15	segname 16
NewCartoon	ColorID 3	segname 2
NewCartoon	ColorID 14	segname 5

Selected Atoms

segname 16

Draw style: Selections Trajectory Periodic

Coloring Method: ColorID 15 Material: Opaque

Drawing Method: NewCartoon Default

Spline Style: Catmull-Rom

Aspect Ratio: 4:10

Thickness: 0.20

Resolution: 6

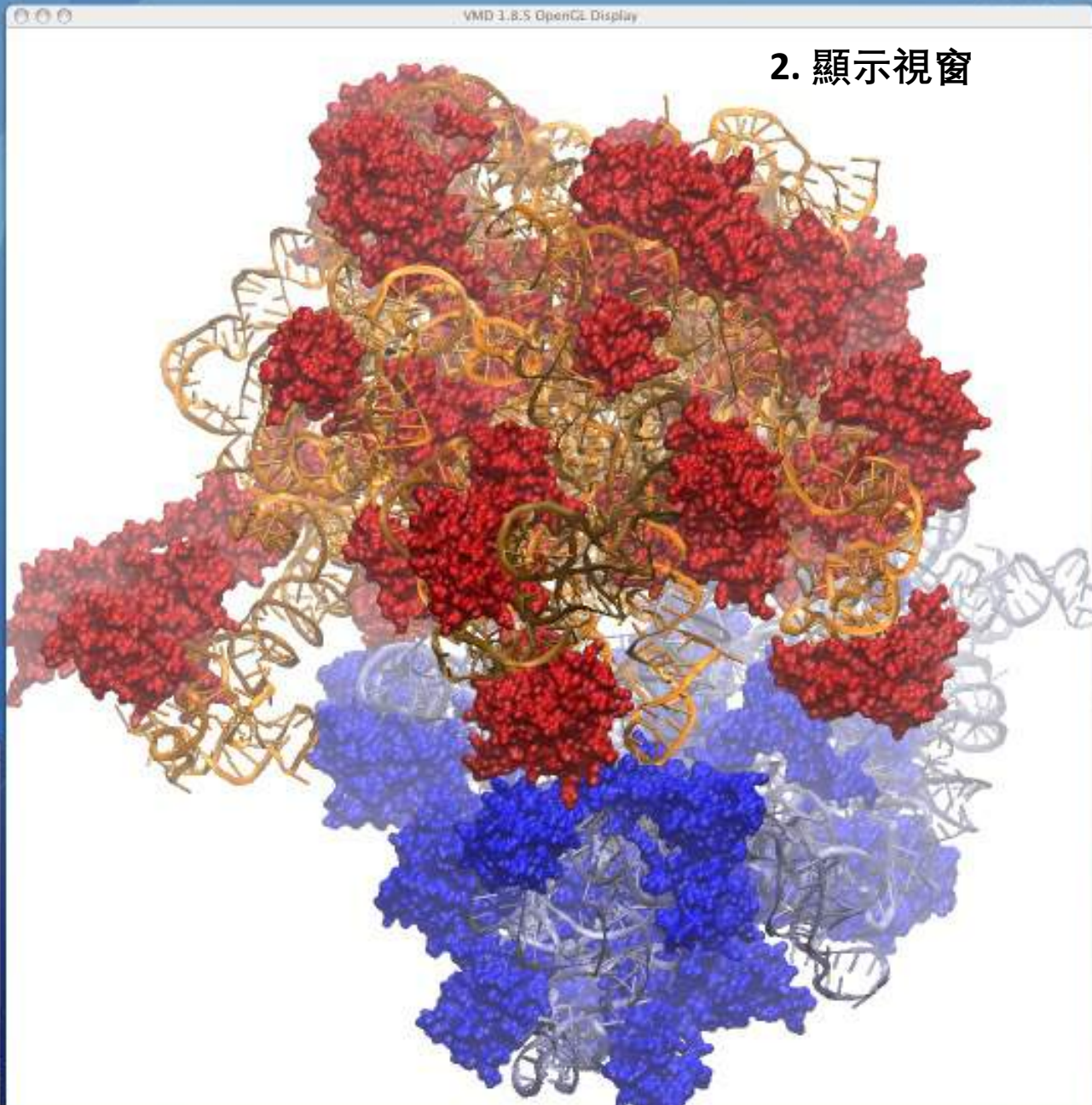
Apply Changes Automatically Apply

80x24

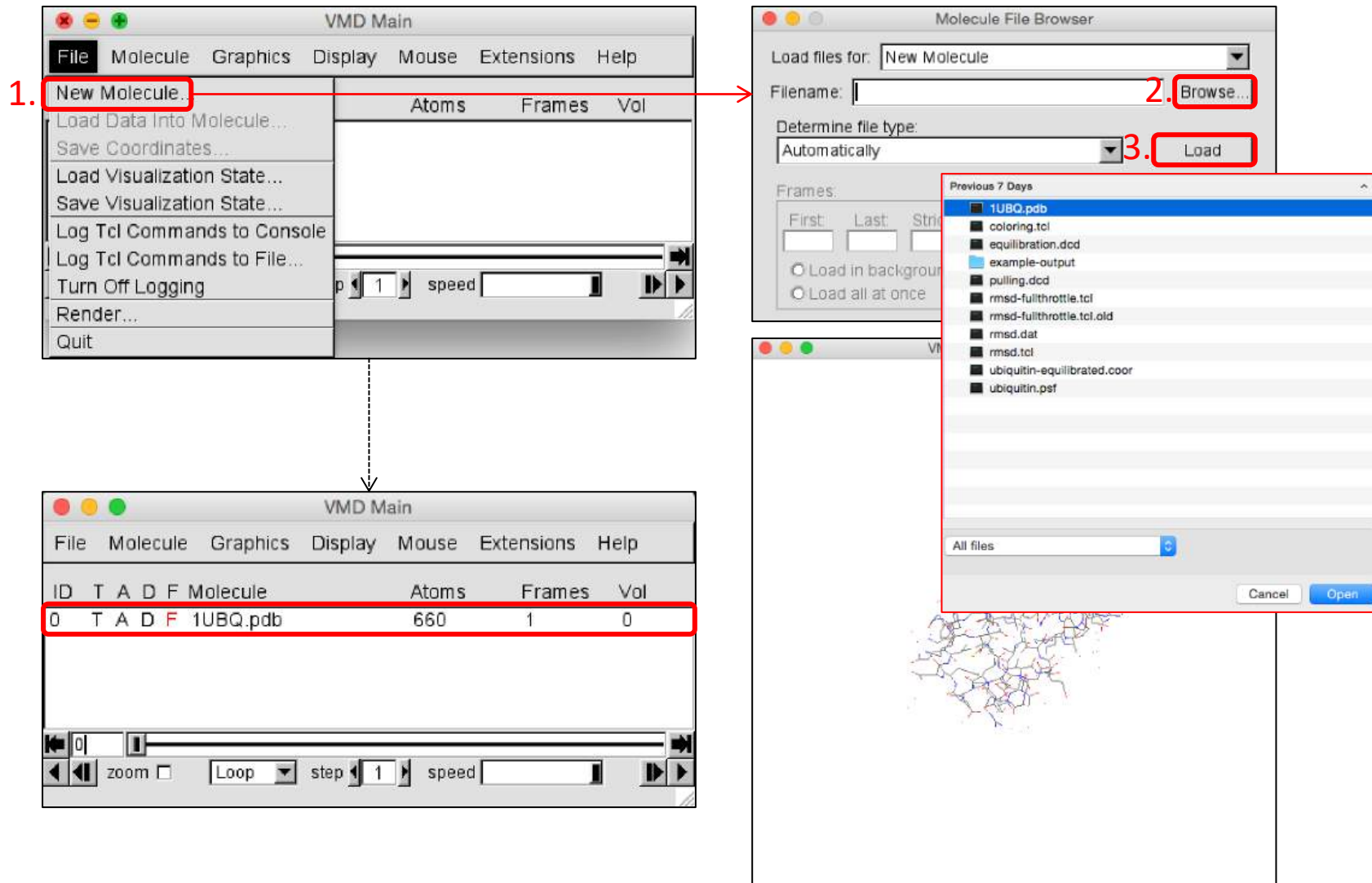
```
Total Triangles 1249631 Total constraints 582998
Max Neighbors per atom 68 Average Neighbors per atom 31.3773
Surface construction + writing time 26.69 seconds
done
[Info] Reading Surf geometry output file...
[Info] Read Surf output file, processing geometry...
[Info] Done.
[Info] Done.
[Info] Done.
(1.000000 0.000000 0.000000 18.449358) (0.000000 1.000000 0.000000 11.89570) (
0.000000 0.000000 1.000000 7.246444) (0.000000 0.000000 0.000000 1.000000) ((-0
-5.28306 0.677526 0.568119 0.000000) (0.825471 0.676590 -0.735918 0.000000) (-0.8
5.9551 -0.369491 -0.369327 0.000000) (0.000000 0.000000 0.000000 1.000000)) ((0.8
0.000000 0.000000 0.000000) (0.000000 0.000000 0.000000 0.000000) (0.000000
0.000000 0.000000 0.000000) (0.000000 0.000000 0.000000 1.000000)) ((1.000000
0.000000 0.000000 0.000000) (0.000000 1.000000 0.000000 -0.830000) (0.000000 0.8
00000 1.000000 0.190000) (0.000000 0.000000 0.000000 1.000000))
0
0
vid > [Info] Done.
[Info] Done.
vid >
```

4. 命令視窗

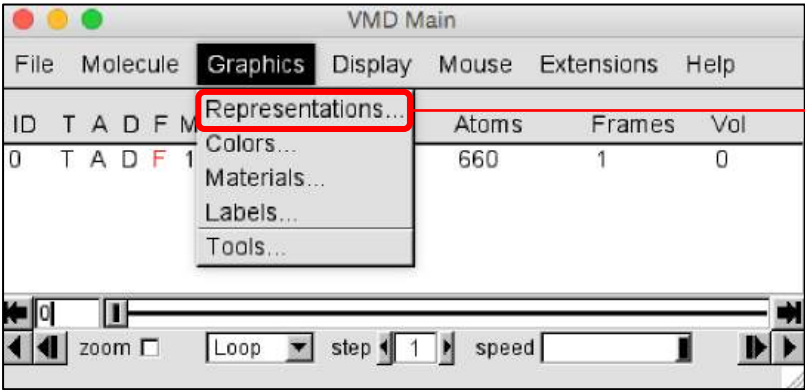
2. 顯示視窗



Loading a *Molecule*, Exploring it and Choosing Representations



Loading a Molecule, Exploring it and Choosing *Representations*



VMD Main

File Molecule **Graphics** Display Mouse Extensions Help

Representations...
Colors...
Materials...
Labels...
Tools...

ID	T	A	D	F	M	Atoms	Frames	Vol
0	T	A	D	F	1	660	1	0

zoom ☐ Loop step 1 speed

Graphical Representations

Selected Molecule
0: 1UBQ.pdb

Create Rep Delete Rep

Style	Color	Selection
Lines	Name	all

Selected Atoms
all

Draw style | Selections | Trajectory | Periodic

Coloring Method
Name

Material
Opaque

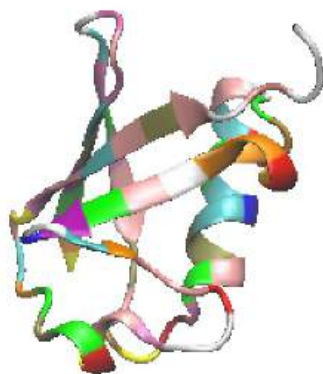
Drawing Method
Lines

Thickness 1

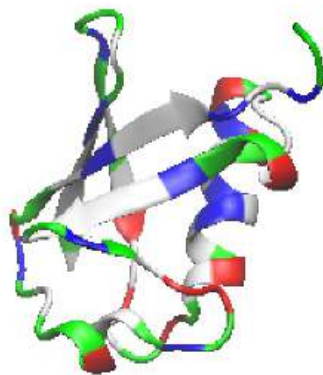
Apply Changes Automatically Apply

Lines VDW CPK Licorice NewCartoon

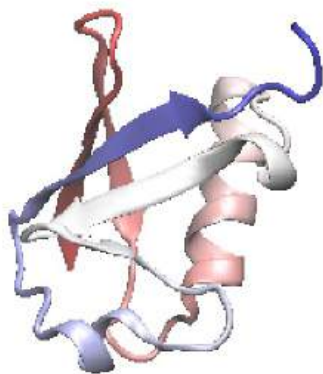
Loading a Molecule, Exploring it and Choosing *Representations*



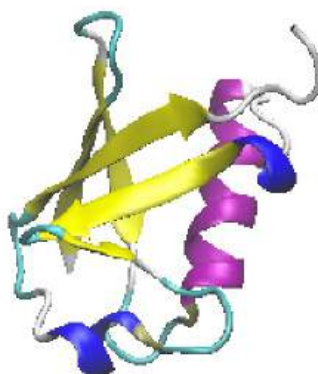
ResName



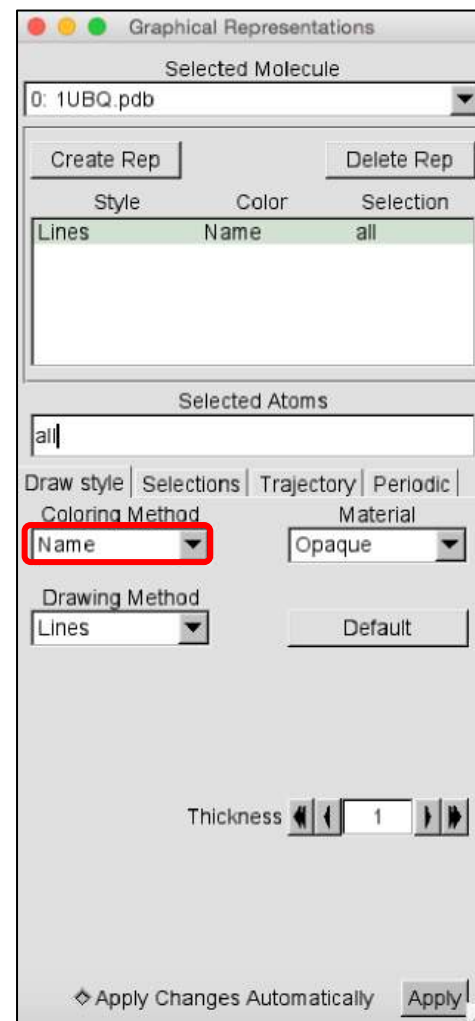
ResType



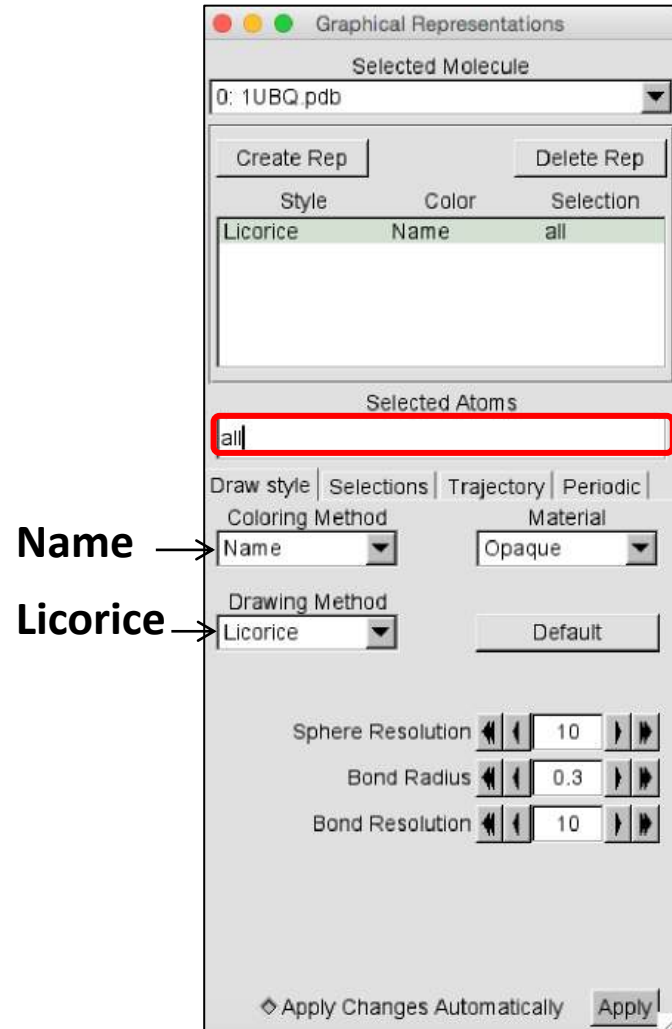
Index



Secondary Structure



Exercise: Selected Atoms

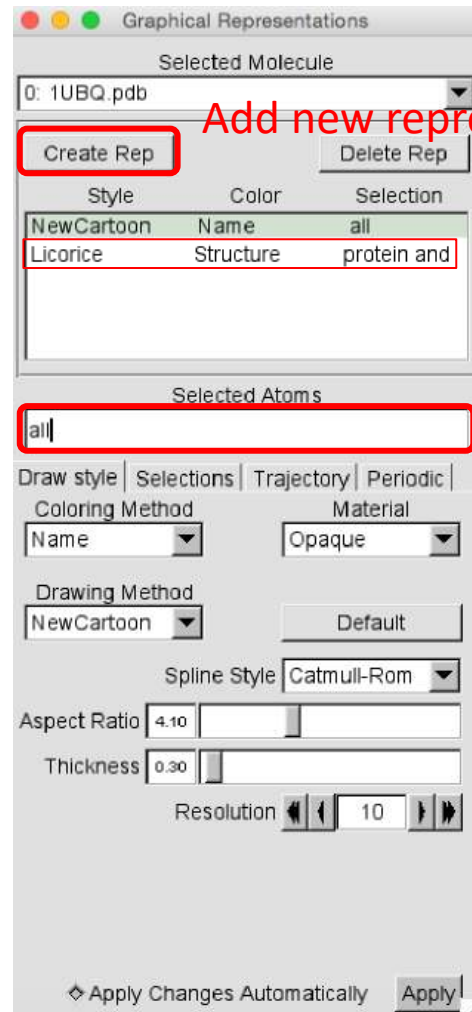


Selection	Action
protein	Shows the Protein
resid 1	The first residue
(resid 1 76) and (not water)	The first and last residues
(resid 23 to 34) and (protein)	The α -helix

In **Drawing Method**, switch to “NewCartoon” to see the helix

Also try,
protein and resname LYS ARG
protein and resname GLU ASP

Displaying Different Selections



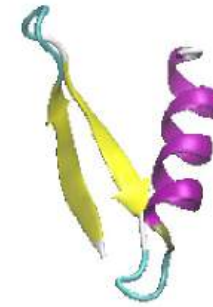
Add new representation



“protein and *resid 1 to 17*” with **Licorice**

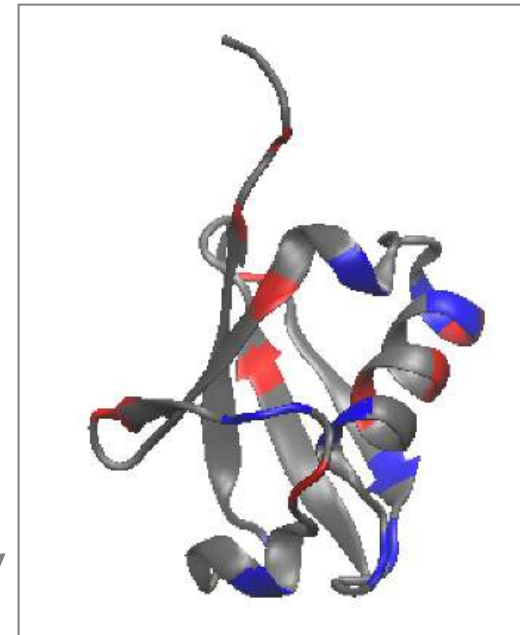
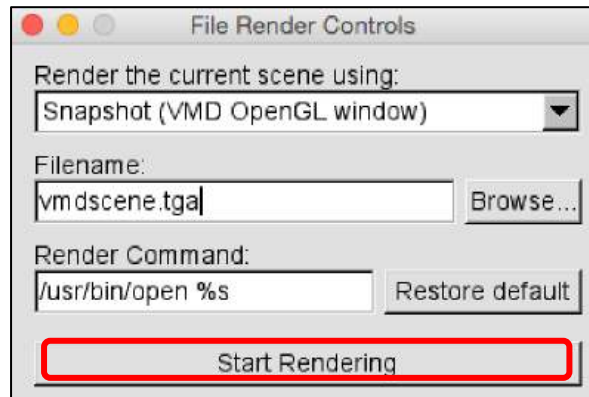
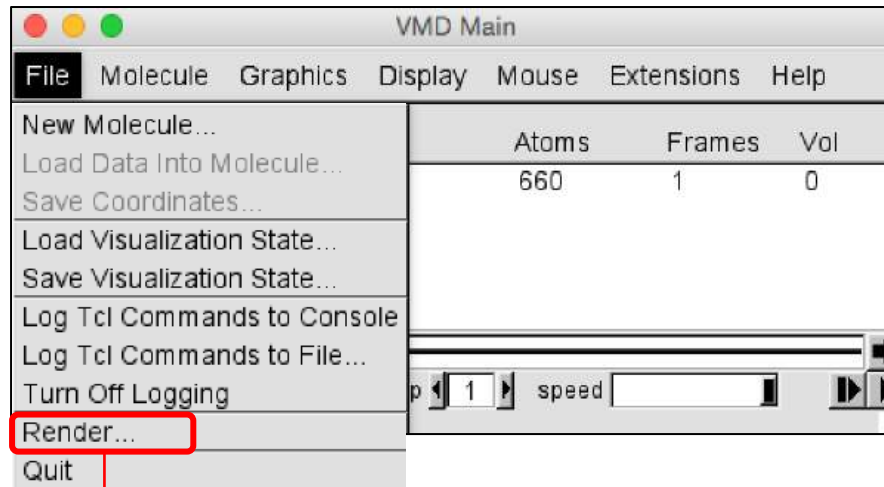


“protein and *resid 1 to 17*”



“protein and *resid 1 to 36*”

Rendering an Image



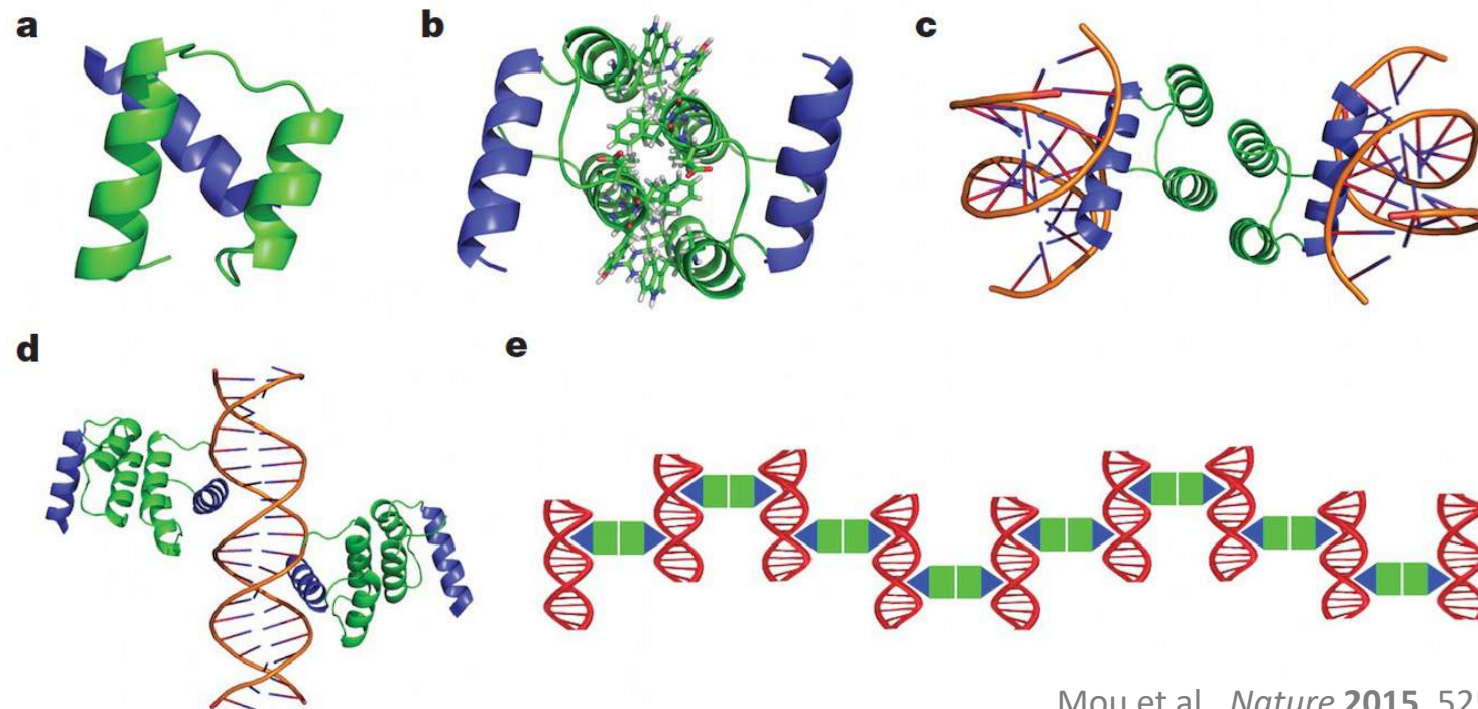
File: vmdscene.tga

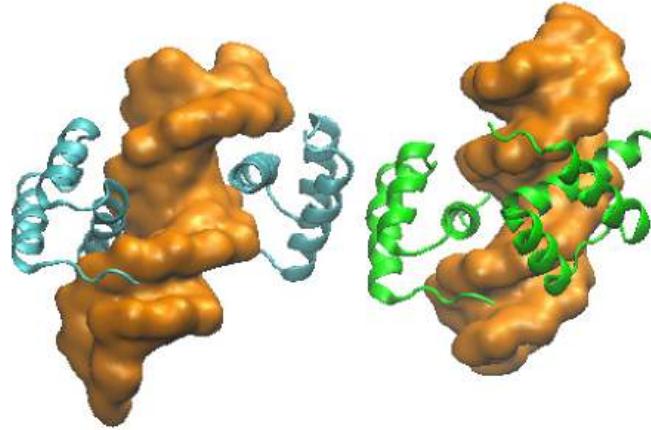


Computational design of co-assembling protein–DNA nanowires

Yun Mou¹, Jiun-Yann Yu², Timothy M. Wannier², Chin-Lin Guo³ & Stephen L. Mayo^{1,2}

230 | NATURE | VOL 525 | 10 SEPTEMBER 2015





Co-crystal structure for dualENH complexed with dsDNA
(PDB ID: 4QTR)

Demo

- Measure the distance between two atoms (Dimension of the molecular assembly)
- Visualize the distribution of charged residues on proteins
- Extract the structure of protein binding interface
- Calculate the total energy of the binding interface

Extended Data Table 1 | Sequences of wild-type ENH and dualENH

ENH (WT)	EKRPRTAFSSEQLARLKREFNENRYLTERRRQQLSSELGLNEAQIKIWFQNKRAKIKKST
dualENH	-----E--KKA-DLA-YFD----PEW-RY--QR-----

Change Directory to Home “Desktop”



```
( ) 32 % pwd  
/  
( ) 33 % cd ~  
(mytsai) 34 % pwd  
/Users/mytsai  
(mytsai) 35 % cd Desktop/  
(Desktop) 36 % pwd  
/Users/mytsai/Desktop  
(Desktop) 37 % cd vmd  
(vmd) 38 %
```

→ Check where you are in the path

→ Change to ‘home’ directory

→ Check the path again

→ Change to ‘home’ Desktop

→ Check the path again

→ OK!

“~” (curly brace): SHIFT + the key next to “1”

Extract the Structure at Binding Interface

VMD Main

ID	T	A	D	F	Molecule	Atoms
0	T	A	D	F	4qtr.pdb	2903

Extensions

- Analysis
- BioCoRE
- Data
- Modeling
- Simulation
- Visualization
- Tk Console**

VMD TkConsole

```
(vmd) 50 % set sel [atomselect top "chain B C"]
atomselect0
(vmd) 51 % $sel num
838
(vmd) 52 % $sel writepdb dimer_mutated.pdb
(vmd) 53 %
```

Molecule File Browser

Load files for: New Molecule

Filename: Browse...

Determine file type: Automatically Load

Frames: First Last Stride

☐ Load in background ☐ Load all at once

Volumetric Datasets

Rep2

- ... Chain A B
- ... ColorID → 10
- ... NewCartoon

Rep1

Selected Atoms: nucleic

Coloring Method: ColorID → 3

Drawing Method: QuickSurf

Rep3

- ... Chain C D
- ... ColorID → 7
- ... NewCartoon

Load with File → New Molecule

Use PSF Builder to Generate All-Atom Model

VMD Main

ID	T	A	D	F	Molecule	Atoms
0		A	D	F	4qtr.pdb	2903
1	T	A	D	F	dimer_mutated.pdb	838

Extensions

- Analysis
- Data
- Modeling
- Simulation
- Visualization
- Tk Console
- VMD Preferences

Automatic PSF Builder

AutoPSF

Options Help

Step 1: Input and Output Files

Molecule: dimer_mutated.pdb Output basename: dimer_mutated_autopsf

Topology: 0 4qtr.pdb

C:/Program Files/VMD/plugins/noarch/tcl/readcharmm1.2/top_all36

C:/Program Files/VMD/plugins/noarch/tcl/readcharmm1.2/top_all36

C:/Program Files/VMD/plugins/noarch/tcl/readcharmm1.2/top_all36

C:/Program Files/VMD/plugins/noarch/tcl/readcharmm1.2/top_all36

Load input files

Step 2: Mariano Selections to include in PSF/PDB

Everything Protein Nucleic Acid

Other: protein or nucleic or glycan

Add a new chain

Edit chain

Delete chain

Add patch

Delete patch

PDB

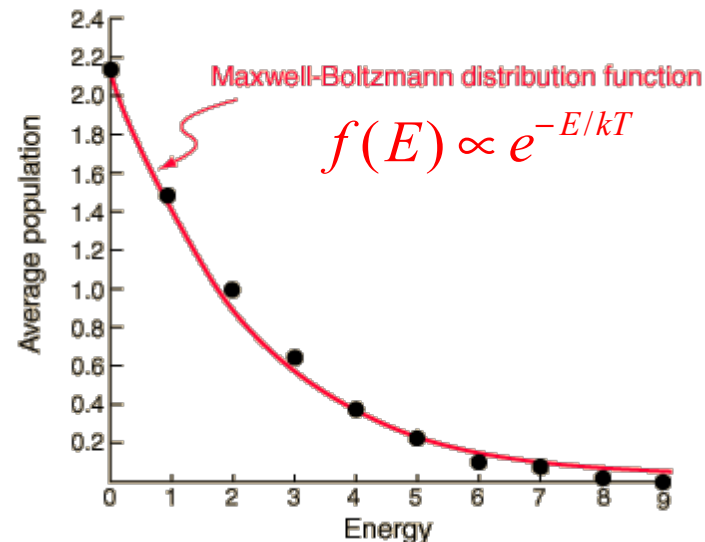
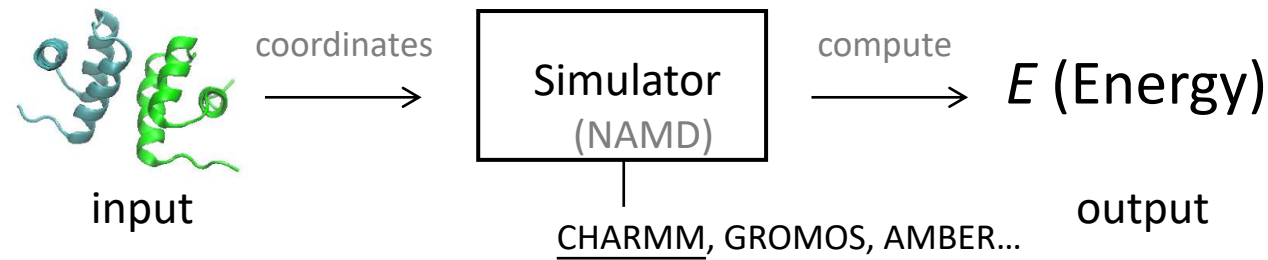
I'm feeling lucky

VMD 1.9.1 OpenGL Display

Add all hydrogen atoms back

NAMD Energy Plug-in

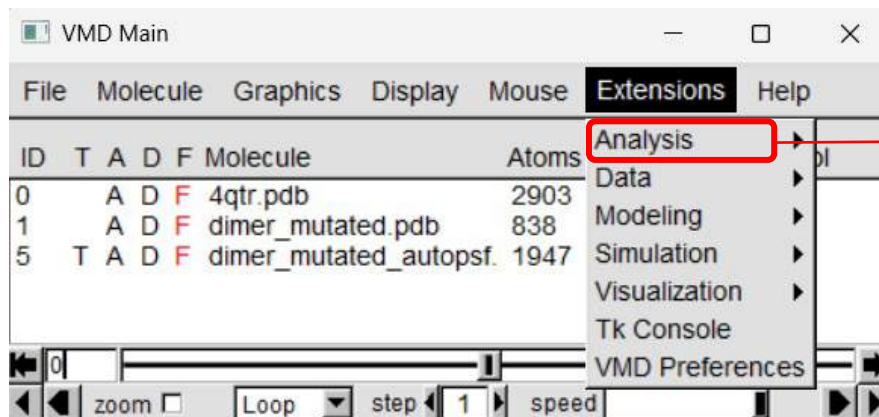
The **NAMD Energy** plugin provides an interface for evaluating energies using *NAMD*. *NAMD Energy* will apply the desired energy calculations to each frame of the selected molecule.



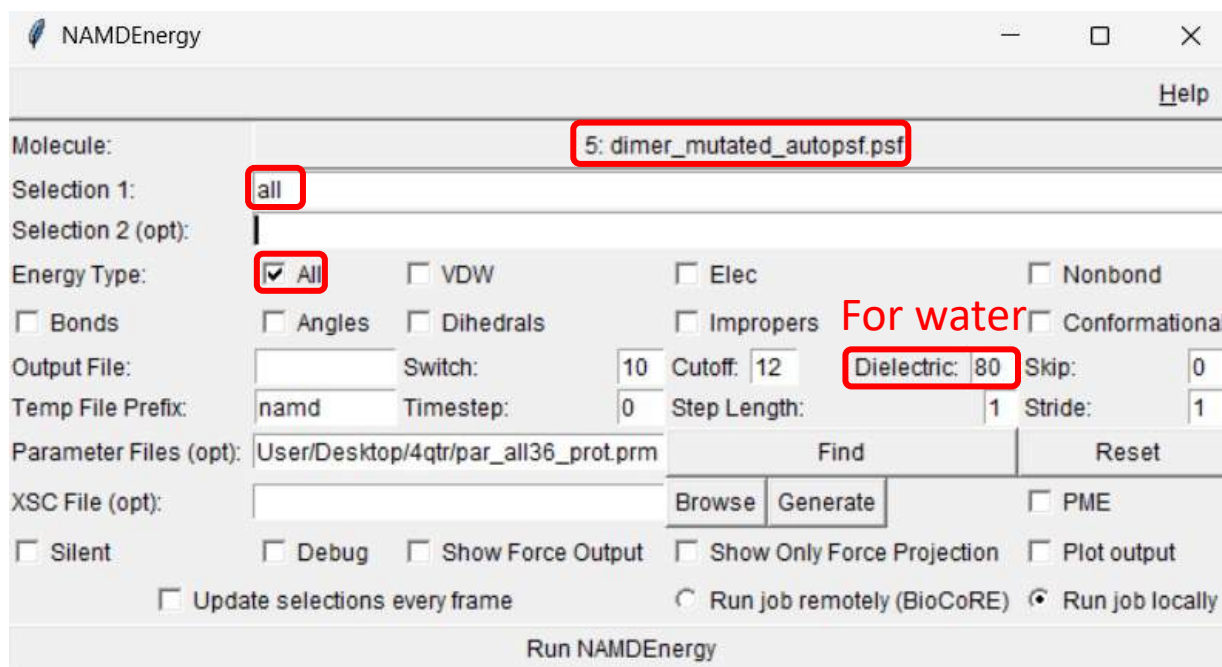
The energy of the system correlates with the system's population through **Boltzmann distribution**.

<http://hyperphysics.phy-astr.gsu.edu>

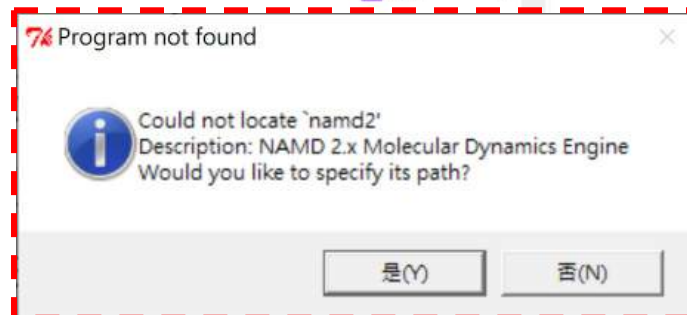
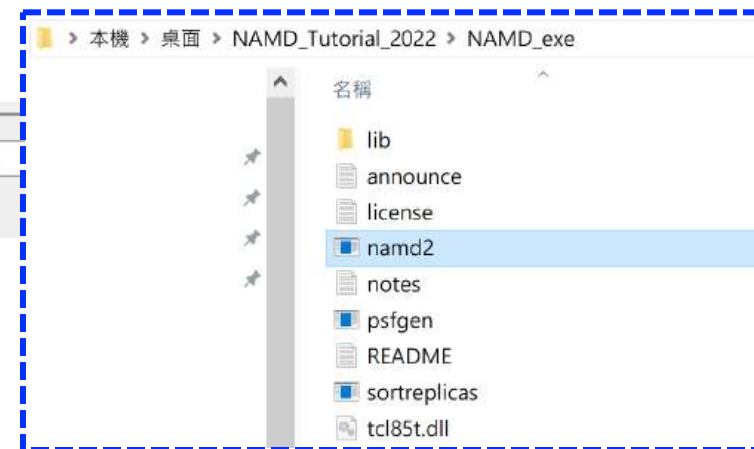
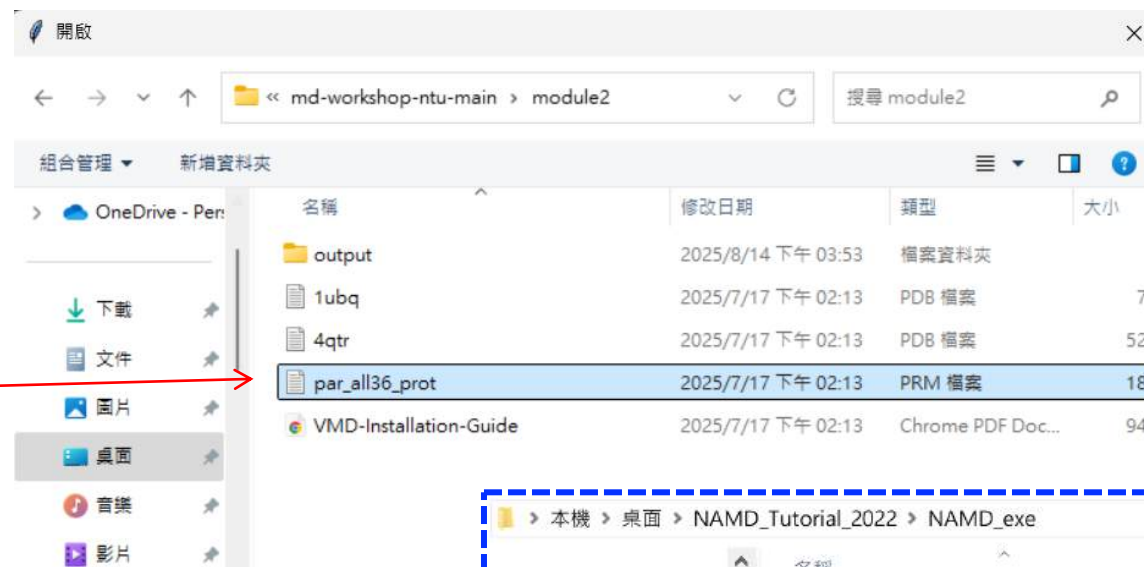
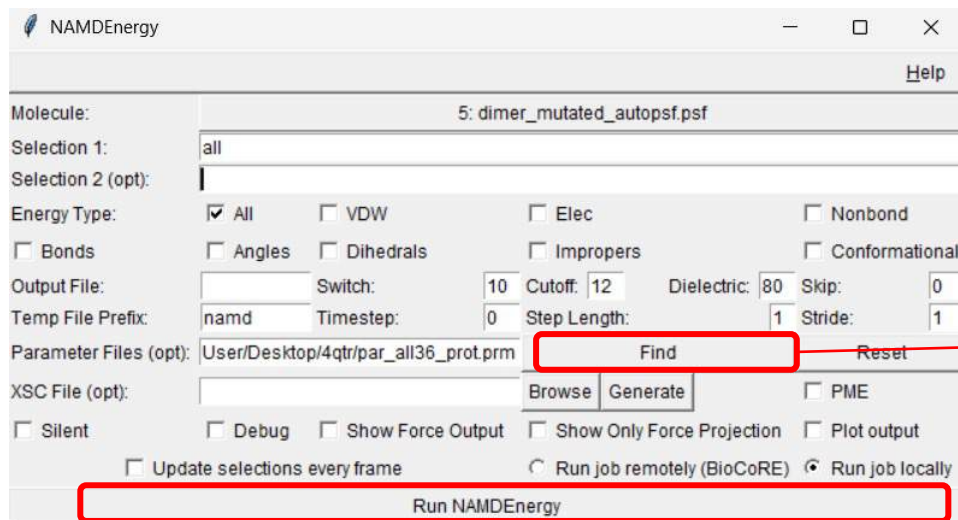
NAMD Energy Plug-in



- Analyze FEP Simulation
- APBS Electrostatics
- Collective variable analysis (PLUMED)
- Contact Map
- Heat Mapper
- Hydrogen Bonds
- Implicit Ligand Sampling
- IR Spectral Density Calculator
- MultiSeq
- NAMD Energy**
- NAMD Plot
- NetworkView
- Normal Mode Wizard
- PME Electrostatics
- PropKa
- Radial Pair Distribution Function g(r)
- Ramachandran Plot
- RMSD Calculator
- RMSD Trajectory Tool
- RMSD Visualizer Tool
- Salt Bridges
- Symmetry Tool
- Sequence Viewer
- Timeline
- VolMap Tool



NAMD Energy Plug-in



```
namdEnergy) Computing energy for selection:
namdEnergy) all
```

```
namdEnergy) Running:
namdEnergy) C:/Users/User/Desktop/NAMD_Tutorial_2022/NAMD_exe/namd2.exe namd-temp.namd
```

Frame	Time	Bond	Angle	Dihed	Impr	Elec	VdW	Conf	Nonbond	Total
0	0	+317.717	+352.285	+1090.8	+13.8173	-32.3035	+1e+11	+1774.62	+1e+11	+1e+11

(4qtr) 53

CHARMM Force Field Files (if necessary)

The screenshot shows the GitHub interface for the `prody/coMD` repository. At the top, the repository name and 'Public' status are visible. Below the navigation bar, the file list for the `master` branch is displayed. The file `par_all36_prot.prm` is highlighted with a red box, and a red arrow points to it with the word 'Download' written in red. The repository statistics on the right indicate 1 star, 6 watchers, and 6 forks. The README section at the bottom provides a brief description of the project and a link to the homepage.

File	Commit Message	Time Ago
LICENSE	Initial commit	8 years ago
README.md	Updated README because homepage is case sensitive	6 years ago
anmmc.py	Revert "added pbc unwrap in case of pbc problems"	4 years ago
comd.tcl	Revert "added pbc unwrap in case of pbc problems"	3 years ago
par_all36_prot.prm	Updated top and par files, and sorted out commandline args	5 years ago
par_all36m_prot.prm	Updated top and par files, and sorted out commandline args	5 years ago
pkgIndex.tcl	initial commit - Force	7 years ago
top_all36_carb.rtf	added modified carb forcefield that works with autopsf	5 years ago
top_all36_prot.rtf	Updated top and par files, and sorted out commandline args	5 years ago
toppar_water_ions.str	Updated top and par files, and sorted out commandline args	5 years ago

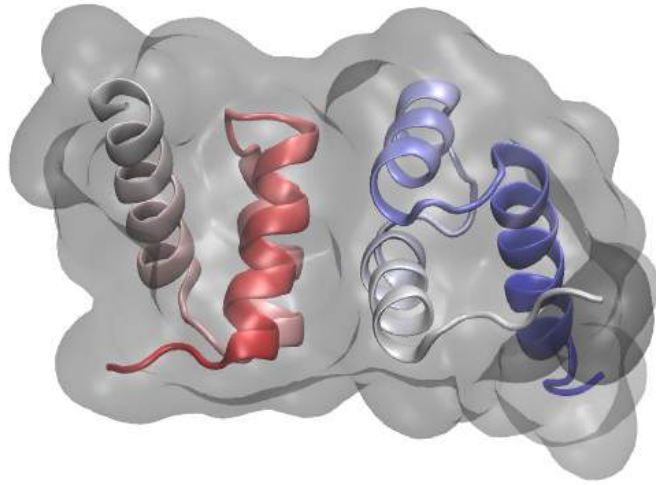
coMD

coMD is a VMD plugin GUI and a Python module developed for setup and analysis of simulations described in Gur et al., 2013.

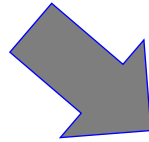
[Homepage](#)

<https://github.com/prody/coMD>

NAMD Energy Plug-in

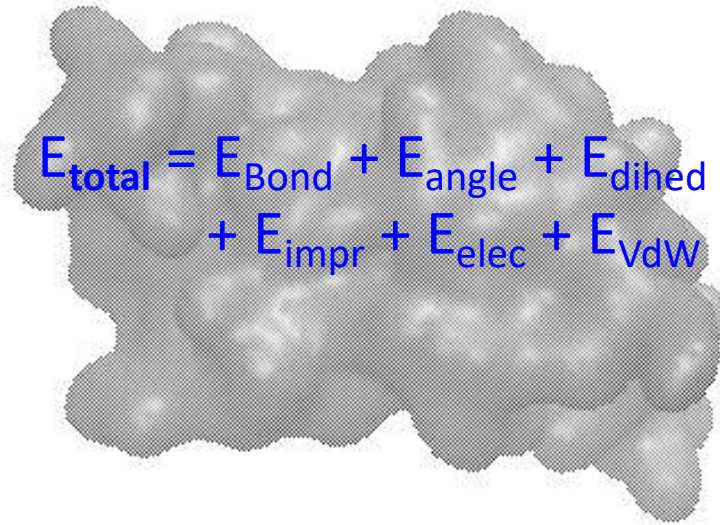


Total Energy



Electronic energy as the
effective potential
experienced by the nuclei
→ Molecular Mechanics

$$E_{\text{total}} = E_{\text{Bond}} + E_{\text{angle}} + E_{\text{dihed}} \\ + E_{\text{impr}} + E_{\text{elec}} + E_{\text{VdW}}$$



NAMD Energy Plug-in

Check the result with

Extensions → Tk console

```
namdEnergy) Computing energy for selection:  
namdEnergy) all
```

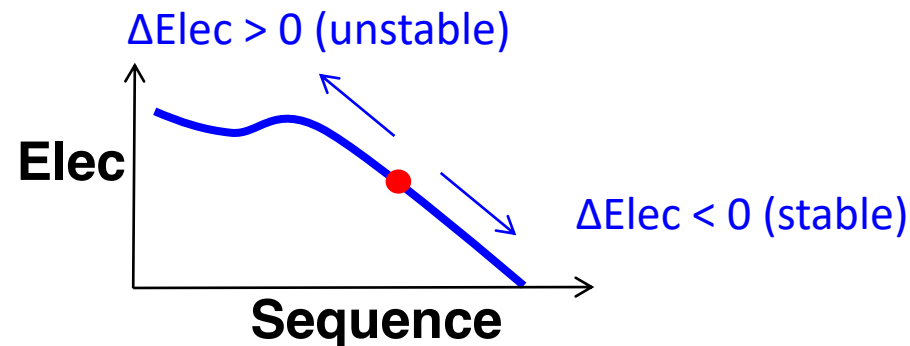
```
namdEnergy) Running:  
namdEnergy) C:/Users/User/Desktop/NAMD_Tutorial_2022/NAMD_exe/namd2.exe namd-temp.namd
```

Frame	Time	Bond	Angle	Dihed	Impr	Elec	VdW	Conf	Nonbond	Total
0	0	+317.717	+352.285	+1090.8	+13.8173	-32.3035	+1e+11	+1774.62	+1e+11	+1e+11

(4qtr) 53 %

In kcal/mol

Evaluate the contribution of electrostatic energy from all the charged residues in the molecule.

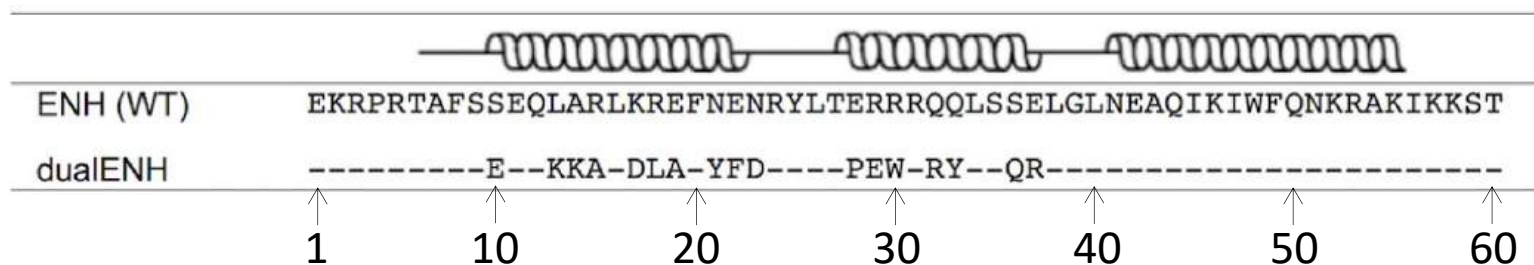
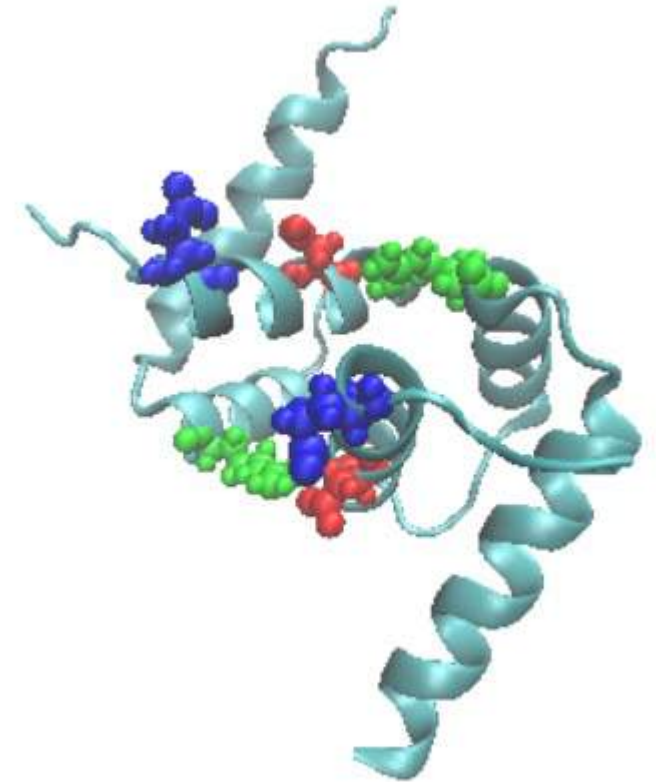


Homework

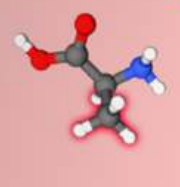
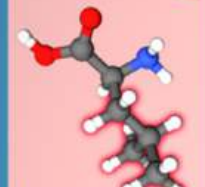
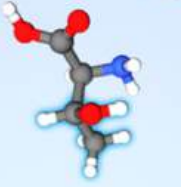
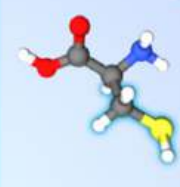
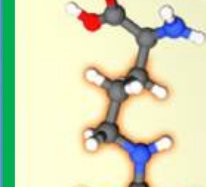
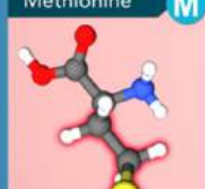
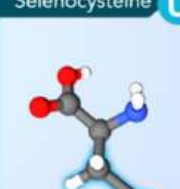
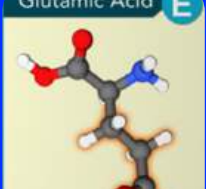
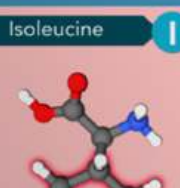
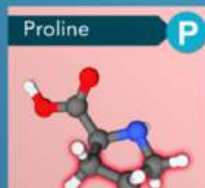
For the sequence of the engineered dimer structure, use the **NAMD Energy** plug-in to calculate its energy and compare the results with those of the individual mutation versions (**E10S**, **D17K**, and **R37E**; see the table below). Which of the sequences is electrostatically more favorable?
(Use *dimer_mutated_autopsf.pdb* and *dimer_mutated_autopsf.psf*)
Then, use VMD to highlight these residues and render an image.

[Hint] Use the NAMD Energy plug-in to compute the electrostatic energy. In VMD, add a new representation to highlight the selected residues, choose a coloring method of interest, and then render the resulting image.

What do you think is the disadvantage of using this approach in designing a new protein dimer? Please comment on it.

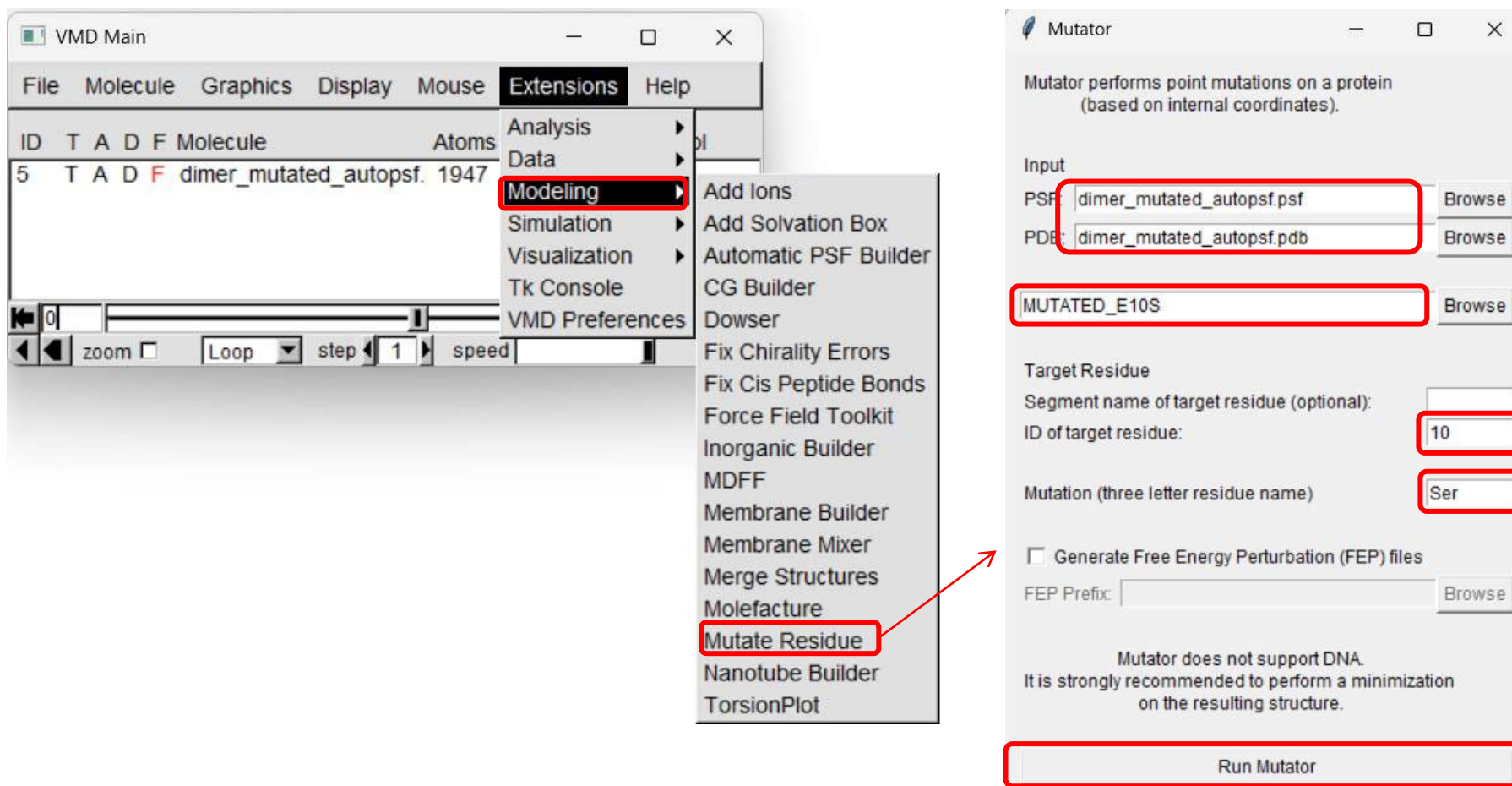


Blue: GLU10
Red: ASP17
Green: ARG37

STANDARD AMINO ACIDS	Alanine A  Ala	Leucine L  Leu	Phenylalanine F  Phe	Threonine T  Thr	Cysteine C  Cys	Arginine R  Arg	Aspartic Acid D  Asp
	Valine V  Val	Methionine M  Met	Tryptophan W  Trp	Asparagine N  Asn	Selenocysteine U  Sec	Histidine H  His	Glutamic Acid E  Glu
	Isoleucine I  Ile	Proline P  Pro	Serine S  Ser	Glutamine Q  Gln	Tyrosine Y  Tyr	Lysine K  Lys	Glycine G  Gly
PDB-101		Hydrophobic		Hydrophilic or polar		Charged	No side chain

Mutator Plug-in

The ***mutator*** plugin provides a very simple method for mutating a target residue selected by its ID, and the three character residue code for the mutant amino acid.



Resources

VMD Tutorials

<http://www.ks.uiuc.edu/Research/vmd/current/docs.html#tutorials>

VMD plugins

<http://www.ks.uiuc.edu/Research/vmd/plugins/>

Tutorials and Scientific Case Studies

<http://www.ks.uiuc.edu/Training/Tutorials/>



I. Simple MD

- Python
- JupyterLab

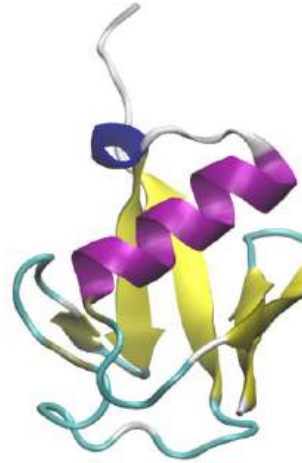
Part III

(MD in Bio)



II. Visualization

- VMD



III. MD (Bio)

- NAMD/VMD
 - Protein
 - Solvation
 - Add ions



IV. MD (Nano)

- NAMD/VMD
 - Graphene
 - Small molecules

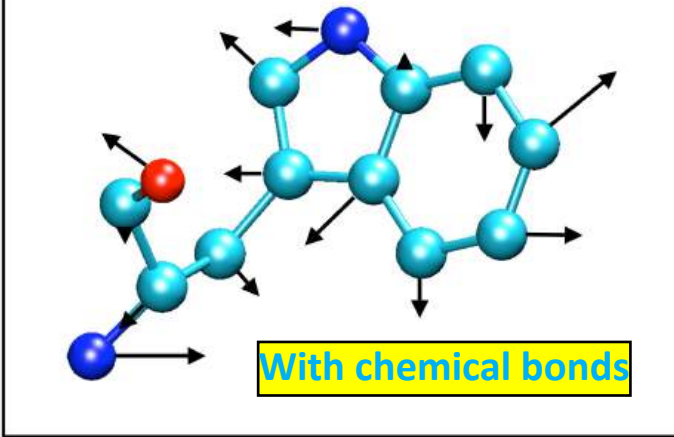


NAMD

Scalable Molecular Dynamics

分子力場/Molecular Force Fields

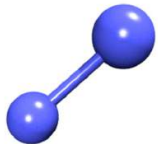
Velocity vectors of atoms in a molecule



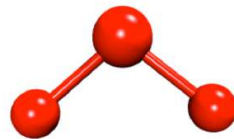
Bond definitions, atom types, atom names, parameters,
Coulomb interaction van der Waals interaction

$$\left[\frac{ij}{6} \right]$$

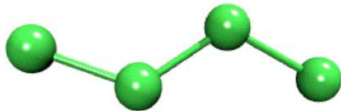
Bond



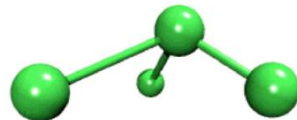
Angle



Dihedral



Improper



Electronic energy as the effective potential experienced by the nuclei

$$U(\vec{R}) = \underbrace{\sum_{bonds} k_i^{bond} (r_i - r_0)^2}_{U_{bond}} + \underbrace{\sum_{angles} k_i^{angle} (\theta_i - \theta_0)^2}_{U_{angle}} + \underbrace{\sum_{dihedrals} k_i^{dihe} [1 + \cos(n_i \phi_i + \delta_i)]}_{U_{dihedral}} + \underbrace{\sum_i \sum_{j \neq i} 4\epsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right] + \sum_i \sum_{j \neq i} \frac{q_i q_j}{\epsilon r_{ij}}}_{U_{nonbond}}$$

U_{bond} = oscillations about the equilibrium bond length

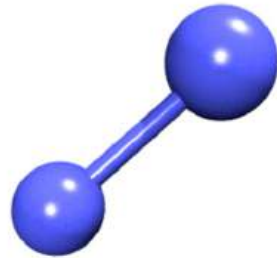
U_{angle} = oscillations of 3 atoms about an equilibrium bond angle

$U_{dihedral}$ = torsional rotation of 4 atoms about a central bond

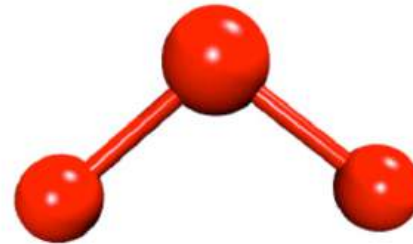
$U_{nonbond}$ = non-bonded energy terms (electrostatics and Lenard-Jones)

能量項 / CHARMm Force Field

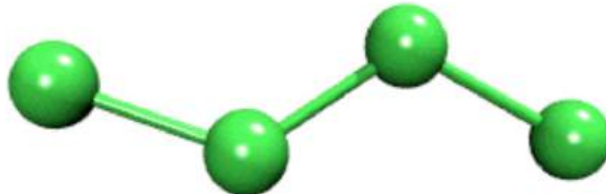
Bond



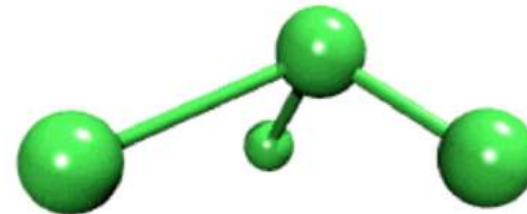
Angle



Dihedral



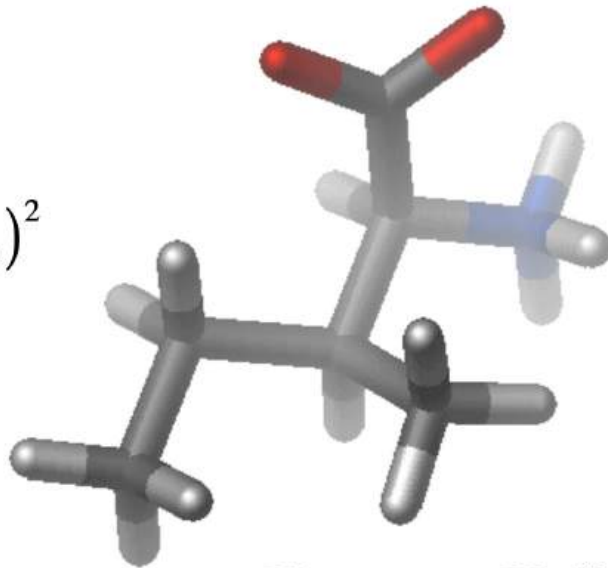
Improper



透過化學鍵的作用力 / Bonded Interactions

$$V_{angle} = K_{\theta} (\theta - \theta_o)^2$$

$$V_{bond} = K_b (b - b_o)^2$$

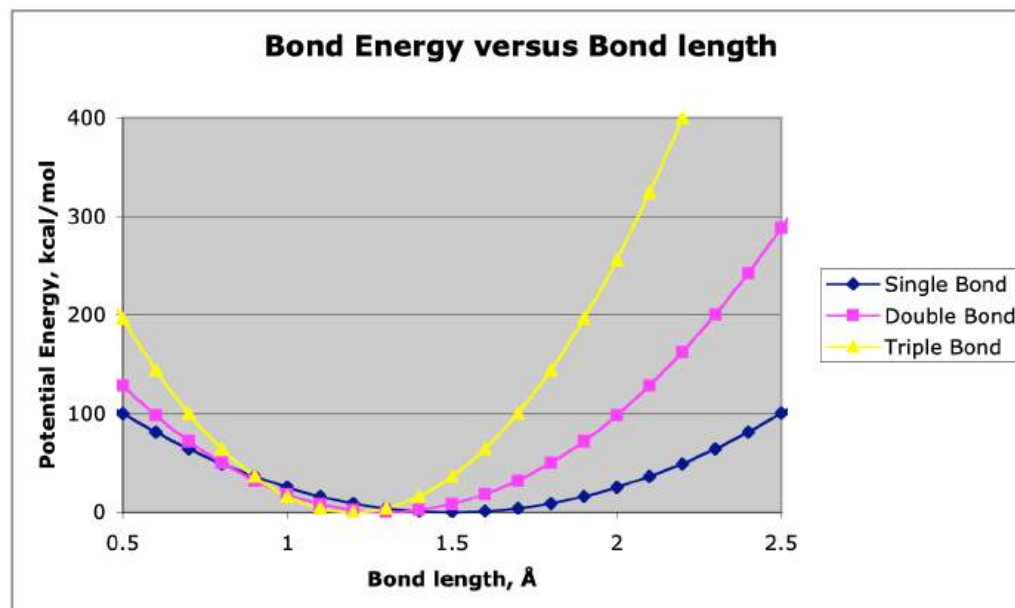
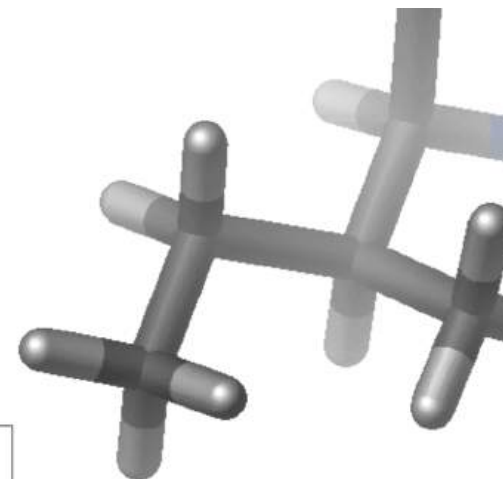


$$V_{dihedral} = K_{\phi} (1 + \cos(n\phi - \delta))$$

鍵長 / Bond Length

$$V_{bond} = K_b (b - b_o)^2$$

Chemical type	K_{bond}	b_o
C-C	100 kcal/mole/Å ²	1.5 Å
C=C	200 kcal/mole/Å ²	1.3 Å
C≡C	400 kcal/mole/Å ²	1.2 Å

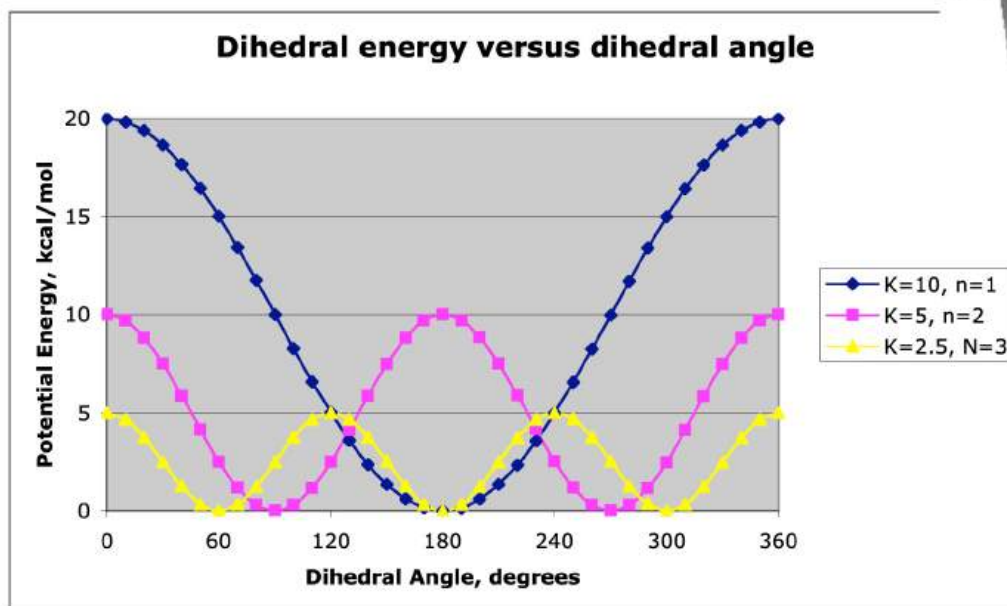
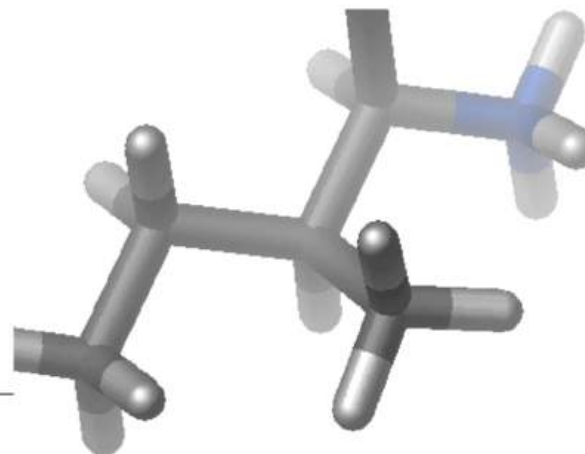


Bond angles and *improper* terms have similar quadratic forms, but with softer spring constants. The force constants can be obtained from vibrational analysis of the molecule (experimentally or theoretically).

雙面角 / Dihedral Angles

Dihedral Potential

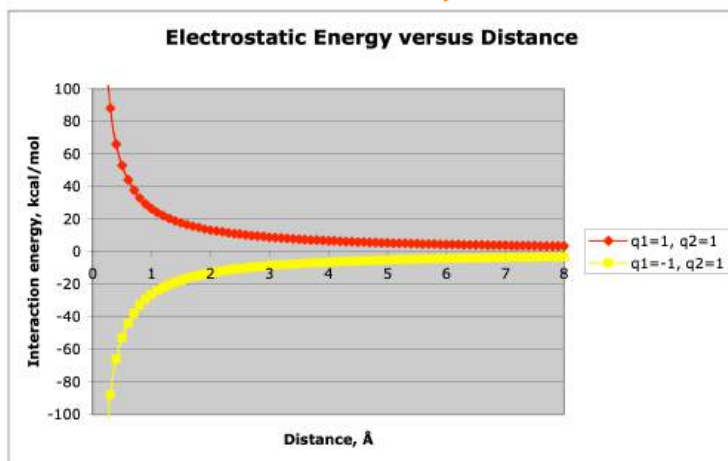
$$V_{dihedral} = K_{\phi} (1 + \cos(n\phi - \delta))$$



$$\delta = 0^\circ$$

非鍵結作用力 / Non-bonded Interactions

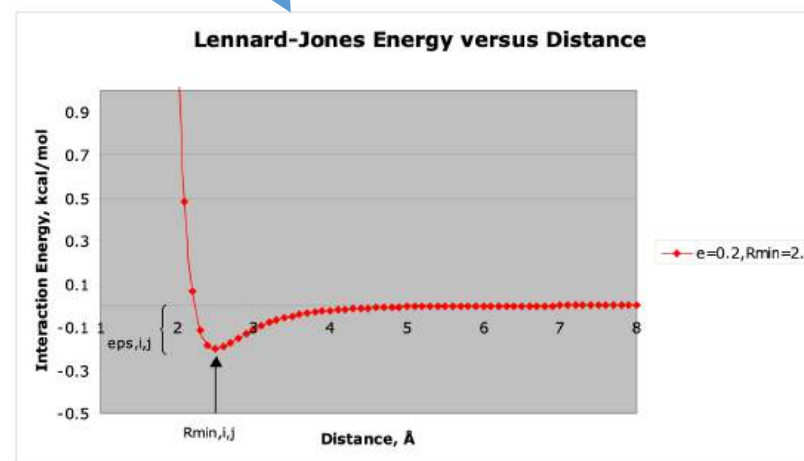
靜電作用力 $\sum_{nonbonded} \frac{q_i q_j}{4\pi D r_{ij}} + \epsilon_{ij} \left[\left(\frac{R_{min,ij}}{r_{ij}} \right)^{12} - 2 \left(\frac{R_{min,ij}}{r_{ij}} \right)^6 \right]$ 凡德瓦力



Note that the effect is long range.

D : dielectric constant

q_i : partial atomic charge



$\epsilon_{i,j}$: Lennard–Jones (LJ), van der Waals well–depth

$$\epsilon_{i,j} = \sqrt{\epsilon_i \cdot \epsilon_j}$$

R_{min} : LJ radius ($R_{min,ij}/2$ in CHARMM)

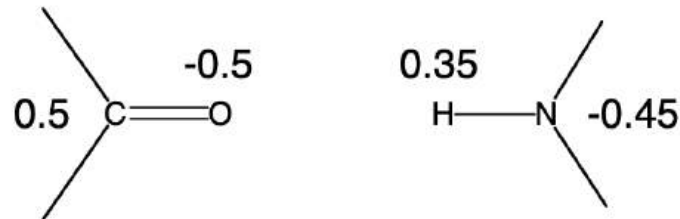
$$R_{min,ij} = R_{min,i} + R_{min,j}$$

電荷擬合方法/Charge Fitting Strategy

CHARMM- Mulliken*

AMBER(ESP/RESP)

Partial atomic charges



*Modifications based on interactions with TIP3 water

主流生物分子力場/Popular Force Fields

CHARMM

Chemistry at HARvard Macromolecular Mechanics

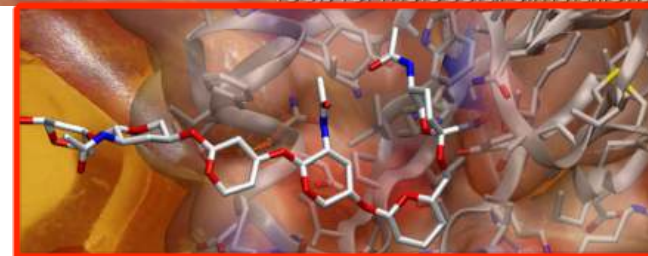
<https://www.charmm.org/>

CHARMM是一種用於分子動力學的分子力場，同時，採用這種力場的分子動力學軟體包也採用了這個名稱。CHARMM開發項目包括一個與馬丁·卡普拉斯及其在哈佛大學的研究小組合作的開發者網絡，這個網絡包含了美國等地的開發者，協同開發及維護CHARMM軟體包。這個軟體允許進行學術研究的個人及小組在支付一定費用之後得到使用許可。

The Amber Home Page

Tools for Molecular Simulations

AMBER力場是在生物大分子的模擬計算領域有著廣泛應用的一個分子力場。開發這個力場的是Peter Kollman課題組(UCSF)，最初AMBER力場是專門為了計算蛋白質和核酸體系而開發的，計算其力場參數的數據均來自實驗值，後來隨著AMBER力場的廣泛應用，包括Kollman在內的很多課題組對AMBER力場的內容不斷進行豐富，逐漸開發出了一個可以用於生物大分子、有機小分子和高分子模擬計算的力場體系。但是總體來講，AMBER力場的優勢在於對生物大分子的計算，其對小分子體系的計算結果常常不能令人滿意。



<https://ambermd.org/>

GROMACS

FAST. FLEXIBLE. FREE.



<http://www.gromacs.org/>

GROMACS是一套分子動力學模擬程序包，主要用來模擬研究蛋白質、脂質、核酸等生物分子的性質。它起初由荷蘭格羅寧根大學生物化學系開發，目前由來自世界各地的大學和研究機構維護。GROMACS是開源免費的軟體，4.6版之前的版本使用GNU通用公共許可證發行，而4.6版之後使用GNU寬通用公共許可證發行。

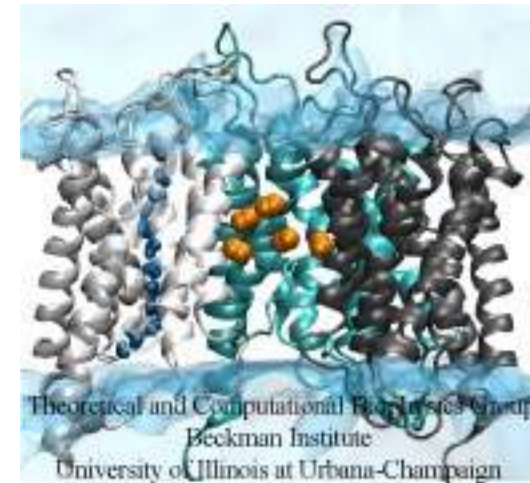
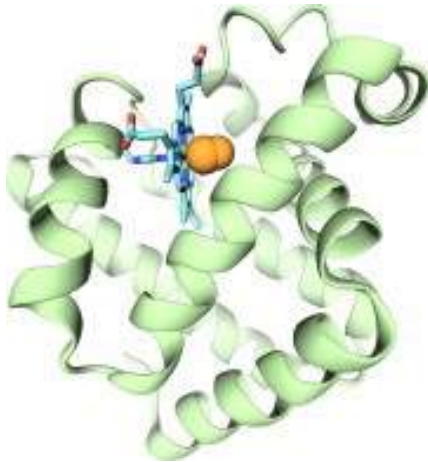
Visual Molecular Dynamics



<http://www.ks.uiuc.edu/Research/vmd/>

VMD is designed for **visualization**, **analysis**, and modeling of biological systems such as proteins, nucleic acids, lipid bilayer assemblies, etc.

VMD can read standard Protein Data Bank (**PDB**) files and display the contained structure.



- Support all major computer platforms (ex. MacOS X, Windows, Linux)
- Many molecular rendering and coloring methods

CHARMM Potential Function

$$\begin{aligned}
 U(\vec{R}) = & \underbrace{\sum_{bonds} k_i^{bond} (r_i - r_0)^2}_{U_{bond}} + \underbrace{\sum_{angles} k_i^{angle} (\theta_i - \theta_0)^2}_{U_{angle}} + \\
 & \underbrace{\sum_{dihedrals} k_i^{dihe} [1 + \cos(n_i \phi_i + \delta_i)]}_{U_{dihedral}} + \underbrace{\sum_i \sum_{j \neq i} 4\epsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right]}_{U_{nonbond}} + \sum_i \sum_{j \neq i} \frac{q_i q_j}{\epsilon r_{ij}}
 \end{aligned}$$

PDB file → **geometry** (幾何構型)
 Topology PSF file → **parameters** (參數) → Parameter file

File Format/Structure

- The structure of a pdb file
- The structure of a psf file
- The topology file
- The parameter file
- Connection to potential energy terms

Structure of a PDB File

index	resname			resid			x	y	z	segname	
	name		chain								
ATOM	22	N	ALA	B	3	-4.073	-7.587	-2.708	1.00	0.00	BH
ATOM	23	HN	ALA	B	3	-3.813	-6.675	-3.125	1.00	0.00	BH
ATOM	24	CA	ALA	B	3	-4.615	-7.557	-1.309	1.00	0.00	BH
ATOM	25	HA	ALA	B	3	-4.323	-8.453	-0.704	1.00	0.00	BH
ATOM	26	CB	ALA	B	3	-4.137	-6.277	-0.676	1.00	0.00	BH
ATOM	27	HB1	ALA	B	3	-3.128	-5.950	-0.907	1.00	0.00	BH
ATOM	28	HB2	ALA	B	3	-4.724	-5.439	-1.015	1.00	0.00	BH
ATOM	29	HB3	ALA	B	3	-4.360	-6.338	0.393	1.00	0.00	BH
ATOM	30	C	ALA	B	3	-6.187	-7.538	-1.357	1.00	0.00	BH
ATOM	31	O	ALA	B	3	-6.854	-6.553	-1.264	1.00	0.00	BH
ATOM	32	N	ALA	B	4	-6.697	-8.715	-1.643	1.00	0.00	BH
ATOM	33	HN	ALA	B	4	-6.023	-9.463	-1.751	1.00	0.00	BH
ATOM	34	CA	ALA	B	4	-8.105	-9.096	-1.934	1.00	0.00	BH
ATOM	35	HA	ALA	B	4	-8.287	-8.878	-3.003	1.00	0.00	BH
ATOM	36	CB	ALA	B	4	-8.214	-10.604	-1.704	1.00	0.00	BH
ATOM	37	HB1	ALA	B	4	-7.493	-11.205	-2.379	1.00	0.00	BH
ATOM	38	HB2	ALA	B	4	-8.016	-10.861	-0.665	1.00	0.00	BH
ATOM	39	HB3	ALA	B	4	-9.245	-10.914	-1.986	1.00	0.00	BH
ATOM	40	C	ALA	B	4	-9.226	-8.438	-1.091	1.00	0.00	BH
ATOM	41	O	ALA	B	4	-10.207	-7.958	-1.667	1.00	0.00	BH
000											

>>> It is an ascii, fixed-format file <<<

“No connectivity information”

拓樸/Topology File

From top_all22_model.inp

```
RESI PHEN          0.00    ! phenol, adm jr.
GROUP
ATOM CG    CA    -0.115    !
ATOM HG    HP     0.115    !
GROUP
ATOM CD1    CA    -0.115    !
ATOM HD1    HP     0.115    !
GROUP
ATOM CD2    CA    -0.115    !
ATOM HD2    HP     0.115    !
GROUP
ATOM CE1    CA    -0.115    !
ATOM HE1    HP     0.115
GROUP
ATOM CE2    CA    -0.115
ATOM HE2    HP     0.115
GROUP
ATOM CZ     CA     0.110
ATOM OH     OH1    -0.540
ATOM HH     H      0.430
BOND CD2 CG CE1 CD1 CZ CE2 CG HG CD1 HD1
BOND CD2 HD2 CE1 HE1 CE2 HE2 CZ OH OH HH
DOUBLE CD1 CG CE2 CD2  CZ CE1
```

Top_all22_model.inp contains all protein model compounds. Lipid, nucleic acid and carbohydrate model compounds are in the full topology files.

HG will ultimately be deleted. Therefore, move HG (hydrogen) charge into CG, such that the CG charge becomes 0.00 in the final compound.

Use remaining charges/atom types without any changes.

Do the same with indole

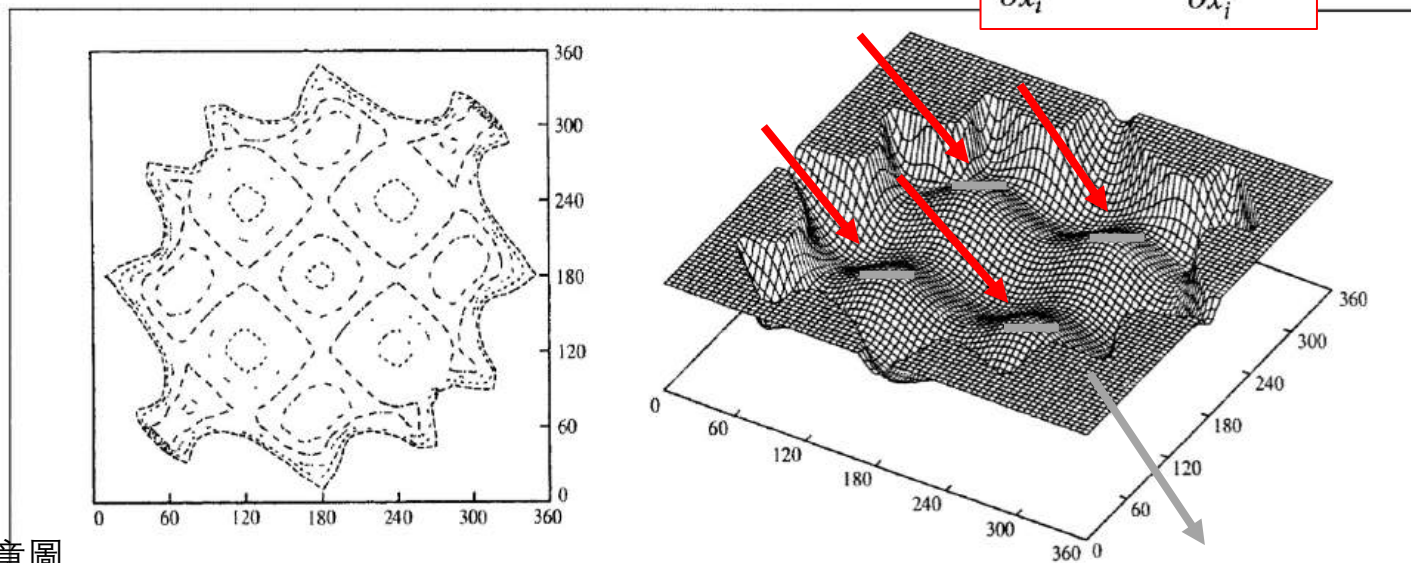
From MacKerell

能量最佳化/Energy Minimization

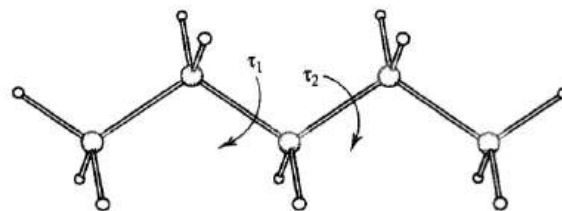
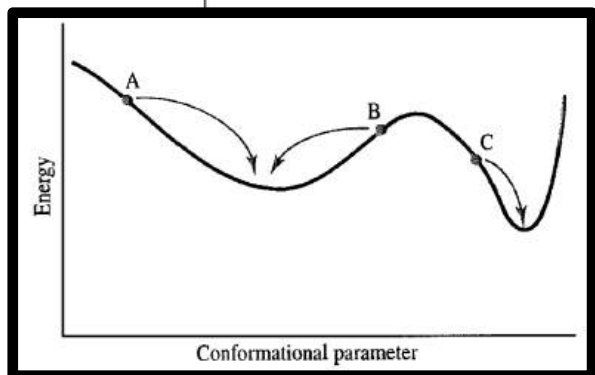
- Energy landscape of pentane with two torsion angles

Energy minima (能量極小值)

$$\frac{\partial f}{\partial x_i} = 0; \quad \frac{\partial^2 f}{\partial x_i^2} > 0$$



1D 能量示意圖



Energy maxima (能量極大值)

$$\frac{\partial f}{\partial x_i} = 0; \quad \frac{\partial^2 f}{\partial x_i^2} < 0$$

能量最佳化/Energy Minimization

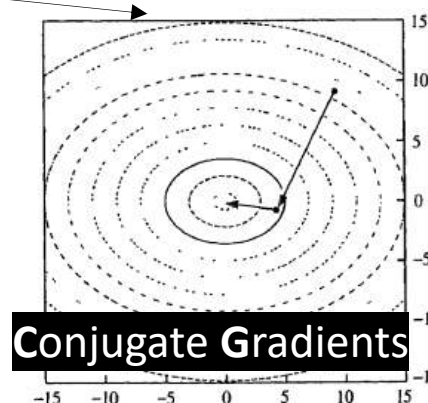
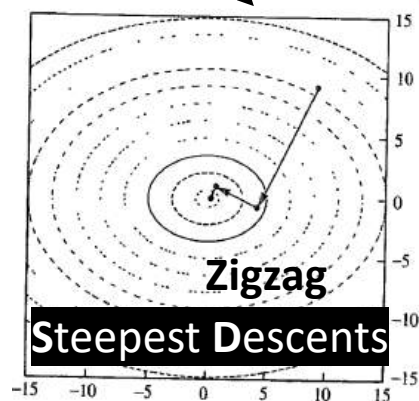
- Compare two different methods



[WikiWand:DNA](#)

最速下降法(SD) →
共軛梯度法(CG) →

	初步結構		細部結構	
	Initial refinement (Av. gradient $< 1 \text{ kcal } \text{\AA}^{-2}$)		Stringent minimisation (Av. gradient $< 0.1 \text{ kcal } \text{\AA}^{-2}$)	
Method	CPU time (s)	Number of iterations	CPU time (s)	Number of iterations
Steepest descents	67	98	1405	1893
Conjugate gradients	149	213	257	367



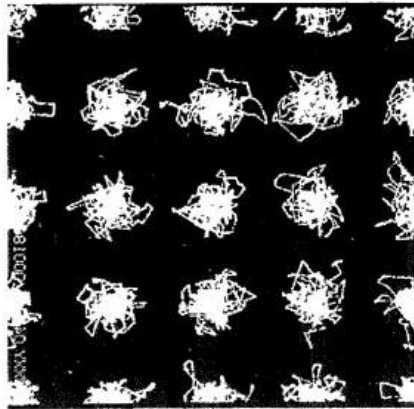
- Steepest Descents → 穩定、慢，但適合「救急」去掉高能接觸。
- Conjugate Gradients → 高效、快，但需要已經「不太差」的初始結構。
- 實務上常常兩者 搭配使用，形成「粗略 → 精細」的最佳化流程。

Molecular Dynamics Simulation Method

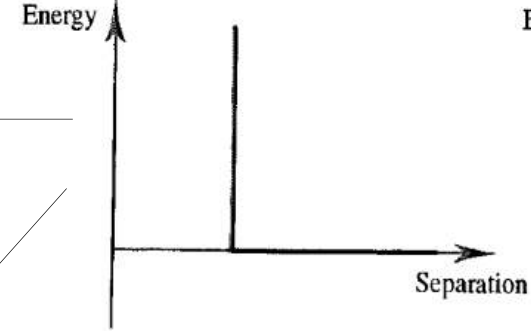
- Integrating Newton's law of motion

$$\frac{d^2 x_i}{dt^2} = \frac{F_{x_i}}{m_i}$$

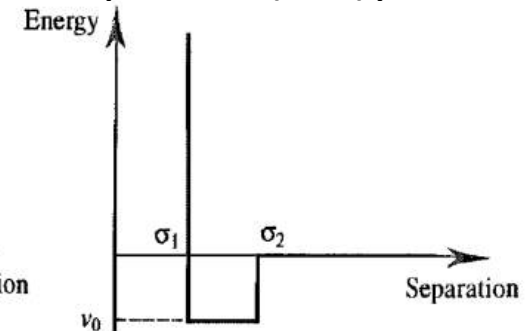
固態 Hard-sphere 模擬



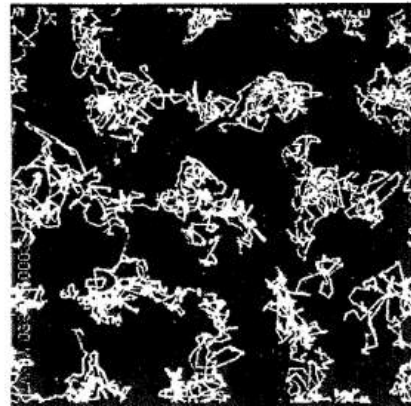
Hard-sphere (硬球) potential



Square-well (方井) potential



液態 Hard-sphere 模擬



Alder and Wainwright, JCP 1959

Molecular Dynamics Simulation Method

- Position and dynamic properties can be approximated as Taylor series expansions
(位置以及動態性質可用泰勒展開式近似)

Position(位置)→ $\mathbf{r}(t + \delta t) = \mathbf{r}(t) + \delta t \mathbf{v}(t) + \frac{1}{2} \delta t^2 \mathbf{a}(t) + \frac{1}{6} \delta t^3 \mathbf{b}(t) + \frac{1}{24} \delta t^4 \mathbf{c}(t) + \dots$

Velocity(速度)→ $\mathbf{v}(t + \delta t) = \mathbf{v}(t) + \delta t \mathbf{a}(t) + \frac{1}{2} \delta t^2 \mathbf{b}(t) + \frac{1}{6} \delta t^3 \mathbf{c}(t) + \dots$

Acceleration(加速度)→ $\mathbf{a}(t + \delta t) = \mathbf{a}(t) + \delta t \mathbf{b}(t) + \frac{1}{2} \delta t^2 \mathbf{c}(t) + \dots$

$\mathbf{b}(t + \delta t) = \mathbf{b}(t) + \delta t \mathbf{c}(t) + \dots$

Truncated
(截斷)

越高次項越準喔~

Verlet algorithm

$$\mathbf{r}(t + \delta t) = \mathbf{r}(t) + \delta t \mathbf{v}(t) + \frac{1}{2} \delta t^2 \mathbf{a}(t) + \dots$$

$$\mathbf{r}(t - \delta t) = \mathbf{r}(t) - \delta t \mathbf{v}(t) + \frac{1}{2} \delta t^2 \mathbf{a}(t) - \dots$$

↓ 相加

Truncated at $O(\delta t^4)$

→ Fourth-order method, Why?

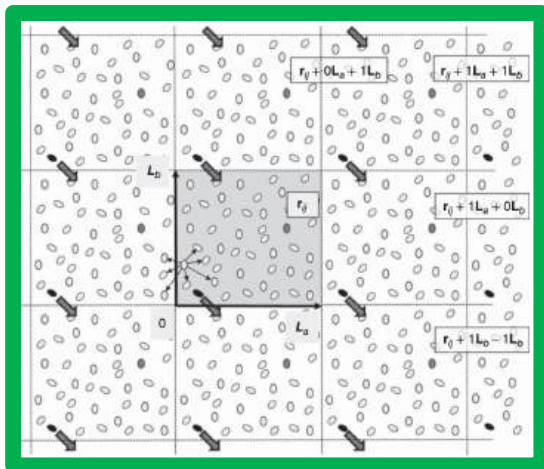
(仍準確至三次項)

$$\mathbf{r}(t + \delta t) = 2\mathbf{r}(t) - \mathbf{r}(t - \delta t) + \delta t^2 \mathbf{a}(t)$$

[補充] 分子模擬的技術細節

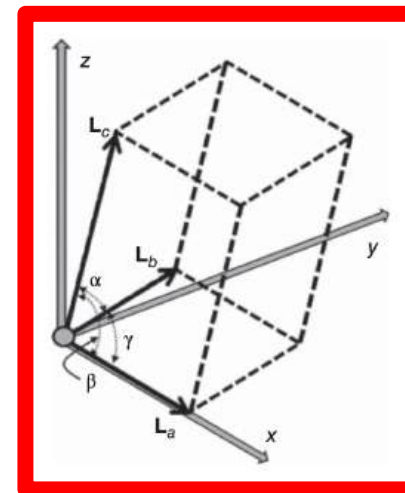
1. Initial Setup of the System

- Determining the shape and size of the container to confine the molecules of interest
- Defining the initial positions and velocities of the atoms
- Creating the **simulation cell**



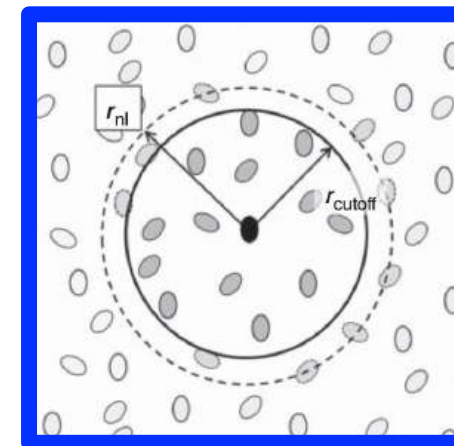
2. Boundary Conditions

- Modeling a macroscopic (bulk) system with a single *simulation cell* (10^6 instead of 10^{23})
- Applying **Periodic boundary conditions (PBC)** to simulate an infinite system

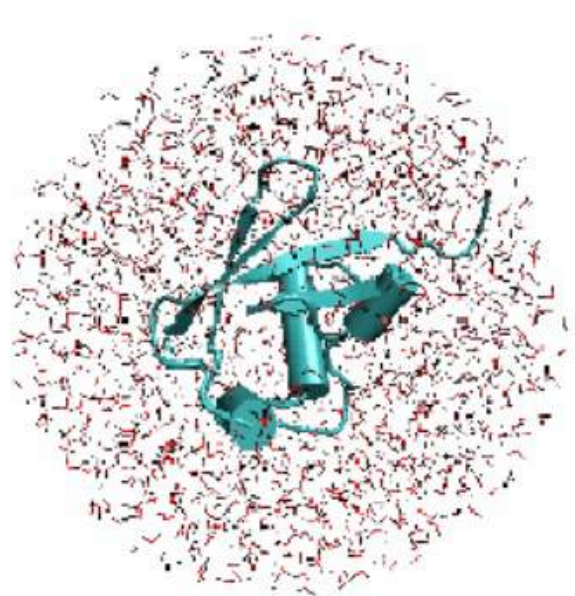


3. Treatment of Short-Range and Long-Range Forces

- Handling short-range forces between nearby atoms using cutoff distances or **neighbor lists**
- Accounting for long-range forces through methods like Ewald summation or *particle mesh Ewald (PME)*

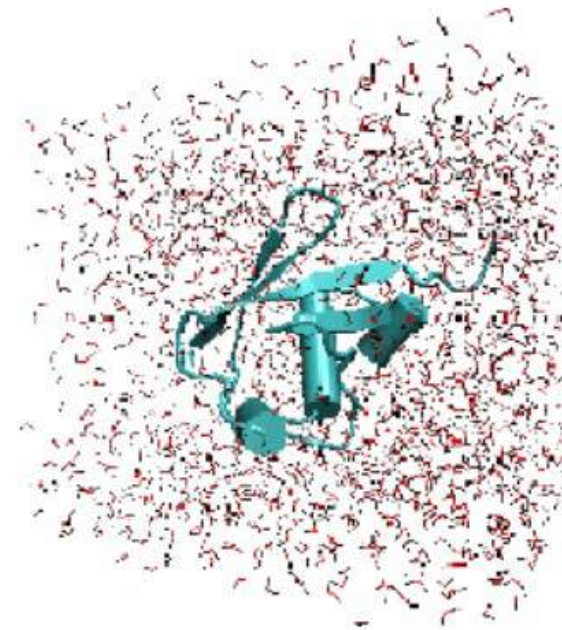


[補充] 表面張力 & 週期邊界條件



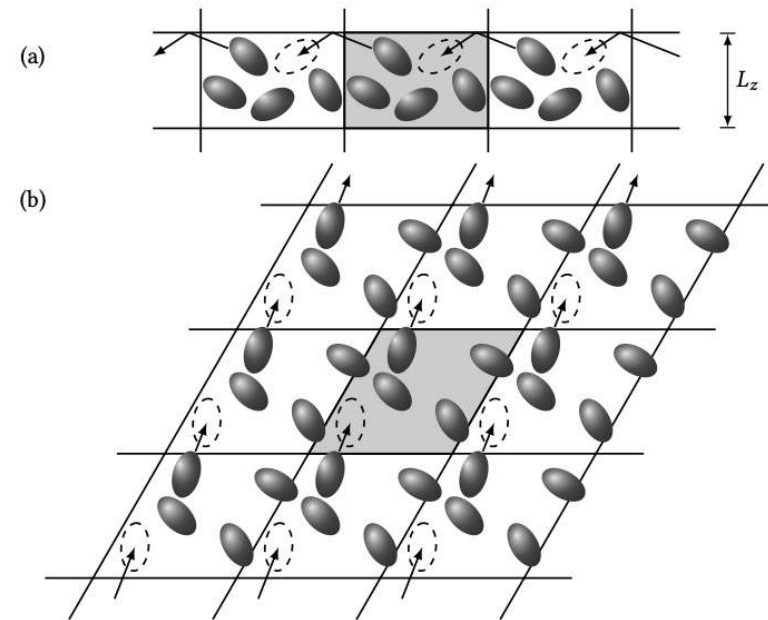
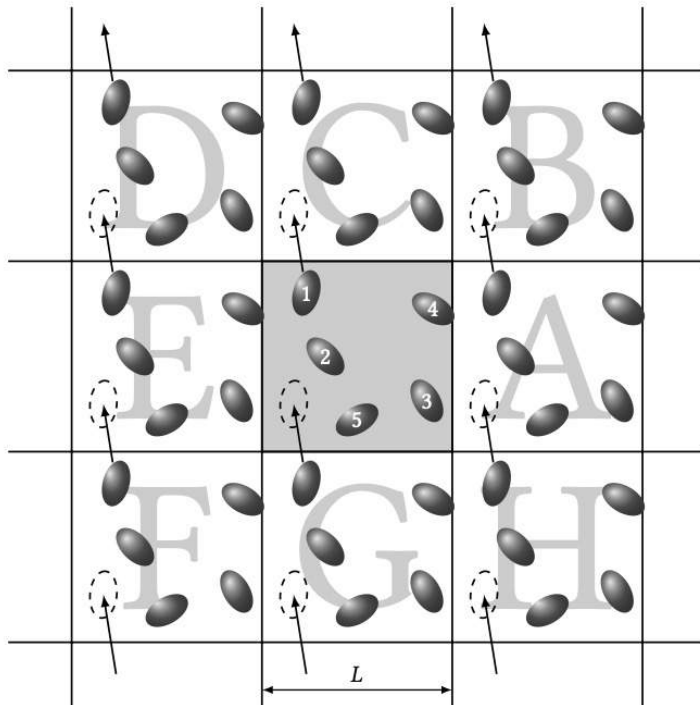
Water sphere

VS



Water box

[補充] 表面張力 & 週期邊界條件



Tutorial: 蛋白質模擬 (Protein)

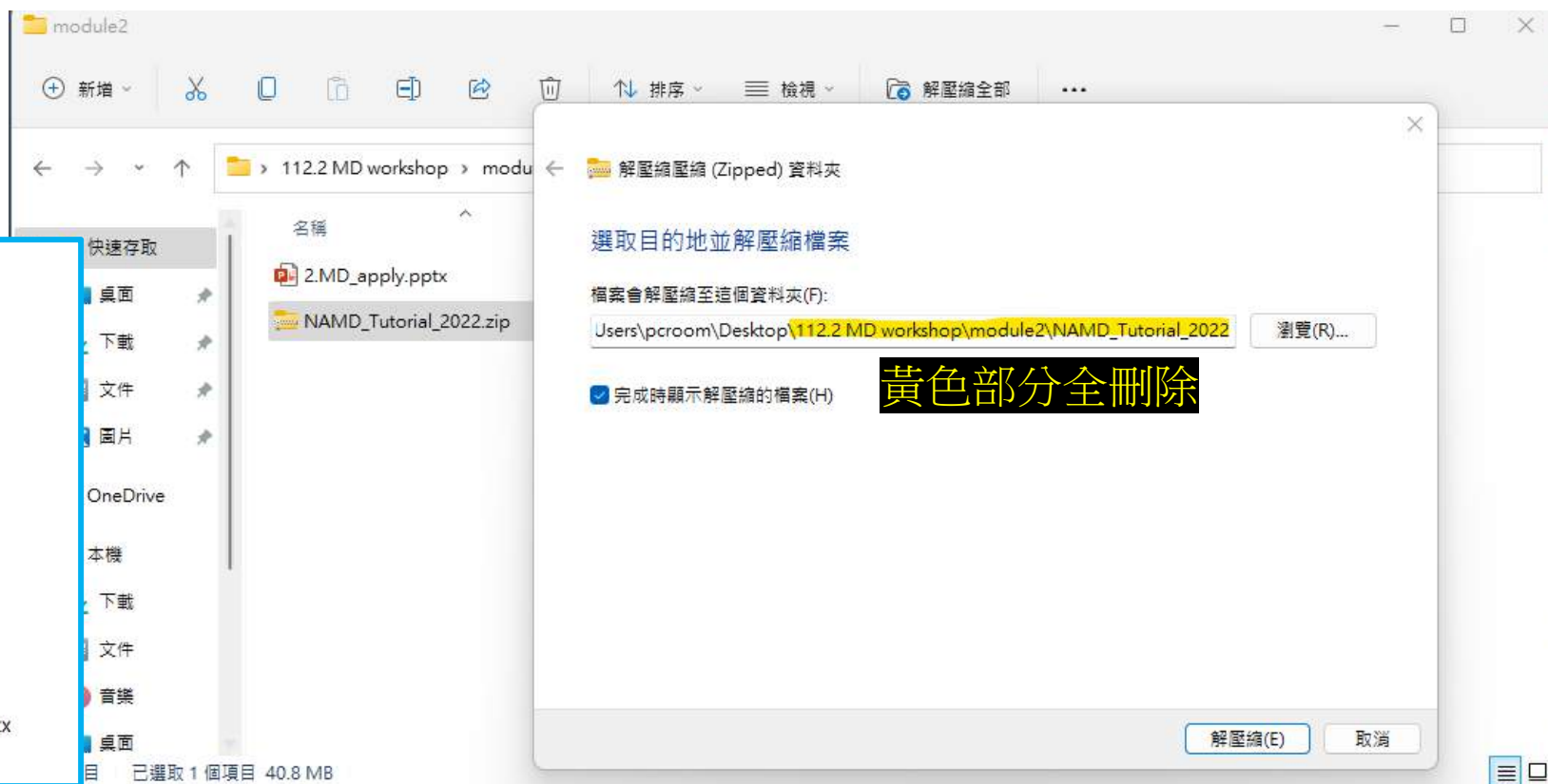
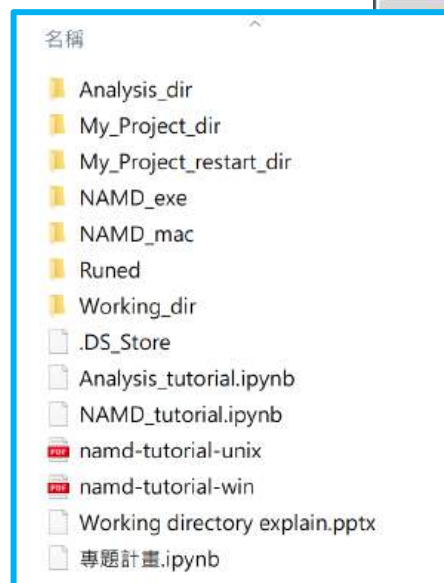
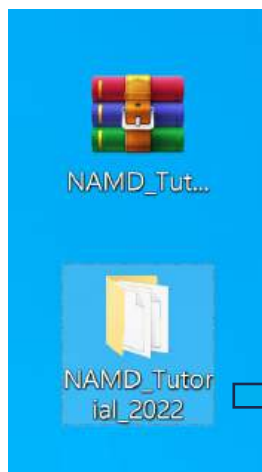
方法一：

1. 從 113.2 MD workshop/module3 中，將 **NAMD_Tutorial_2025.zip** 資料夾拖曳至桌面
2. 按滑鼠右鍵 → 解壓縮檔案... → 目的地路徑 \.....\.....\Desktop\ (此處全刪除)
3. 解壓縮

方法二：

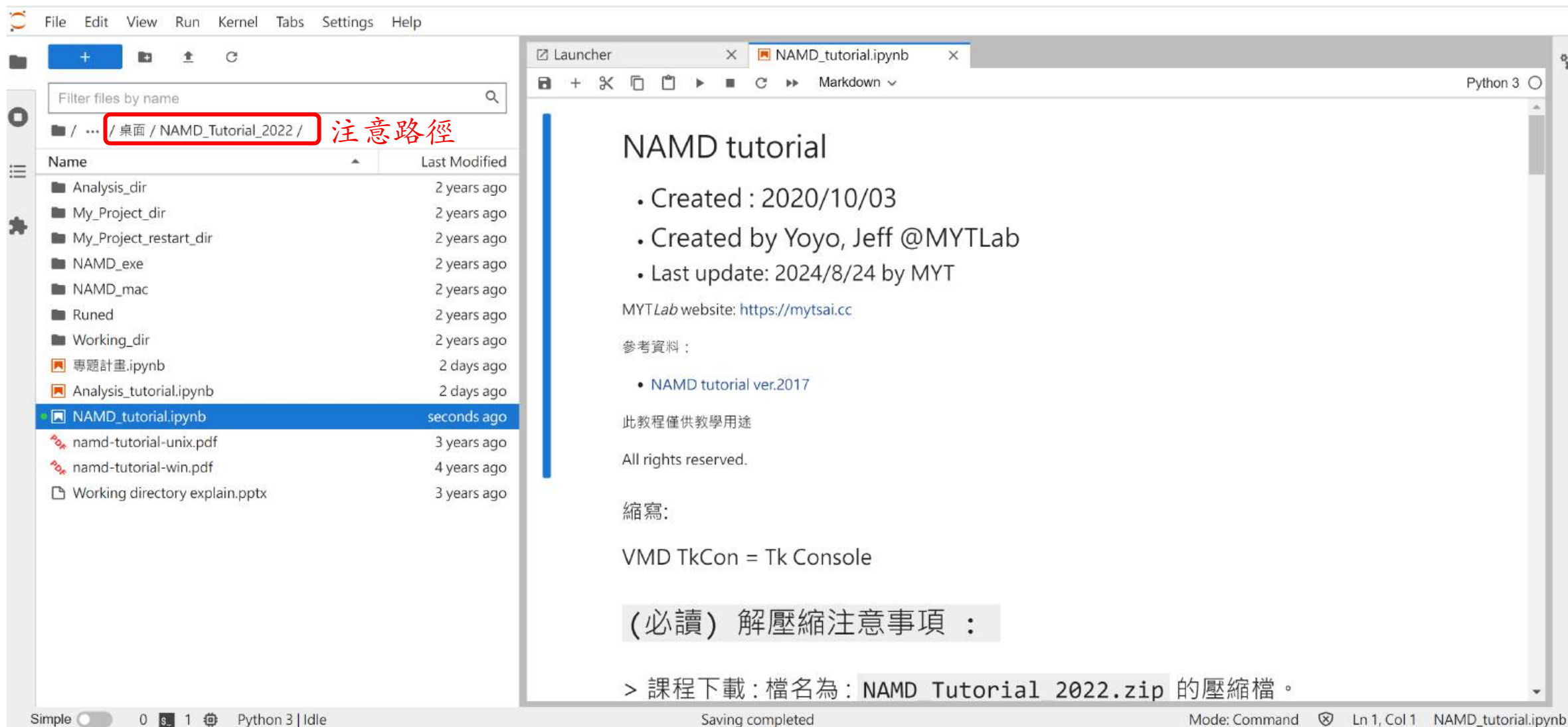
或是直接從 **NAMD_Tutorial_2025.zip** 上按右鍵 → 解壓縮檔案...

桌面



Tutorial: 蛋白質模擬 (Protein)

開啟 JupyterLab



The screenshot shows the JupyterLab interface. On the left, the file browser displays the directory structure. The path `/ 桌面 / NAMD_Tutorial_2022 /` is highlighted with a red box, and the text **注意路徑** (Note path) is written next to it. The file `NAMD_tutorial.ipynb` is selected. On the right, the notebook content is displayed, titled "NAMD tutorial".

NAMD tutorial

- Created : 2020/10/03
- Created by Yoyo, Jeff @MYT Lab
- Last update: 2024/8/24 by MYT

MYT Lab website: <https://mytsai.cc>

參考資料 :

- [NAMD tutorial ver.2017](#)

此教程僅供教學用途

All rights reserved.

縮寫:

VMD TkCon = Tk Console

(必讀) 解壓縮注意事項 :

> 課程下載: 檔名為: `NAMD Tutorial 2022.zip` 的壓縮檔。

補充 **configuration file** 參數說明
(以 NAMD_Tutorial_2022/Working_dir 內檔案為例說明)

Configuration File (ubq_wb_eq.conf)

ubq_wb_0.15

```
structure      ubq_wb.psf
coordinates    ubq_wb.pdb
```

structure: 指定系統的 PSF 檔案 (Protein Structure File)，定義了原子的拓撲結構 (Topology)，如鍵結、角度、質量、原子類型等。

coordinates: 指定系統的 PDB 檔案 (Protein Data Bank)，含有每個原子的初始三維座標 (但不含鍵結資訊)。

```
set temperature      310
set outputname        ubq_wb_0.15_eq
firsttimestep         0
```

temperature: 設定模擬的目標溫度 (單位 K)。

outputname: 定義輸出的檔名前綴。

firsttimestep: 模擬開始的步數。

```
paraTypeCharmm      on
parameters           par_all27_prot_lipid.inp
temperature          $temperature
```

paraTypeCharmm: 是否使用 CHARMM 格式的力場 (此處為 “on”)。

parameters: 指定參數檔 (含力場常數、非鍵結參數等)。

temperature: 設定模擬的初始溫度 (溫度與上方相同)。

```
# Force-Field Parameters
exclude          scaled1-4
1-4scaling       1.0
cutoff           12.0
switching        on
switchdist       10.0
pairlistdist     14.0
```

exclude scaled1-4: 排除共價鍵連接的 1-2, 1-3 原子間交互作用，1-4 原子進行縮放。

1-4scaling: 縮放 1-4 靜電作用力大小 (此處為 1.0，表示不縮放)。

cutoff: 非鍵結力截斷半徑，計算距離內凡德瓦力 (此處為 12 Å)。

switching : 是否啟用 switching function 平滑處理在 cutoff 區間內的力變化 (此處為 **on**)。

switchdist: switching function 開始的距離 (此處為 10 Å)。

pairlistdist: 原子對列表更新的距離 (此處為 14 Å)。

```
# Integrator Parameters
timestep         2.0   ;# 2fs/step
rigidBonds       all   ;# needed for 2fs steps
nonbondedFreq    1
fullElectFrequency 2
stepspcycle      10
```

timestep: 積分步長，單位為 fs (此處為 2 fs)。

rigidBonds: 使用 SHAKE 讓氫鍵長固定 (需使用 2 fs 步長)。

nonbondedFreq: 非鍵結力計算頻率 (此處為 1 step)。

fullElectFrequency: 完全電場計算頻率 (此處為 2 steps)。

stepspcycle: 定義內部時間整合器如何平衡力的更新頻率與效率 (此處為 10 steps)。

```
# Constant Temperature Control
langevin          on      ;# do langevin dynamics
langevinDamping   1       ;# damping coefficient (gamma) of 1/ps
langevinTemp      $temperature
langevinHydrogen  off      ;# don't couple langevin bath to hydrogens
```

langevin : 是否啟用 Langevin 熱浴模擬恆溫條件 (此處為 on)。

langevinDamping: 阻尼係數，控制與熱浴耦合強度 (此處為 1 ps^{-1})。

langevinTemp: 設定目標溫度 (溫度與上方相同)。

langevinHydrogen : 是否對氫原子施加 Langevin 力 (此處為 off)。

```
# Periodic Boundary Conditions
cellBasisVector1  40.6    0.    0.0
cellBasisVector2   0.0   42.2    0.0
cellBasisVector3   0.0    0   46.2
cellOrigin        30.8   29.0   17.7
wrapAll           on
```

cellBasisVector[1-3]: 定義模擬盒的邊界向量，這裡是正交盒。

cellOrigin: 模擬盒的原點位置。

wrapAll: 是否將穿越邊界原子包裹回盒內 (此處為 on)。

```
# PME (for full-system periodic electrostatics)
PME                yes
PMEGridSpacing     1.0
```

PME: 是否使用 Particle Mesh Ewald 計算長距離靜電力 (此處為 yes)。

PMEGridSpacing: 網格間距設定， $1 \text{ \AA}$ 網格點間距通常能提供足夠精度。

```
# Constant Pressure Control (variable volume)
useGroupPressure   yes ;# needed for rigidBonds
useFlexibleCell    no
useConstantArea    no

langevinPiston     on
langevinPistonTarget 1.01325 ;# in bar -> 1 atm
langevinPistonPeriod 100.0
langevinPistonDecay 50.0
langevinPistonTemp  $temperature
```

useGroupPressure: 是否使用群體壓力方法 (此處為 yes)。

useFlexibleCell: 是否讓模擬盒在所有方向均等調整 (此處為 no)。

useConstantArea: 模擬盒是否固定 XY 平面面積、僅允許 Z 軸改變體積 (此處為 no)。

langevinPiston: 啟用 piston 模型模擬等壓系統 (此處為 on)。

langevinPistonTarget: 設定目標壓力 (此處為 1 atm)。

langevinPistonPeriod: 活塞震盪的時間週期 (此處為 100fs)。

langevinPistonDecay: 阻尼時間 (此處為 50fs)。

langevinPistonTemp: 活塞也需與溫度耦合，設為與主模擬相同。

```
# Output
outputName      $outputname

restartfreq      500      ;# 500steps = every 1ps
dcdfreq          250
xstFreq          250
outputEnergies   100
outputPressure   100
```

outputName: 所有輸出檔案（如 DCD, XST, restart, log）之檔名前綴。

restartfreq: 幾步輸出一一次重啟檔 (此處為 500 steps)。

dcdfreq: 幾步輸出一一次 DCD 座標 (此處為 250 steps)。

xstFreq: 幾步輸出一一次盒子大小 (此處為 250 steps)。

outputEnergies: 幾步輸出一一次能量資訊至 log 檔 (此處為 100 steps)。

outputPressure: 幾步輸出一一次壓力資訊至 log 檔 (此處為 100 steps)。

```
# Minimization
minimize          100
reinitvels        $temperature

run 2500 ;# 5ps
```

minimize: 進行幾步的能量最小化 (此處為 100 steps)。

reinitvels: 以目標溫度初始化系統原子速度分布 (溫度與上方相同)。

run: 模擬步數設定 (此處為 2500 steps，5 ps)。

Restart (ubq_wb_eq_restart.conf)

如果要接續先前的模擬工作（restart）需要對 Configuration file (.conf) 進行修改：

Step 1 ~ Step 4

Step 1. 加上下方參數和對應檔案路徑，以讀取前次模擬之結果與系統參數

透過設定參數“1”開啟讀取前此資料；”0”則關閉

輸入前一次模擬的輸出檔路徑與前綴

```
# Continuing a job from the restart files
if {1} {
  set inputname    ../ubq_wb_0.15_eq
  binCoordinates   $inputname.restart.coor
  binVelocities    $inputname.restart.vel  ;# remove the "temperature" entry if you use this!
  extendedSystem   $inputname.restart.xsc
}
```

binCoordinates: 讀取前一次模擬結束時儲存的原子位置 (*.restart.coor)。

binVelocities: 讀取前一次模擬的原子速度 (*.restart.vel)。

extendedSystem: 讀取模擬盒子大小與形狀（PBC）的資訊，保持空間一致性 (*.restart.xsc)。

Step 2. 將力場設定參數下方的 temperature 前方加上 “#” 使其變成註解

```
paraTypeCharmm on
parameters      par_all27_prot_lipid.inp
temperature    $temperature
```

```
paraTypeCharmm on
parameters      par_all27_prot_lipid.inp
#temperature  $temperature
```

Step 3. PBC設定參數則添加 “if” 指令用於控制此參數組

```
# Periodic Boundary Conditions
cellBasisVector1 40.6 0. 0.0
cellBasisVector2 0.0 42.2 0.0
cellBasisVector3 0.0 0 46.2
cellOrigin        30.8 29.0 17.7
wrapAll           on
```

```
# Periodic Boundary Conditions
# if run restart change 1 to 0 #
if {0} {
cellBasisVector1 40.6 0. 0.0
cellBasisVector2 0.0 42.2 0.0
cellBasisVector3 0.0 0 46.2
cellOrigin        30.8 29.0 17.7
}
wrapAll           on
```

透過設定參數 0 關閉讀取此參數；1 則開啟

Step 4. 將minimize 、reinitvels 前方加上 “#” 使其成為註解

```
# Minimization
minimize          100
reinitvels       $temperature

run 2500 ;# 5ps
```

```
# Minimization
# minimize          100
# reinitvels       $temperature

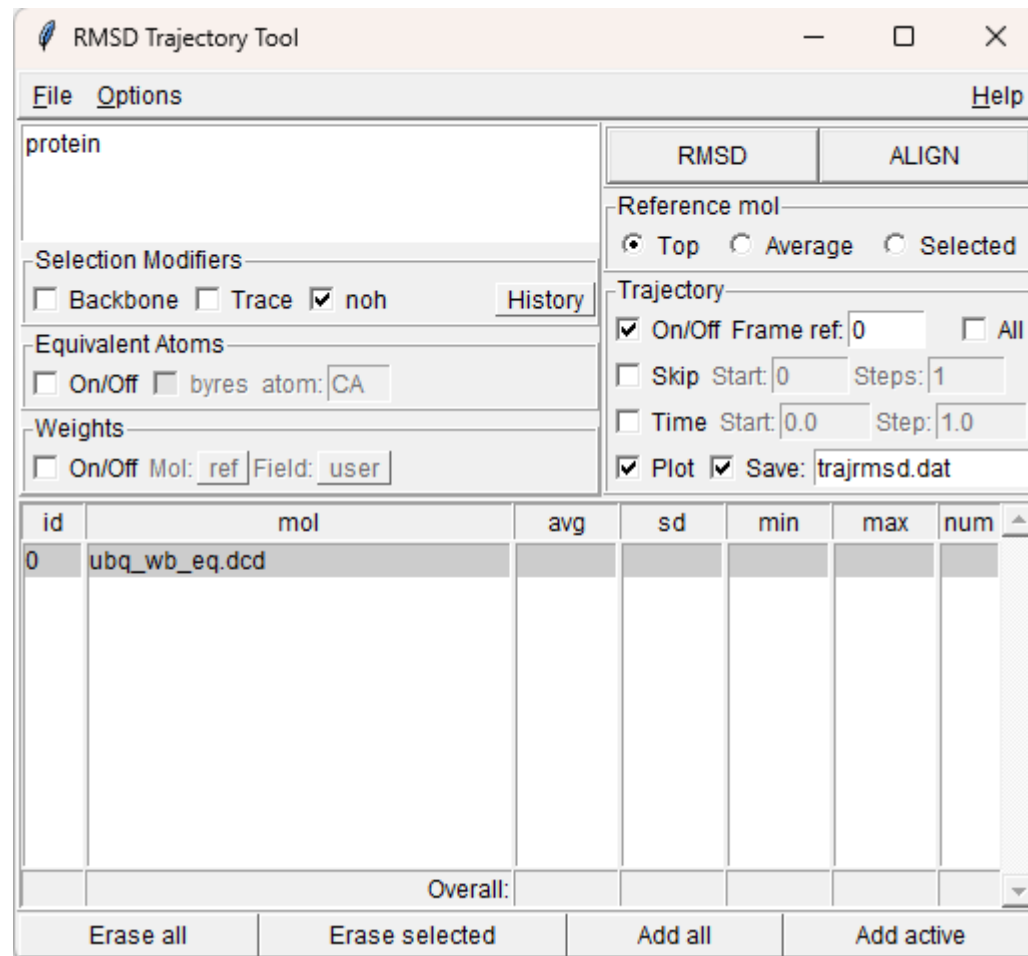
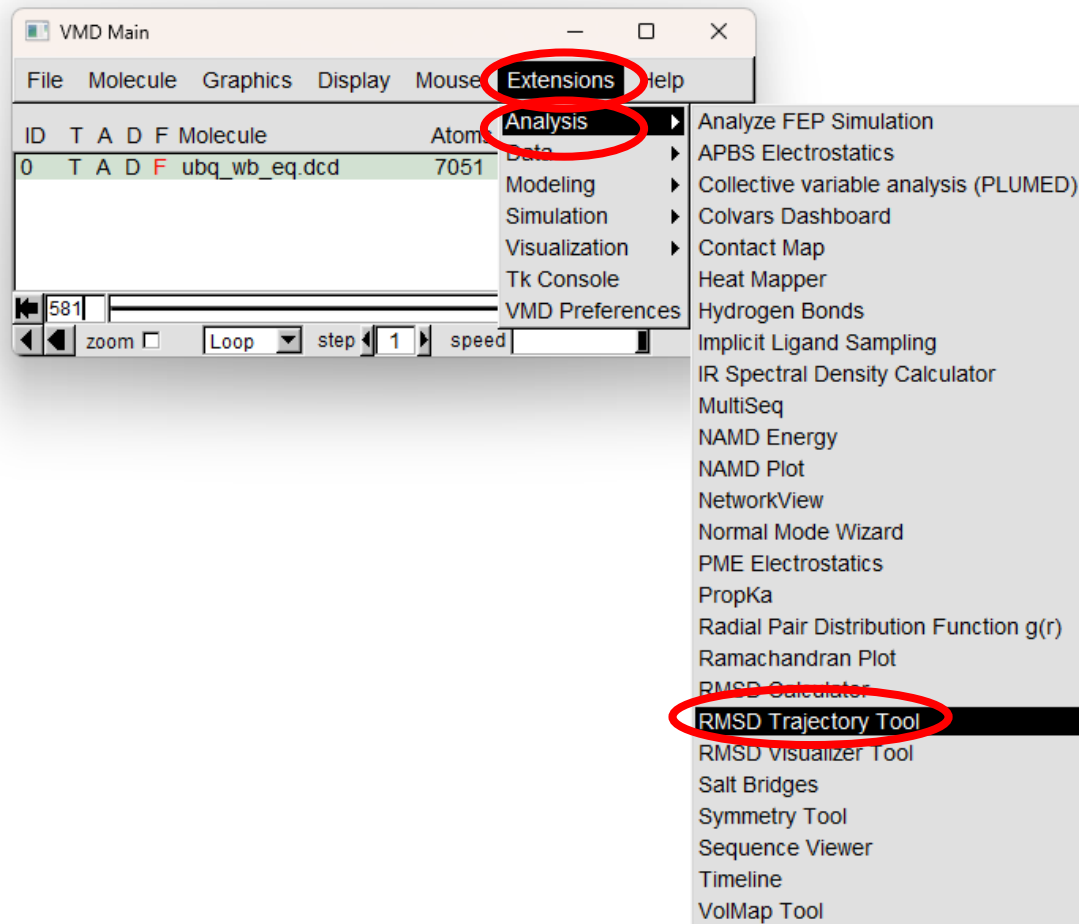
run 50000 ;# 100ps
```

run 2500 ; # 5ps

RMSD Trajectory Tool Plug-in

GUI 視窗化介面

開啟 RMSD Trajectory Tool



介紹 RMSD Trajectory Tool

定義選取分子結構的關鍵字

軟體預先定義的選取分子結構選項

Backbone: 選擇分子骨架的部分

Trace: 選擇所有C α 原子

noh: 排除氫原子

加入的結構會顯示在這裡

去除所有結構

去除選擇結構

加入所有結構

加入新結構

RMSD: 計算RMSD

ALIGN: 將結構對齊至參考分子

選取參考結構:

Top: VMD Main 正在操作的分子ID (前面有個 "T")

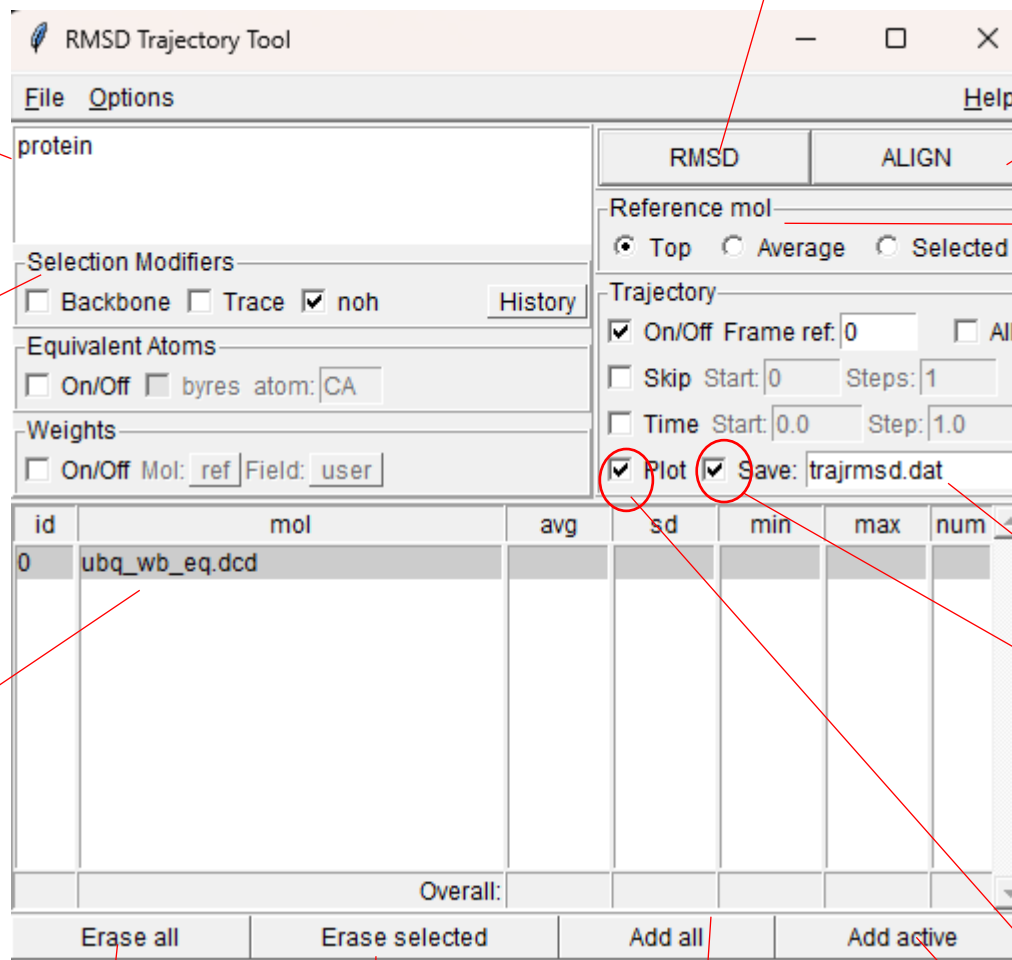
Average: 以所有結構的平均結構為參考 (不適用於 ALIGN)

Selected: 使用表格中選取的第一個分子作為參考。

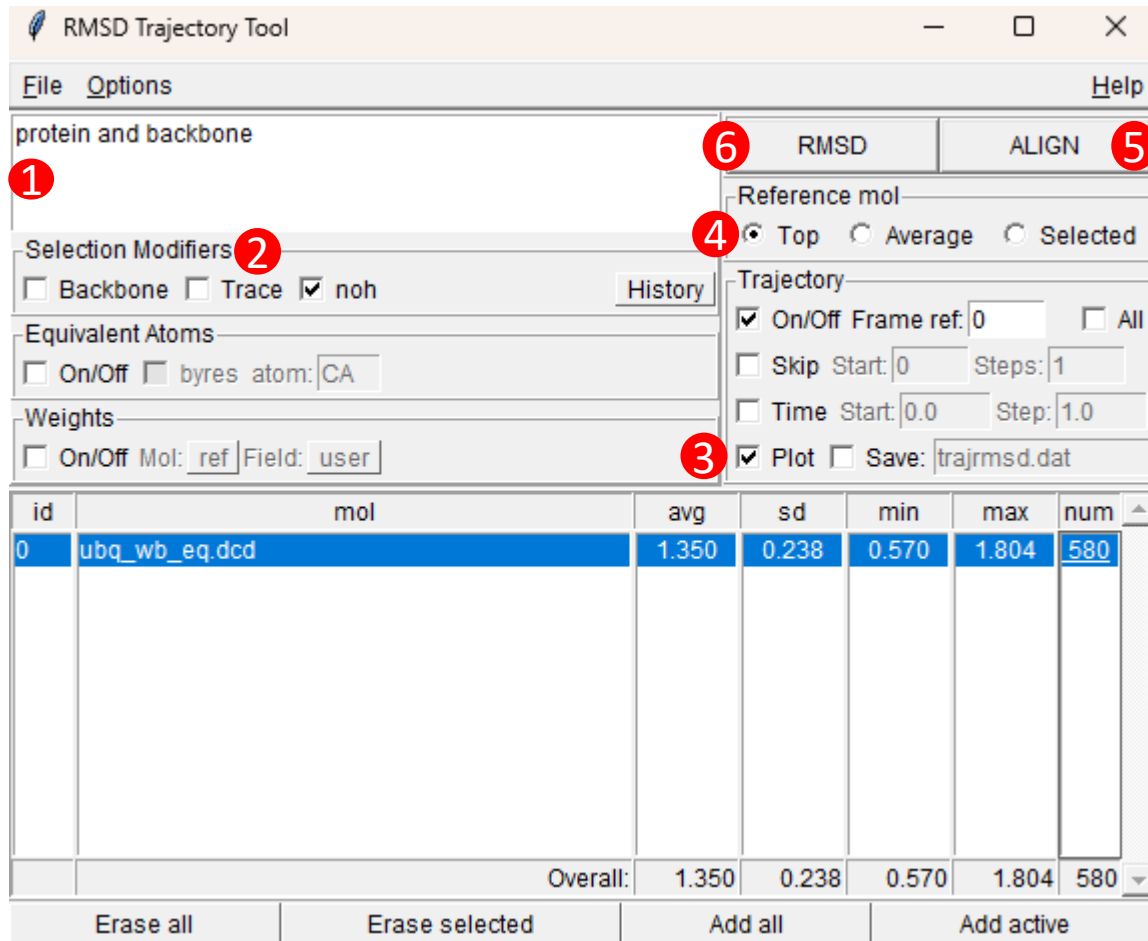
文字檔的檔名

讓插件輸出:文字檔

顯示出RMSD變化的趨勢圖



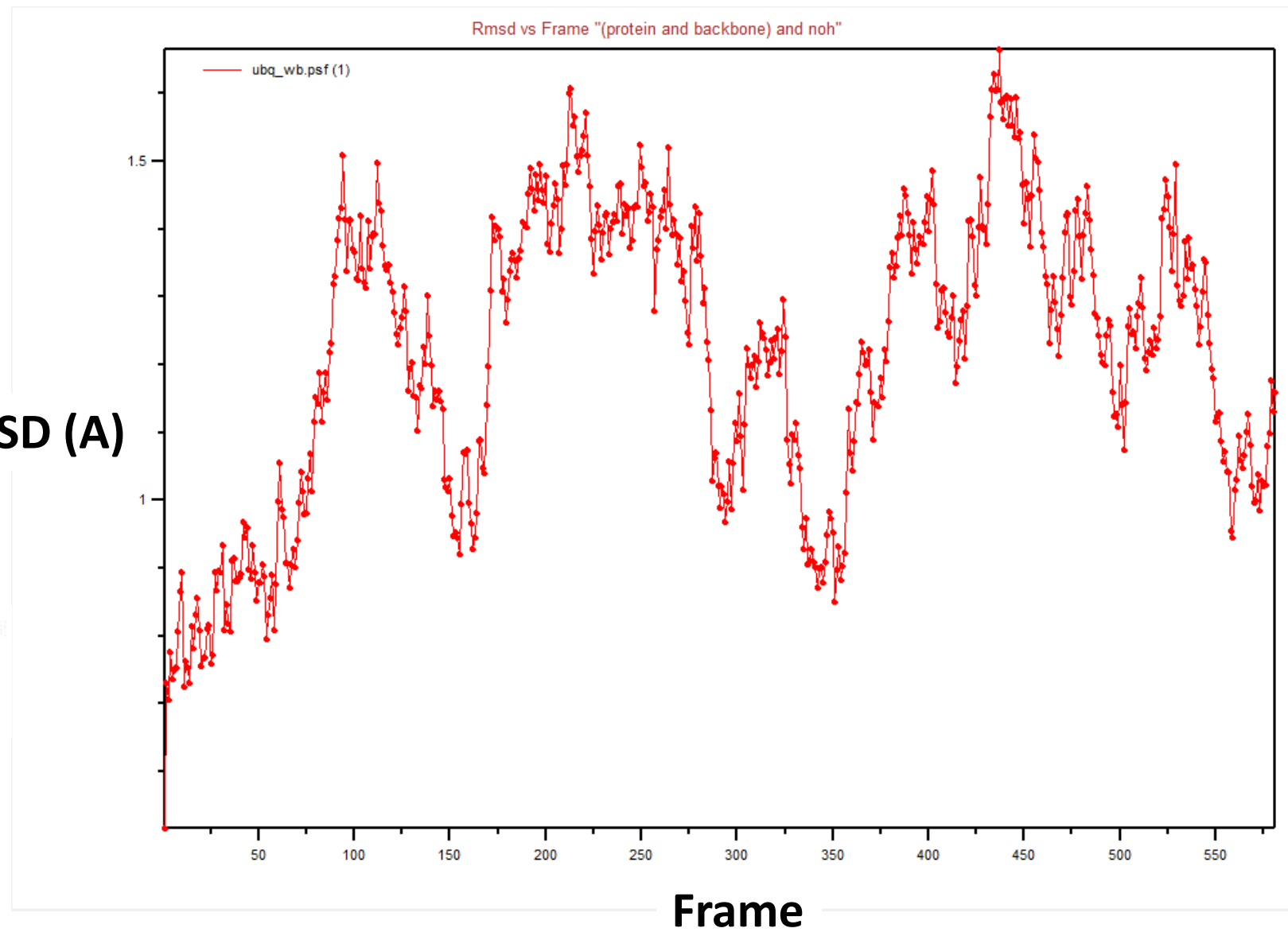
使用範例



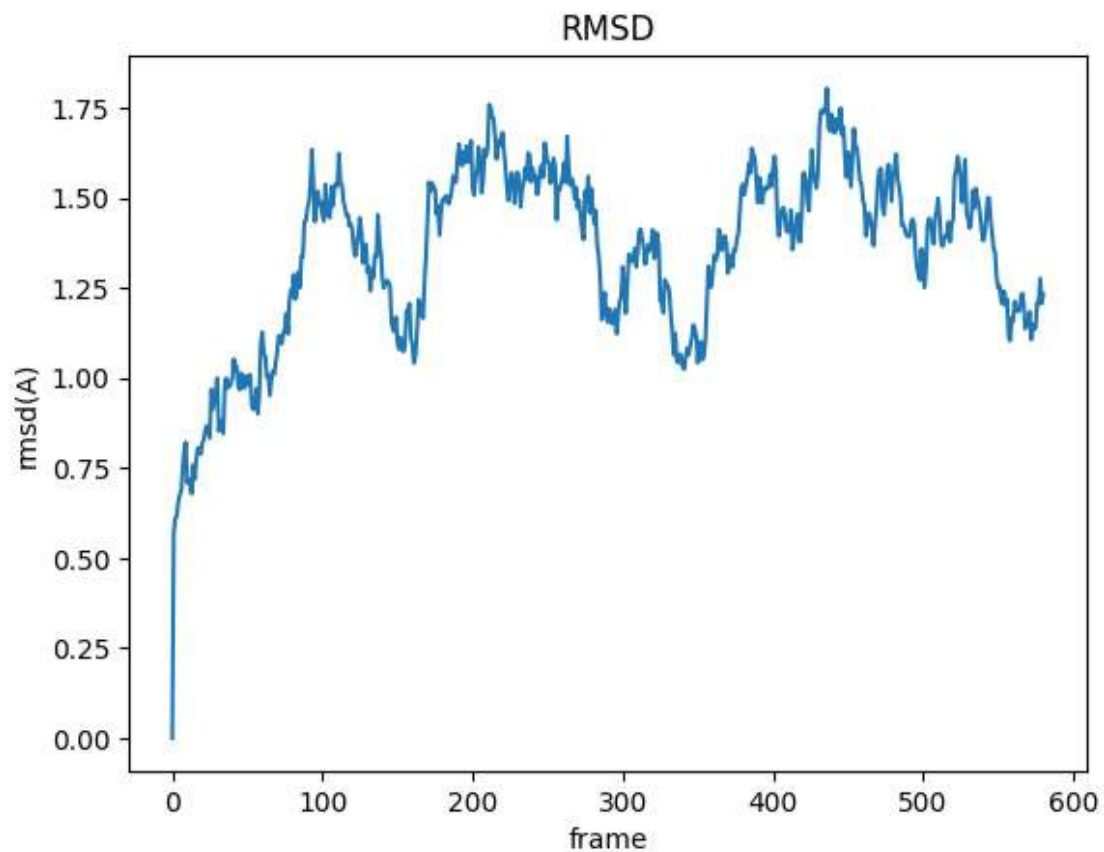
- 1 選擇蛋白質且只選擇骨架的部分
- 2 勾選 noh 是去除氫 (它一次只能選一樣)
- 3 勾選 plot 軟體才會畫出圖來
- 4 點選 Top 的意思是, VMD Main 正在操作的分子 ID (前面有個 "T"), 而下方 Trajectory 的部分,
☒ On/Off Frame ref: 0 ☐ All
則表示我們是選取第 0 個 frame 作為參考結構。
- 5 結構對齊 (以第 4 步的 Frame 0 為參考結構)
- 6 進行 RMSD 計算

結果 → RMSD (Å)

$$RMSD_{\alpha}(t_j) = \sqrt{\frac{\sum_{\alpha=1}^{N_{\alpha}} (\vec{r}_{\alpha}(t_j) - \langle \vec{r}_{\alpha} \rangle)^2}{N_{\alpha}}}$$

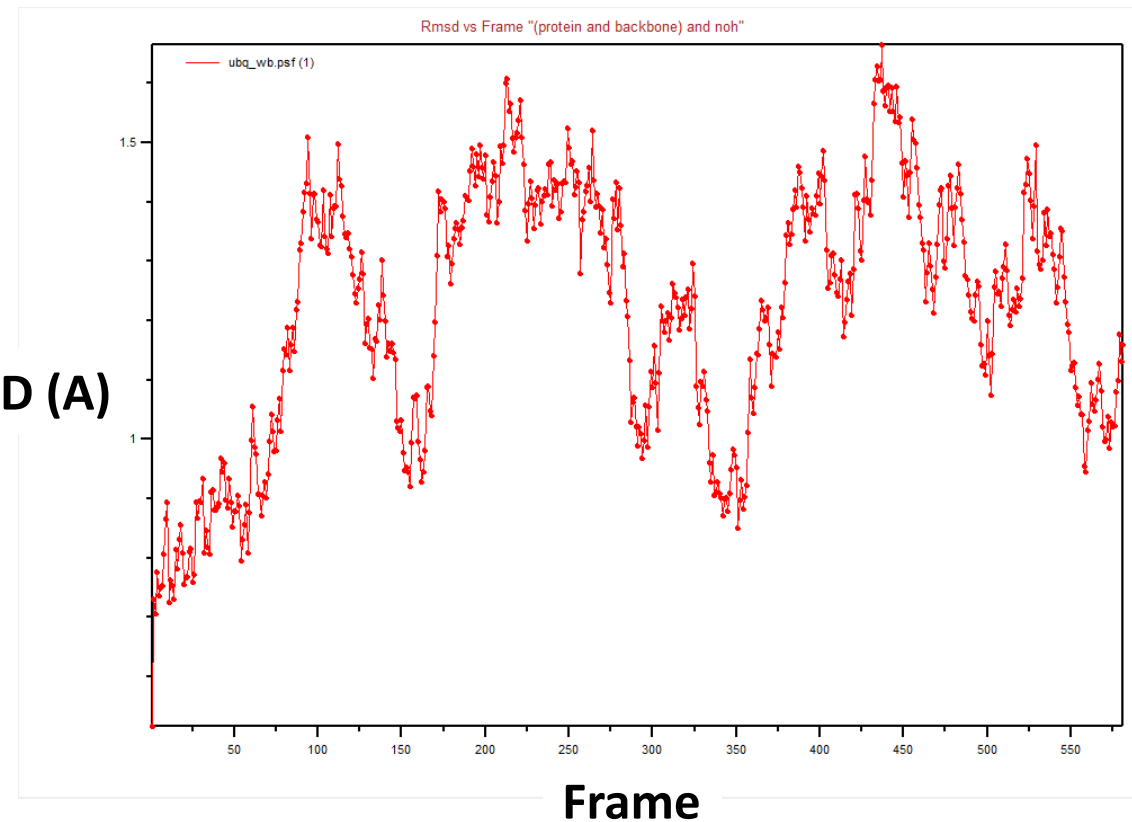


Script vs GUI 結果比較



Python

RMSD (A)



GUI

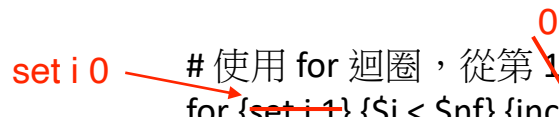
[補充] 程式碼解讀

開啟一個寫入檔案 sample.dat，並將檔案控制權賦值給變數 outfile
set outfile [open sample.dat w]

取得當前(top)分子模型的總幀數，並賦值給變數 nf
set nf [molinfo top get numframes]

選取第 0 幀中的 backbone 非氫原子，作為對齊與計算的參考結構，存為 frame0
set frame0 [atomselect top "protein and backbone and noh" frame 0]

選取整個軌跡中的 backbone 非氫原子，準備對齊與 RMSD 計算
set sel [atomselect top "protein and backbone and noh"]

# 使用 for 迴圈，從第 1 幀開始，逐一與 frame0 對齊並計算 RMSD
for {~~set i 1~~} {\$i < \$nf} {incr i} {
 # 將 sel 切換到第 i 幀
 \$sel frame \$i

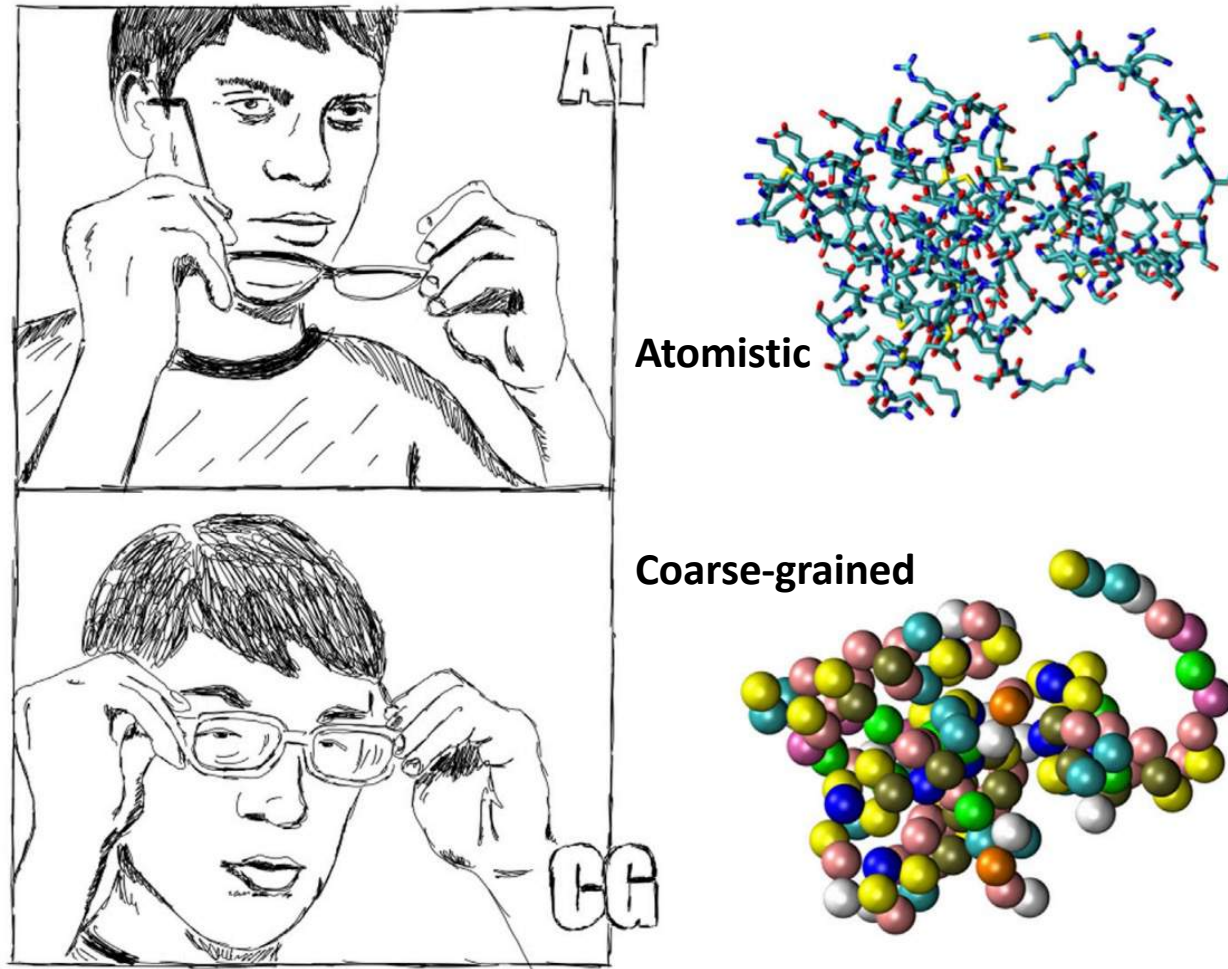
對齊第 i 幀與第 0 幀的 backbone 結構，並將 sel 移動到最佳重疊位置
\$sel move [measure fit \$sel \$frame0]

計算對齊後的 RMSD，並寫入 outfile
puts \$outfile "[measure rmsd \$sel \$frame0]"
}

關閉檔案，完成寫入
close \$outfile

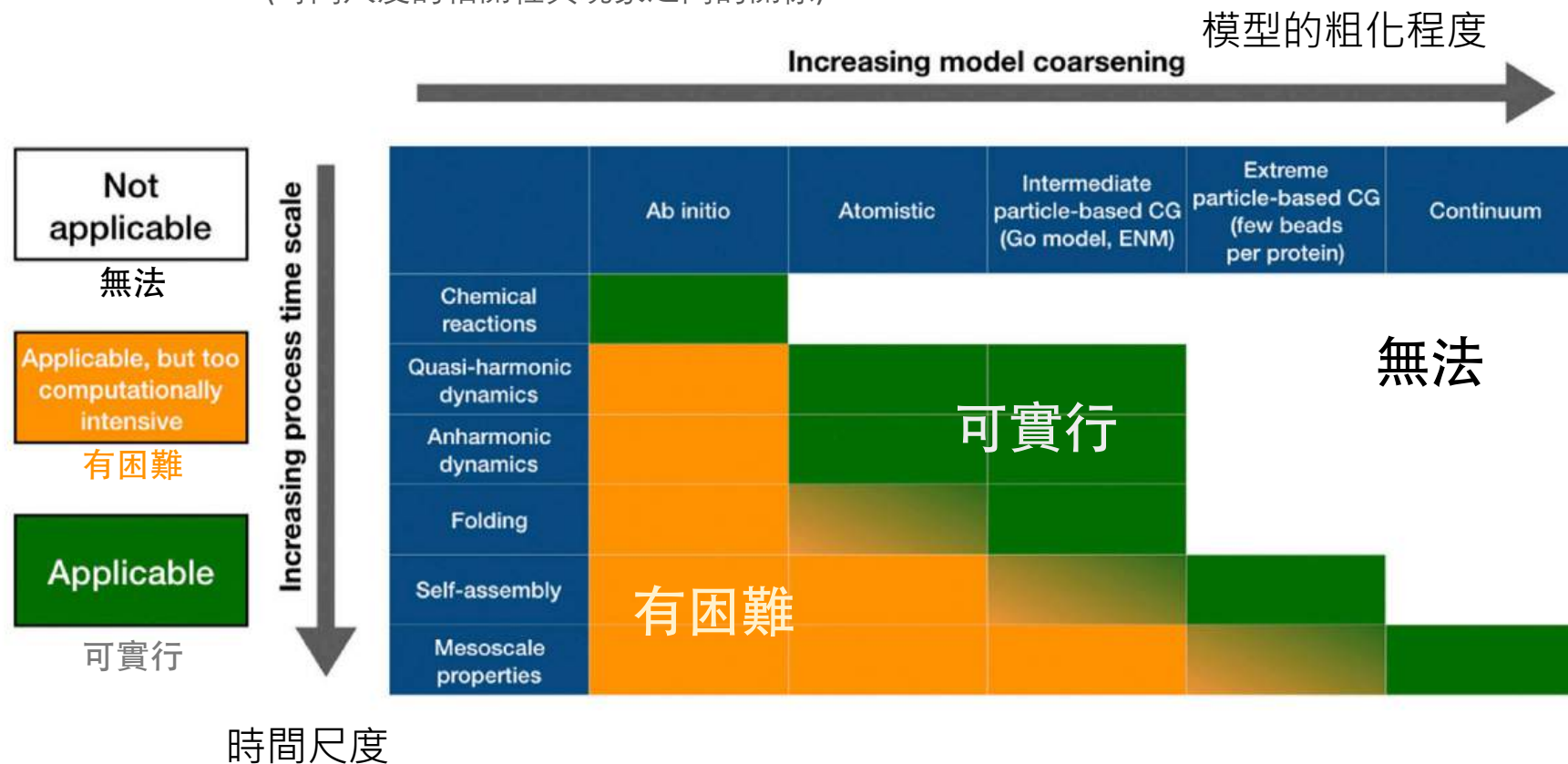
[補充] 粗粒化模型

- The loss of detail is a defect, however, it does have the advantage of simplicity and parsimony
(粗粒化有缺點，但也具有簡化後的優勢)



[補充] 時間尺度與解析程度

- (時間尺度的相關性與現象之間的關係)



Giulini, M.; Rigoli, M.; Mattiotti, G.; Menichetti, R.; Tarenzi, T.; Fiorentini, R.; Potestio, R. *Frontiers in Molecular Biosciences* **2021**, 8. <https://doi.org/10.3389/fmolb.2021.676976>.

[補充] 實驗、理論與模擬之間的連結

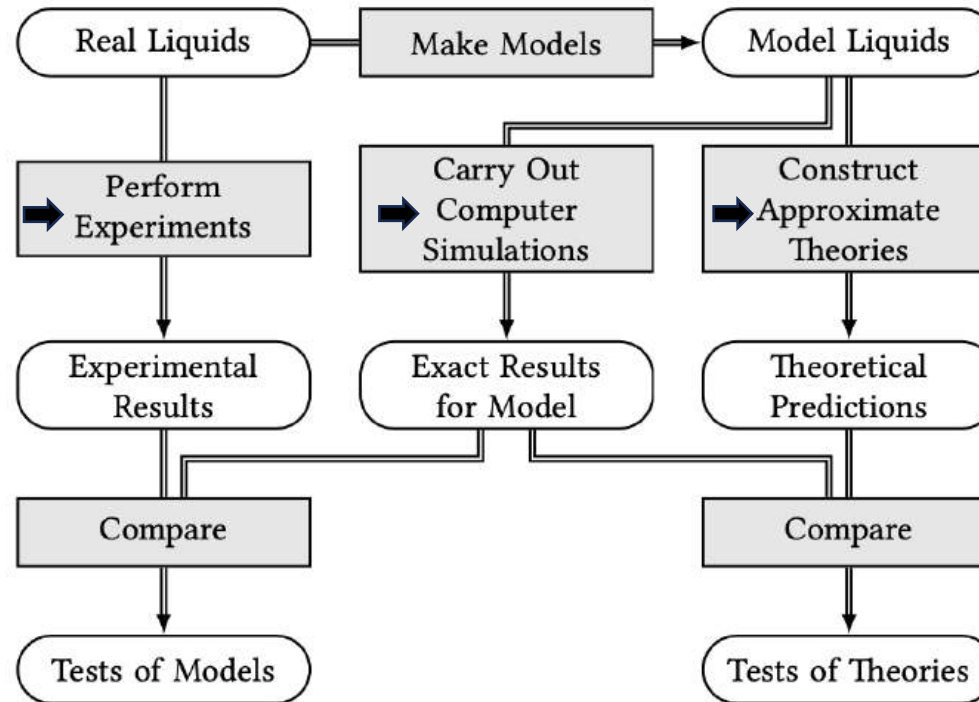


Fig. 1.2 The connection between experiment, theory, and computer simulation.



I. Simple MD

- Python
- JupyterLab



JupyterLab



II. Visualization

- MolView
- VMD

VMD
Visual Molecular Dynamics



Part IV

(MD in Nano)

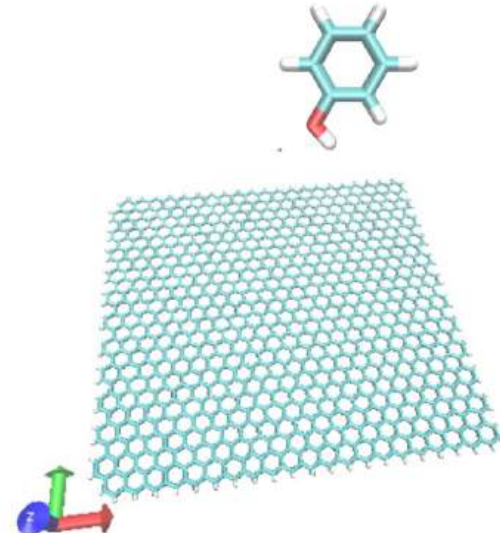


- ### III. MD (Bio)
- NAMD/VMD
 - Protein



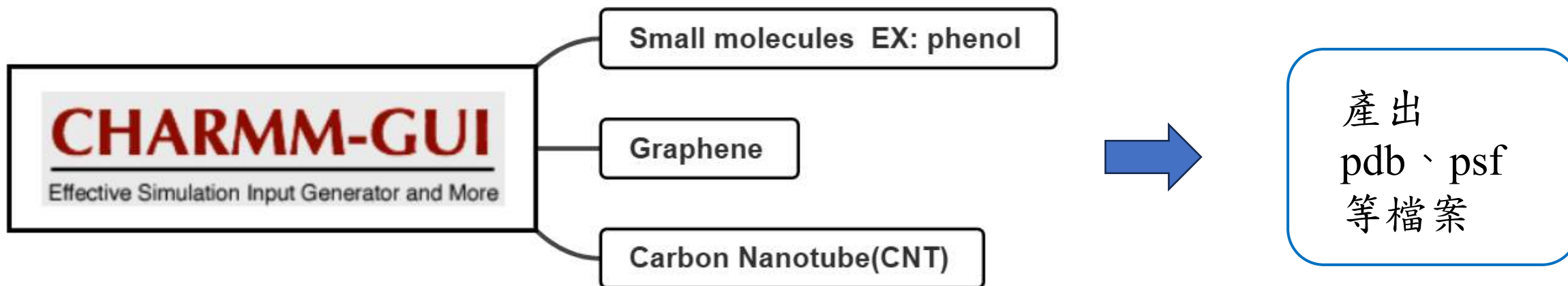
IV. MD (Nano)

- NAMD/VMD
 - Graphene
 - Small molecules



NAMD
Scalable Molecular Dynamics

CHARMM-GUI 建模教學



Small molecules 建置



Input Generator
Job Retriever
Force Field Converter
PDB Reader & Manipulator
Glycan Reader & Modeler
Ligand Reader & Modeler
Glycolipid Modeler
LPS Modeler
Nanomaterial Modeler
Multicomponent Assembler
Solution Builder
Membrane Builder
Martini Maker
PACE CG Builder
Polymer Builder
Drude Prepper
Enhanced Sampler
Constant-pH Simulator
Free Energy Calculator
LBS Finder & Refiner
Ligand Designer
Ligand Docker
EnzyDocker
High-Throughput Simulator
QM/MM Interfacer

Ligand Reader & Modeler

Ligand Reader & Modeler aims to provide:

- residue name in the CHARMM topology files if the ligand is defined,
- if not, the CHARMM-compatible topology and parameter files of the ligand using CgenFF
- PSF / CRD / PDB files of the ligand

Therefore, these outputs can be used in other CHARMM-GUI modules to model a biomolecular system contain

Please note that

- There is a Ligand Reader & Modeler demo in [Video Demo](#) to illustrate how to use this module.
- This page is powered by [Marvin JS](#). Please see its [user's guide](#) for help.
- Only molecules in the HETATM record in a PDB file are recognized.
- Since carbohydrates are automatically recognized in PDB Reader via Glycan Reader, carbohydrates are
- When there are multiple molecules of an identical ligand in a PDB file, only one structure is used for input
- The ligand structure and its topology and parameter files are based on the structure shown on Marvin JS
- In the case of "Load PDB ID" and "Load Ligand ID", the hydrogen atoms are explicitly generated based on Component Dictionary. If the selected ligand structure is the same structure or substructure of the RCSB protonation state if necessary.
- **All hydrogen and missing atoms should be explicitly placed for accurate result.**
- If the uploaded structure is modified using Marvin JS, the coordinates of the unmodified heavy atoms will
- Please draw the ligand in detail, including chirality, bond type, and atom charge using the provided option
- For lipid-like molecules (having long acyl chains), the similar structure search using the maximum common
- The combinatorial structure generation is possible. Please see [R-Group](#).
- If the PDB structure has the multiple chains, non-modified chains will be combined with modified ligand in

Reference for Ligand Reader & Modeler:

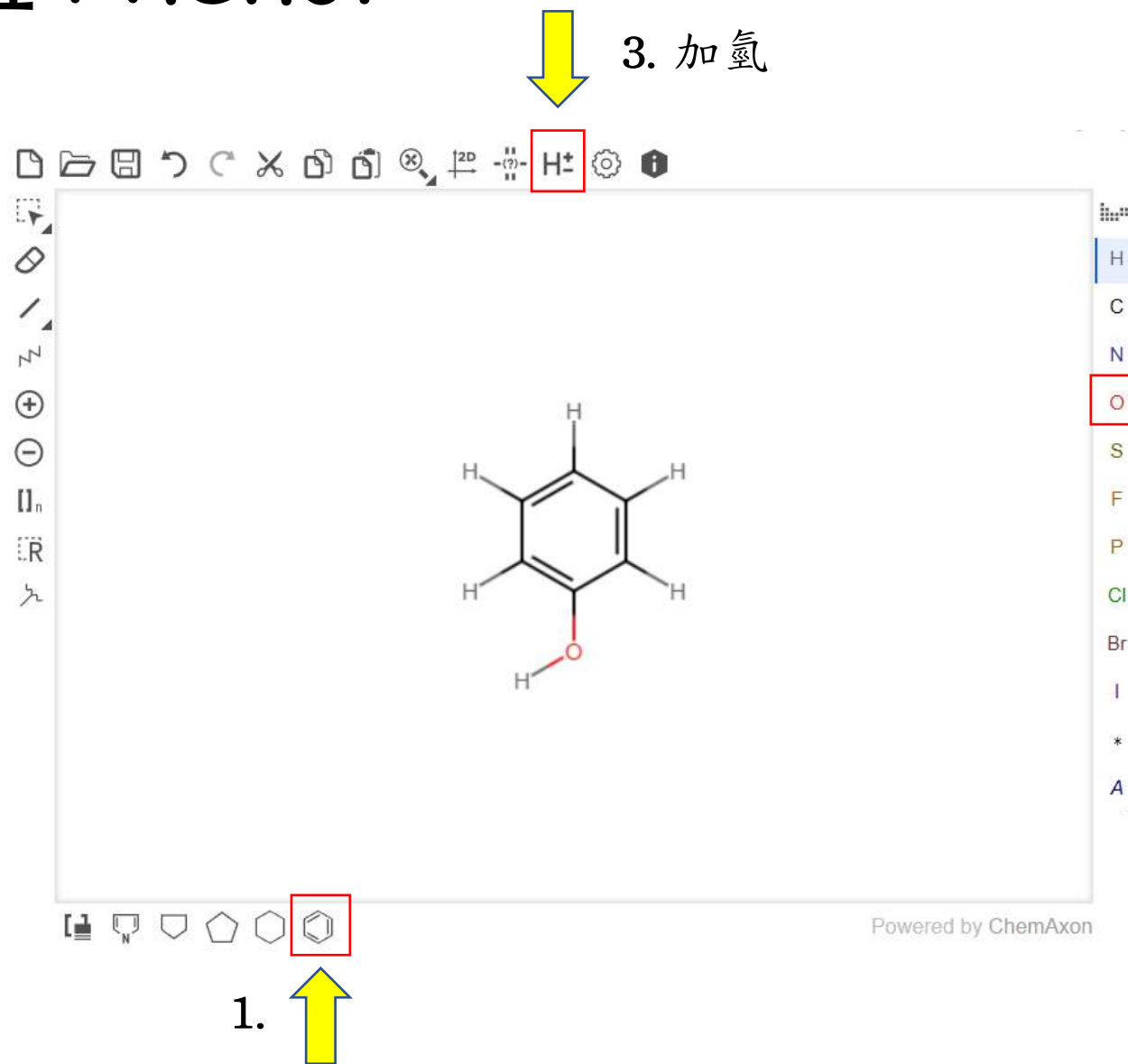
S. Jo, T. Kim, V.G. Iyer, and W. Im (2008)

CHARMM-GUI: A Web-based Graphical User Interface for CHARMM. [J. Comput. Chem. 29:1859-1865](#)

S. Kim, J. Lee, S. Jo, C.L. Brooks III, H.S. Lee, and W. Im (2017)

CHARMM-GUI Ligand Reader and Modeler for CHARMM Force Field Generation of Small Molecules. [J. Comput. Chem.](#)

Ex: 建置 Phenol



2. 從碳上拉出氧

4. 按右下角下一步

Next Step:
Search ligand 

5. 確認為phenol的parameter

Exact

Click the residue name to visualize the structure.

residue name	Charge	Residue in	Comment
PHEN	0.00	toppar_all36_prot_model.str	PHENOL, ADM JR.

Isomer

No isomer in CHARMM forcefield.

6. 按右下角產出
pdb、psf檔

Next Step:
Generate PDB



Ligand Reader & Modeler

User Pro

Bookmark this [link](#), if you want to comeback to this page

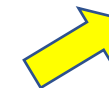
PDB Info

CHARMM Input: [ligandrm.inp](#)
CHARMM Output: [ligandrm.out](#)
CHARMM PDB: [ligandrm.pdb](#) ([view structure](#))
CHARMM CRD: [ligandrm.crd](#)
CHARMM PSF: [ligandrm.psf](#)

JOB ID: 5446577346

[download tgz](#)

7. 點擊下載



8. 將ligandrm.pdb & psf
檔名改為phenol.pdb & psf

phenol

2025/3/3 下午 03:40

PDB 檔案

phenol.psf

2025/3/3 下午 03:40

PSF 檔案

Graphene 建置

1. 

Input Generator

- Job Retriever
- Force Field Converter
- PDB Reader & Manipulator
- Glycan Reader & Modeler
- Ligand Reader & Modeler
- Glycolipid Modeler
- LPS Modeler
- Nanomaterial Modeler**
- Multicomponent Assembler
- Solution Builder
- Membrane Builder
- Martini Maker
- PACE CG Builder
- Polymer Builder
- Drude Prepper
- Enhanced Sampler
- Constant-pH Simulator
- Free Energy Calculator
- LBS Finder & Refiner
- Ligand Designer

2. 

CHARMM-GUI Nanomaterial Modeler for Modeling and Simulation of Nanomaterial Systems. [J. Chem. Theory C.](#)
Download our initial systems and inputs [here](#).

Select a material from the list below. Currently selected material: **None**

Nanomaterial Type

- Metals
- Metal Oxides
- Metal Hydroxides
- Battery Materials
- Clay Minerals
- Rad Mica
- Calcium Sulfates
- Cement Minerals
- Calcium Silicate Hydrates
- Silica
- Phosphate minerals
- Transition Metal Dichalcogenides
- Carbonaceous Materials**
- Carbon Nanotube (CNT)
- Graphene
- Graphite

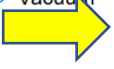
Sphere Options

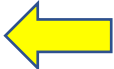
Material Volume

System Volume

System Type

- ☐ Solvated
- ☒ Vacuum

3. 

4. 

References for Nanomaterial Force Fields:

F. S. Emami, V. Puddu, R. J. Berry, V. Varshney, S. V. Patwardhan, C. C. Perry and H. Heinz (2014)
Force Field and a Surface Model Database for Silica to Simulate Interfacial Properties in Atomic Resolution. [Chem](#)

H. Heinz, T.-J. Lin, R. Kishore Mishra and F. S. Emami (2013)
Thermodynamically consistent force fields for the assembly of inorganic, organic, and biological nanostructures: the

T.-J. Lin and H. Heinz (2016)
Accurate force field parameters and pH resolved surface models for hydroxyapatite to understand structure, mecha

5. 設定好Graphene性質

Select a material from the list below. Currently selected material: **Graphene**

Nanomaterial Shape:

Percent Defect

Use virtual electron model ☐

Box Options

Miller Index

X length (Å) (54.3)

Y length (Å) (51.3)

Z length (Å) (3.3)

Material Volume (Å³) 9,317.3

System Volume (Å³) 9,317.3

Unit Cell Info

Unit Cell X 4.934 Å

Unit Cell Y 4.273 Å

Unit Cell Z 3.348 Å

Layer Dimension Z

Number of Layers 1

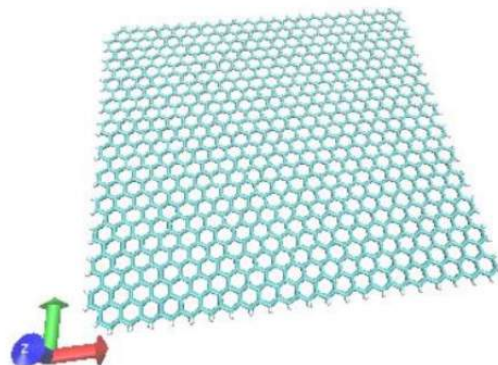
Periodic Options

☒ X ☐ Y ☐ Z

System Type

☐ Solvated

☒ Vacuum



6. 按右下角下一步

Next Step: 
Build Nanomaterial

Building | Size Determination | PBC Setup | Input Generator

JOB ID: 2449995467

Building Input: [step1_nanomaterial.inp](#)
 Building Output: [step1_nanomaterial.out](#)
 Building PSF: [step1_nanomaterial.psf](#)
 Image Patch PSF: [step1_nanomaterial_img.psf](#)
 Building PDB: [step1_nanomaterial.pdb \(view structure\)](#)
 Building Info: [step1_nanomaterial.str](#)

[download .tgz](#)

7. 點擊下載



glycan.yaml	4	Yaml 標識檔案
moleculex.py	9,508	? Python 來源檔案
moleculex.pyc	10,183	? PYC 檔案
rmimg_info.py	10,489	? Python 來源檔案
step1.1_user_input.str	150	? STR 檔案
step1_nanomaterial.crd	274,340	? CRD 檔案
step1_nanomaterial.inp	2,992	? INP 檔案
step1_nanomaterial.out	129,153	? OUT 檔案
step1_nanomaterial.pdb	149,981	? PDB 檔案
step1_nanomaterial.psf	928,180	? PSF 檔案
step1_nanomaterial.str	415	? STR 檔案
step1_nanotube.crd	274,134	? CRD 檔案
step1_nanotube.pdb	157,492	? PDB 檔案
step1_nanotube.rtf	98,373	? RTF 格式
toppar.str	3,670	? STR 檔案

8. 將step1_nanomaterial.pdb & psf
檔名改為graphene.pdb & psf

Carbon nanotube 建置

GCMC/BD Ion Simulator

- The generated structure can be used in [Multicomponent Assembler](#) to model a nano-bio system.

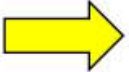
References for Nanomaterial Modeler:

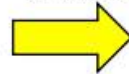
S. Jo, T. Kim, V.G. Iyer, and W. Im (2008)
CHARMM-GUI: A Web-based Graphical User Interface for CHARMM. [J. Comput. Chem. 29:1859-1865](#)

Y.K. Choi, N.R. Kern, S. Kim, K. Kanhaiya, S.H. Jeon, Y. Afshar, S. Jo, B.R. Brooks, J. Lee, E.B. Tadmor, H. Heinz, and W. Im (2022)
CHARMM-GUI Nanomaterial Modeler for Modeling and Simulation of Nanomaterial Systems. [J. Chem. Theory Comput. 18:479-493](#)

Download our initial systems and inputs [here](#).

Select a material from the list below. Currently selected material: **None**

1. 

2. 

Nanomaterial Type

Nanomaterial Shape

Sphere Options

Radius

Material Volume

System Volume

System Type

☒ Solvated

☐ Vacuum

Metals

Metal Oxides

Metal Hydroxides

Battery Materials

Clay Minerals

Mica

Calcium Sulfates

Cement Minerals

Calcium Silicate Hydrates

Silica

Phosphate minerals

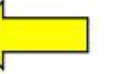
Transition Metal Dichalcogenides

Carbonaceous Materials

Carbon Nanotube (CNT)

Graphene

Graphite

3. 

References for Nanomaterial Force Fields:

F. S. Emami, V. Puddu, R. J. Berry, V. Varshney, S. V. Patwardhan, C. C. Perry and H. Heinz (2014)
Force Field and a Surface Model Database for Silica to Simulate Interfacial Properties in Atomic Resolution. [Chem. Mater. 26:2647-2658](#)

H. Heinz, T.-J. Lin, R. Kishore Mishra and F. S. Emami (2013)
Thermodynamically consistent force fields for the assembly of inorganic, organic, and biological nanostructures: the INTERFACE force field. [Langmuir 29:1754-1765](#)

Next Step:  Build Nanomaterial

4. 設定好CNT性質

Nanomaterial Type

Chirality (n, m) ,

Cell Copies

Use virtual electron model ☐

Calculated Tube Info

Tube Length	81.3 Å
Tube Diameter	7.5 Å
Tube Conductivity	Semiconductive
System Volume	114,212.9 Å ³

Periodic Options

☒ X ☐ Y ☐ Z

System Type

- ☐ Solvated
☒ Vacuum



5. 按右下角下一步

Next Step: 
Build Nanomaterial

Building | Size Determination | PBC Setup | Input Generator

JOB ID: 2449995467

Building Input: [step1_nanomaterial.inp](#)
 Building Output: [step1_nanomaterial.out](#)
 Building PSF: [step1_nanomaterial.psf](#)
 Image Patch PSF: [step1_nanomaterial_img.psf](#)
 Building PDB: [step1_nanomaterial.pdb \(view structure\)](#)
 Building Info: [step1_nanomaterial.str](#)

[download .tgz](#)

6. 點擊下載



glycan.yaml	4	YAML 文件
moleculex.py	9,508	? Python 來源檔案
moleculex.pyc	10,183	? PYC 檔案
rmimg_info.py	10,489	? Python 來源檔案
step1.1_user_input.str	150	? STR 檔案
step1_nanomaterial.crd	274,340	? CRD 檔案
step1_nanomaterial.inp	2,992	? INP 檔案
step1_nanomaterial.out	129,153	? OUT 檔案
step1_nanomaterial.pdb	149,981	? PDB 檔案
step1_nanomaterial.psf	928,180	? PSF 檔案
step1_nanomaterial.str	415	? STR 檔案
step1_nanotube.crd	274,134	? CRD 檔案
step1_nanotube.pdb	157,492	? PDB 檔案
step1_nanotube.rtf	98,373	? RTF 格式
toppar.str	3,670	? STR 檔案

7. 將step1_nanomaterial.pdb & psf
檔名改為nanotube.pdb & psf

補充

改變 cell copy 數可產生不同長度的碳管 (n倍長)

Nanomaterial Type

Chirality (n, m) ,

Cell Copies

Use virtual electron model ☐

Calculated Tube Info

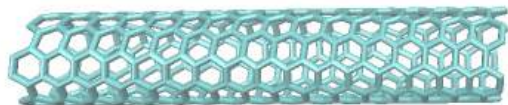
Tube Length	40.7 Å
Tube Diameter	7.5 Å
Tube Conductivity	Semiconductive
System Volume	57,106.4 Å ³

Periodic Options

☒ X ☐ Y ☐ Z

System Type

☐ Solvated
☒ Vacuum



Nanomaterial Type

Chirality (n, m) ,

Cell Copies

Use virtual electron model ☐

Calculated Tube Info

Tube Length	81.3 Å
Tube Diameter	7.5 Å
Tube Conductivity	Semiconductive
System Volume	114,212.9 Å ³

Periodic Options

☒ X ☐ Y ☐ Z

System Type

☐ Solvated
☒ Vacuum



Nanomaterial Type

Chirality (n, m) ,

Cell Copies

Use virtual electron model ☐

Calculated Tube Info

Tube Length	122.0 Å
Tube Diameter	7.5 Å
Tube Conductivity	Semiconductive
System Volume	171,319.3 Å ³

Periodic Options

☒ X ☐ Y ☐ Z

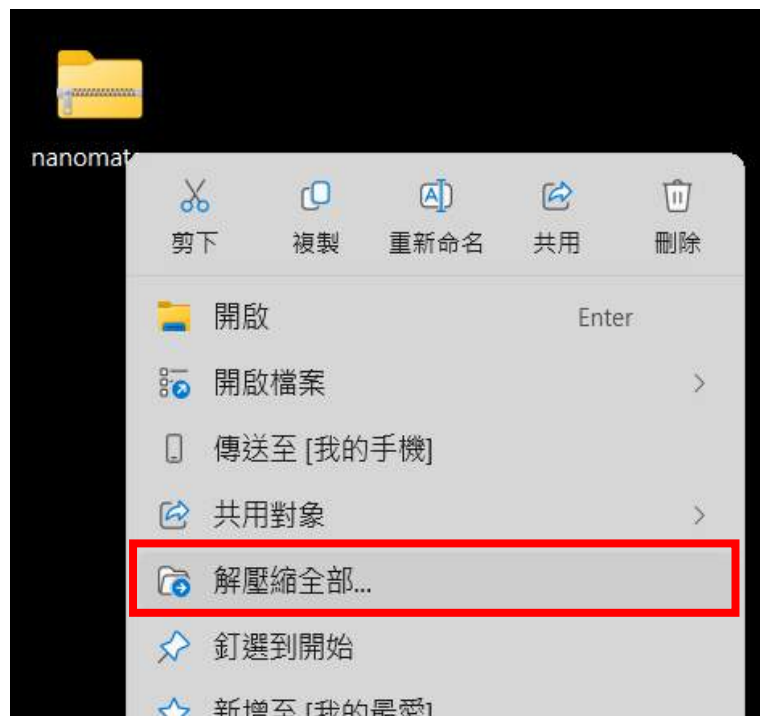
System Type

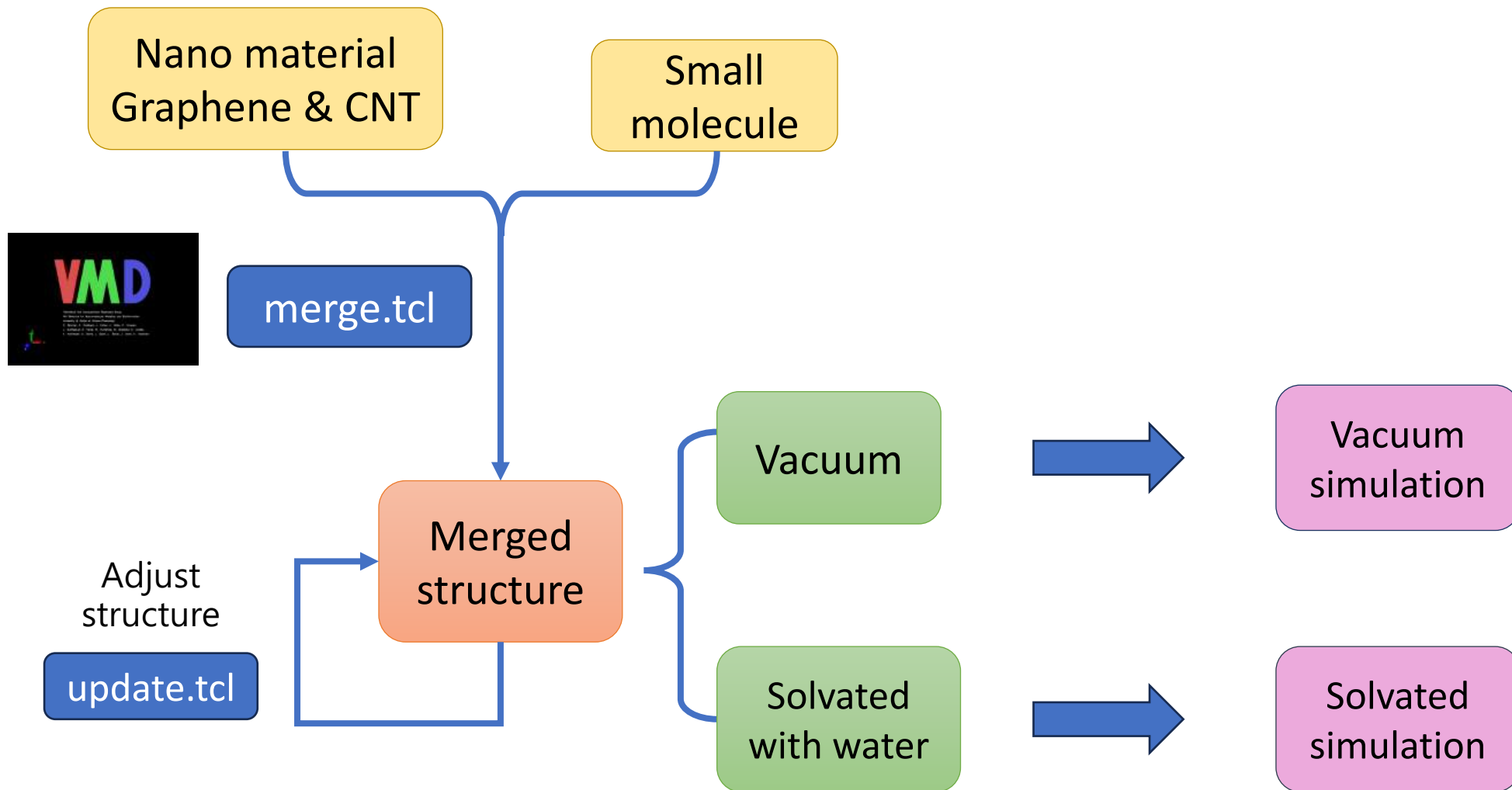
☐ Solvated
☒ Vacuum



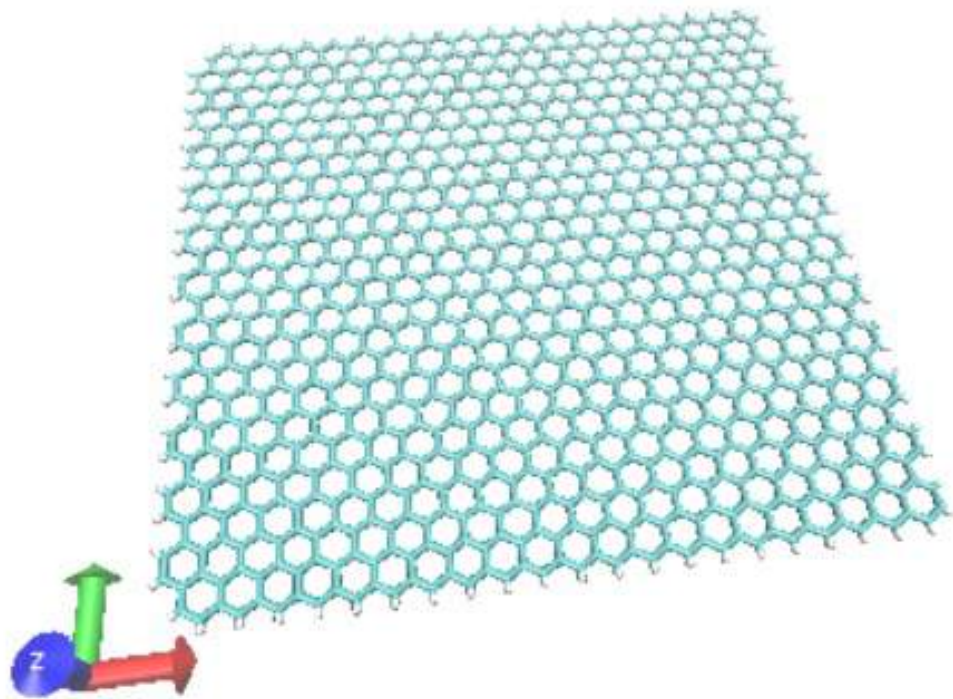
Nanomaterial 解壓縮

1. 從 113.2 MD workshop/module4 中，將 **nanomaterial.zip** 資料夾拖曳至桌面
2. 按滑鼠右鍵 → 解壓縮全部... → 目的地路徑 \.....\.....\Desktop\ **(此處全刪除)**
3. 解壓縮

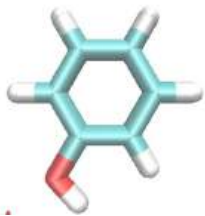




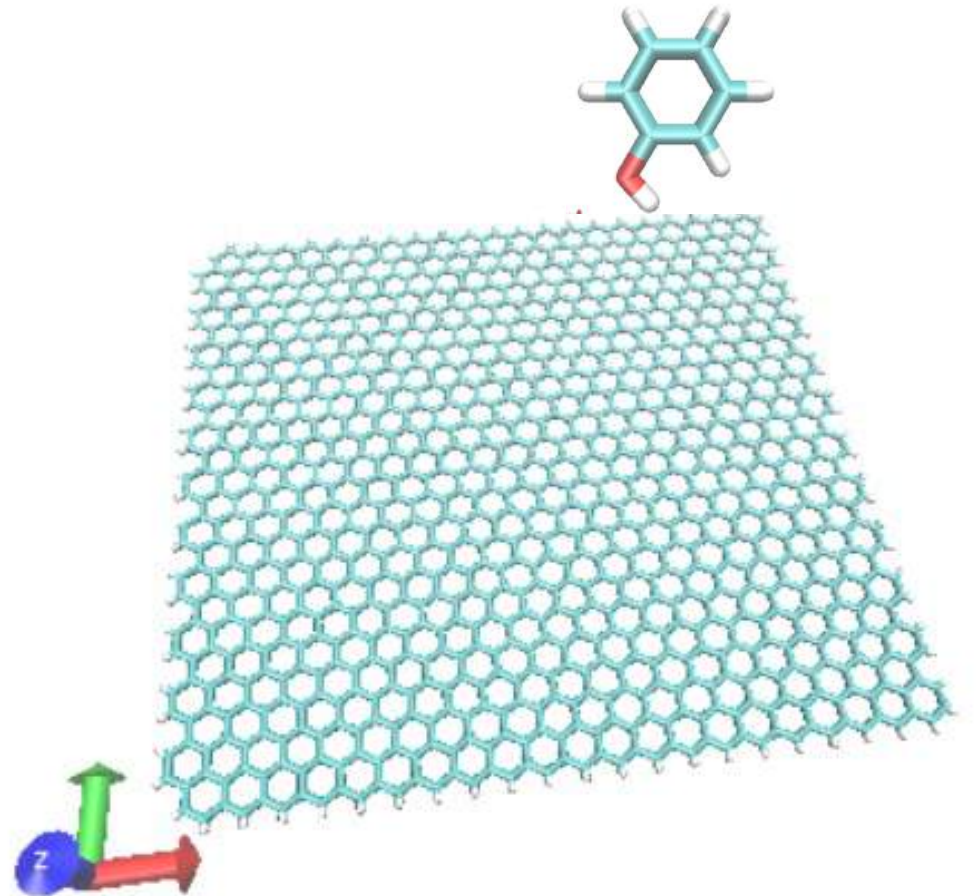
奈米材料模型建立 (1. Merge)



+



Merge



Merge.tcl介紹

```
#load nanomaterial
package require topotools 1.7
set midlist {}
set mol [mol new graphene.psf waitfor all]
mol addfile graphene.pdb

#load molecule in system
lappend midlist $mol
set mol [mol new phenol.psf waitfor all]
mol addfile phenol.pdb $mol

#moving molecule
set sel [atomselect top all]
$sel moveby {0.0 20.0 0.0}
lappend midlist $mol
#do the magic
set mol [::TopoTools::mergemols $midlist]

animate write psf graphene_sim_temp.psf $mol
animate write pdb graphene_sim_temp.pdb $mol
```

搭配VMD 1.9.3
(VMD 1.9.4則使用1.8)

匯入Graphene or CNT structure

匯入phenol structure

選擇merge位置

輸出merge檔案名稱

使用方式(以Graphene為例)

1. 將右側框起檔案放置同個路徑中
(Ex: ~ ./nanomaterial/Working_dir/)

開始備份

>

桌面

>

md-workshop-ntu-main

>

module4

>

working_dir

>

nanomaterial

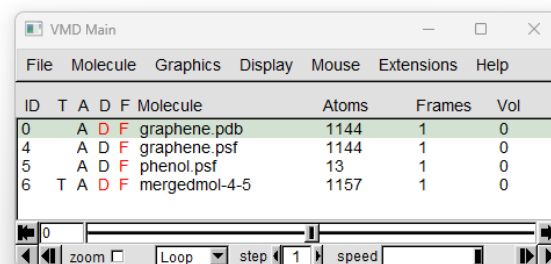
↑ 排序

≡ 檢視

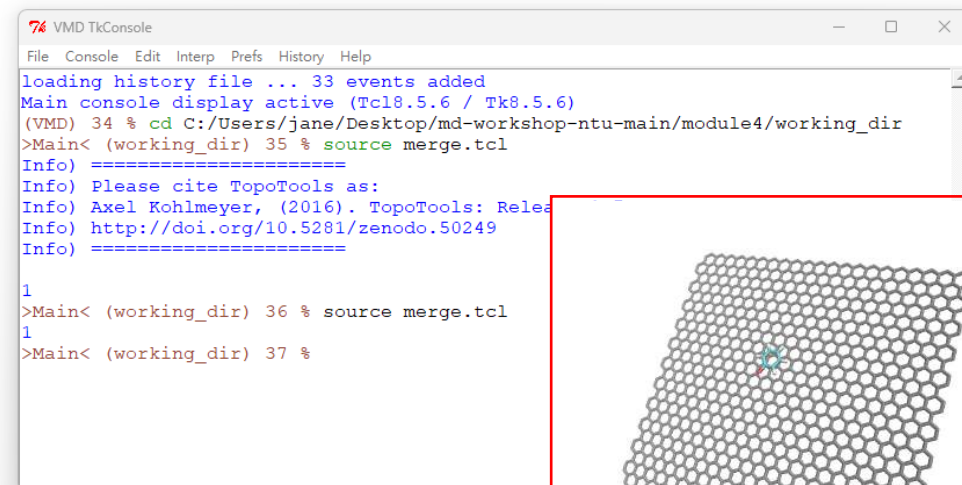
...

名稱	修改日期	類型	大小
.ipynb_checkpoints	2025/8/5 下午 04:20	檔案資料夾	
parafire	2025/3/3 下午 03:40	檔案資料夾	
graphene	2025/3/3 下午 03:40	PDB 檔案	89 KB
graphene.psf	2025/3/3 下午 03:40	PSF 檔案	483 KB
graphene_sim	2025/3/3 下午 03:40	CONF 檔案	4 KB
graphene_sim_wb	2025/3/3 下午 03:40	CONF 檔案	4 KB
merge	2025/8/6 下午 01:21	TCL 檔案	1 KB
module 4---nanomaterial application	2025/8/6 下午 04:29	IPYNB 檔案	10 KB
nanotube	2025/8/1 下午 01:14	PDB 檔案	57 KB
nanotube.psf	2025/8/1 下午 01:14	PSF 檔案	353 KB
nanotube_sim_temp	2025/8/6 下午 01:21	PDB 檔案	60 KB
nanotube_sim_temp.psf	2025/8/6 下午 01:21	PSF 檔案	339 KB
phenol	2025/3/3 下午 03:40	PDB 檔案	2 KB
phenol.psf	2025/3/3 下午 03:40	PSF 檔案	4 KB

2. 於vmd中匯入graphene.pdb後
打開tk console
cd至放置檔案的路徑後按enter
再鍵入source merge.tcl後按enter

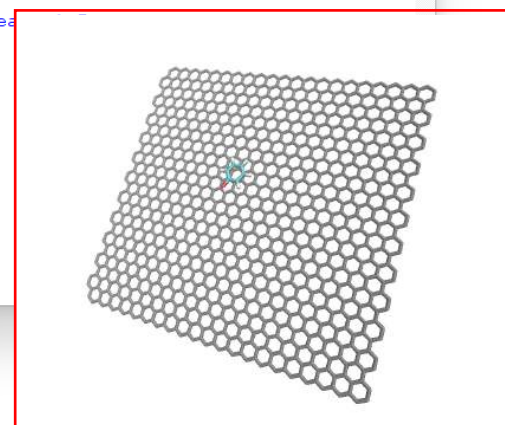


ID	T	A	D	F	Molecule	Atoms	Frames	Vol
0	A	D	F		graphene.pdb	1144	1	0
4	A	D	F		graphene.psf	1144	1	0
5	A	D	F		phenol.psf	13	1	0
6	T	A	D	F	mergedmol-4-5	1157	1	0



```
loading history file ... 33 events added
Main console display active (Tcl8.5.6 / Tk8.5.6)
(VMD) 34 % cd C:/Users/jane/Desktop/md-workshop-ntu-main/module4/working_dir
>Main< (working_dir) 35 % source merge.tcl
Info) =====
Info) Please cite TopoTools as:
Info) Axel Kohlmeyer, (2016). TopoTools: Release
Info) http://doi.org/10.5281/zenodo.50249
Info) =====
1
>Main< (working_dir) 36 % source merge.tcl
1
>Main< (working_dir) 37 %
```

3. 獲得merged structure



補充：Merge 不同位置示範

```
#load nanomaterial
package require topotools 1.7
set midlist {}
set mol [mol new graphene.psf waitfor all]
mol addfile graphene.pdb

#load molecule in system
lappend midlist $mol
set mol [mol new phenol.psf waitfor all]
mol addfile phenol.pdb $mol

#moving molecule
set sel [atomselect top all]
$sel moveby {0.0 0.0 10.0}
lappend midlist $mol
#do the magic
set mol [::TopoTools::mergemols $midlist]

animate write psf graphene_sim_temp.psf $mol
animate write pdb graphene_sim_temp.pdb $mol
```

將 x 值改成15

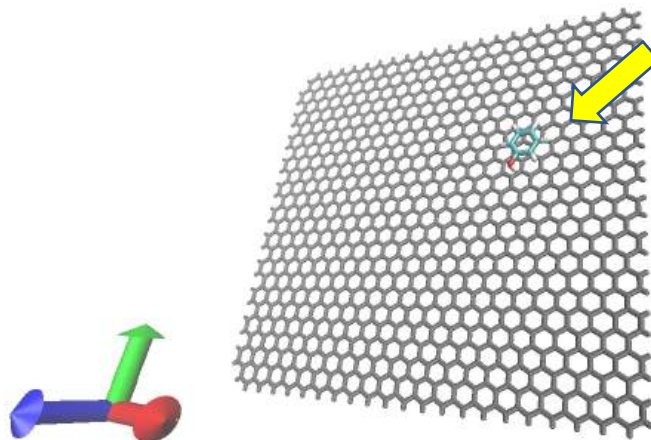
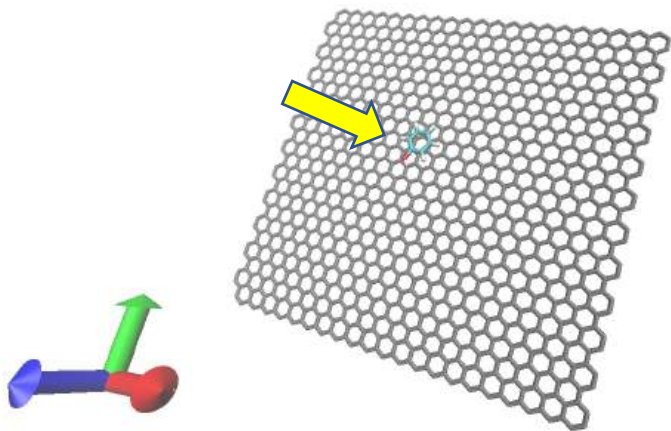


```
#load nanomaterial
package require topotools 1.7
set midlist {}
set mol [mol new graphene.psf waitfor all]
mol addfile graphene.pdb

#load molecule in system
lappend midlist $mol
set mol [mol new phenol.psf waitfor all]
mol addfile phenol.pdb $mol

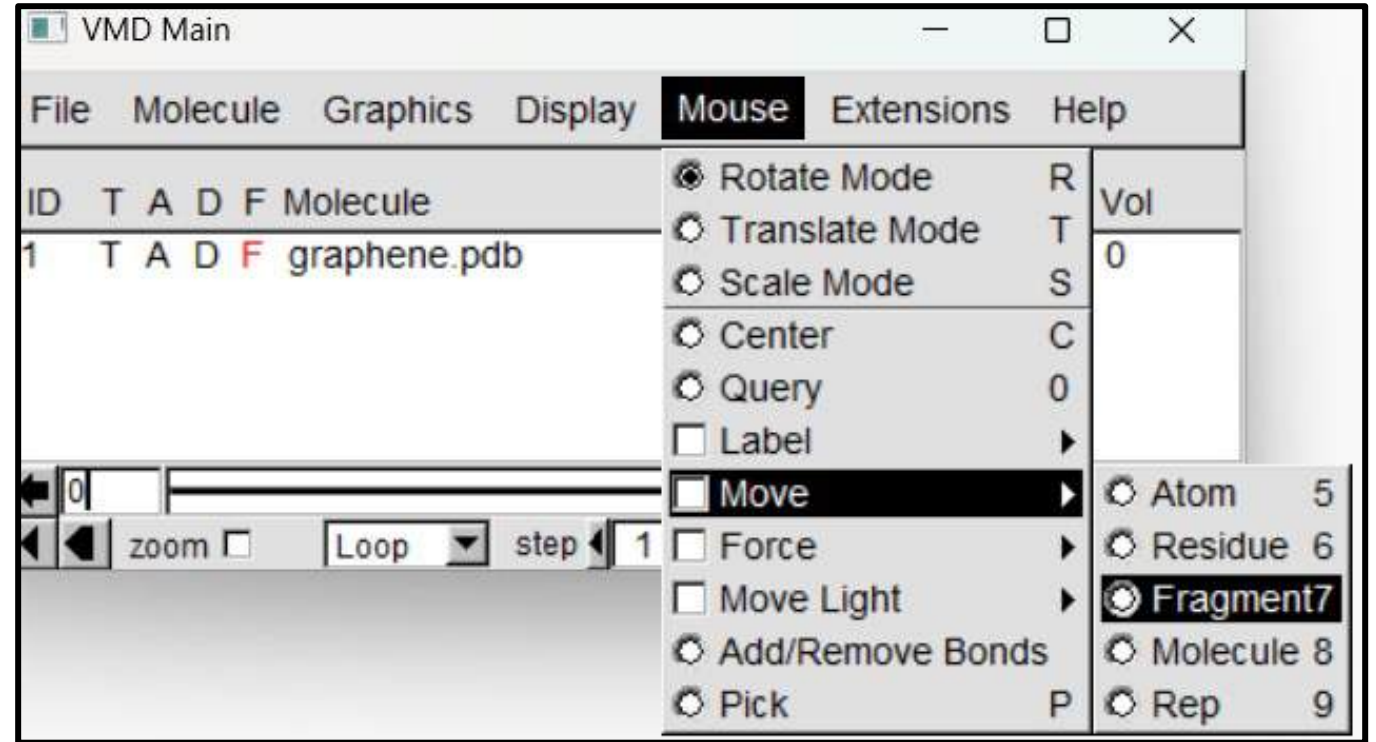
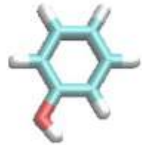
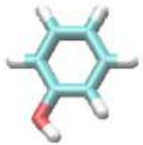
#moving molecule
set sel [atomselect top all]
$sel moveby {15.0 0.0 10.0}
lappend midlist $mol
#do the magic
set mol [::TopoTools::mergemols $midlist]

animate write psf graphene_sim_temp_1.psf $mol
animate write pdb graphene_sim_temp_1.pdb $mol
```



Adjustment

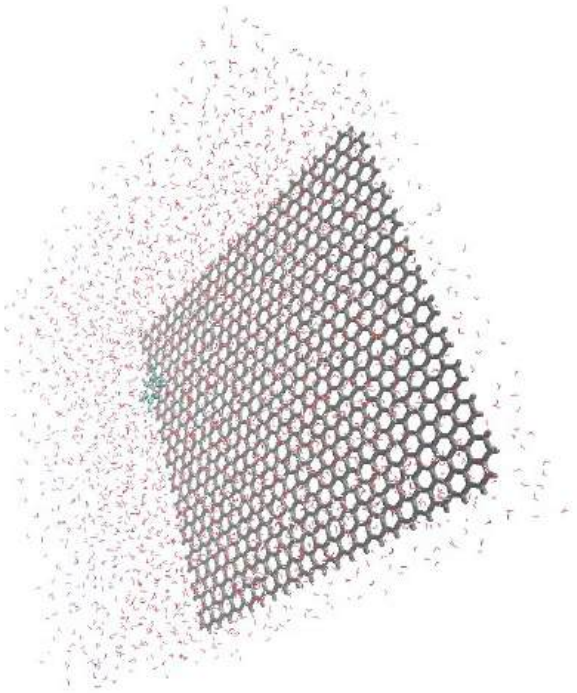
按住**shift**與左鍵可以
旋轉分子



完成旋轉與調整後，在 tk console 輸入
source update.tcl

Solvated simulation (Graphene)

將: graphene_sim_wb.conf graphene_sim_wb.pdb
graphene_sim_wb.psf 放入/graphene/solvated/資料夾內



graphene_sim_wb.conf ↓

```
# if run restart change 1 to 0 #
temperature      $temperature

# Force-Field Parameters
exclude          scaled1-4
1-4scaling       1.0
cutoff           12.0
switching        on
switchdist       10.0
pairlistdist     14.0
margin           5

# Integrator Parameters
timestep         2.0      ;# 2fs/step
rigidBonds       all      ;# needed for 2fs steps
nonbondedFreq    1
fullElectFrequency 2
stepspercycle    10

# Constant Temperature Control
langevin         on       ;# do langevin dynamics
langevinDamping  1        ;# damping coefficient (gamma) of 1/ps
langevinTemp     $temperature
langevinHydrogen off      ;# don't couple langevin bath to hydrogens

# Periodic Boundary Conditions
cellBasisVector1 18.0     0.0     0.0
cellBasisVector2 0.0      37.0    0.0
cellBasisVector3 0.0      0.0     93.0
cellOrigin       0.0      10.4    0.0

wrapAll          on

# PME (for full-system periodic electrostatics)
PME              yes
PMEGridSpacing   1.0

#manual grid definition
#PMEGridSizeX    45
#PMEGridSizeY    45
#PMEGridSizeZ    48

# Constant Pressure Control (variable volume)
useGroupPressure yes ;# needed for rigidBonds
useFlexibleCell  no
useConstantArea  no

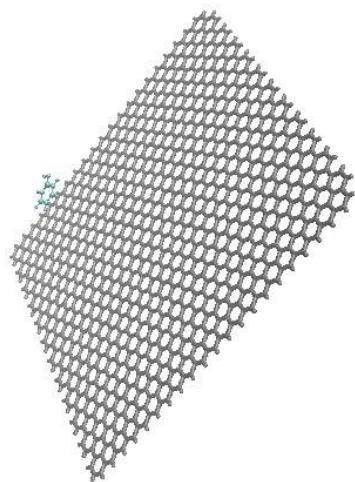
langevinPiston   on
langevinPistonTarget 1.01325 ;# in bar -> 1 atm
langevinPistonPeriod 100.0
langevinPistonDecay 50.0
langevinPistonTemp $temperature

# Output
outputName       $outputname

restartfreq      250      ;# 500steps = every 1ps
dcdfreq          250
xstfreq          250
outputEnergies   100
outputPressure   100
```

Vacuum simulation (Graphene)

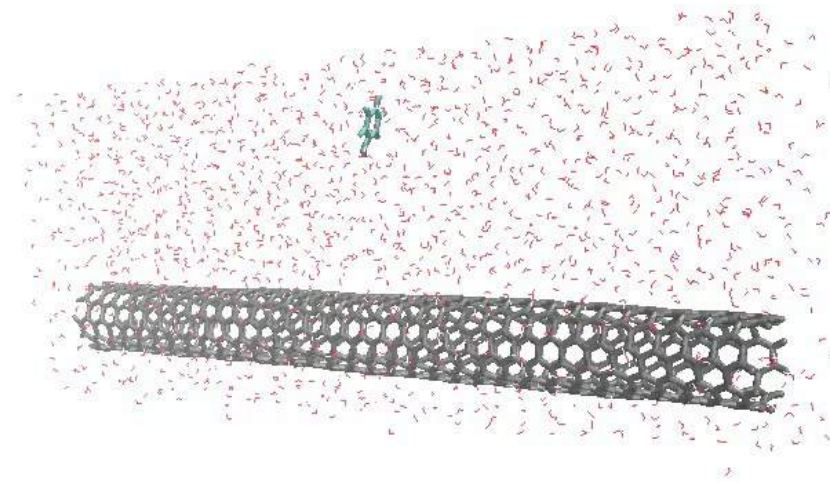
將: graphene_sim.conf graphene_sim_temp.pdb
graphene_sim_temp.psf 放入/graphene/vacuum/資料夾內



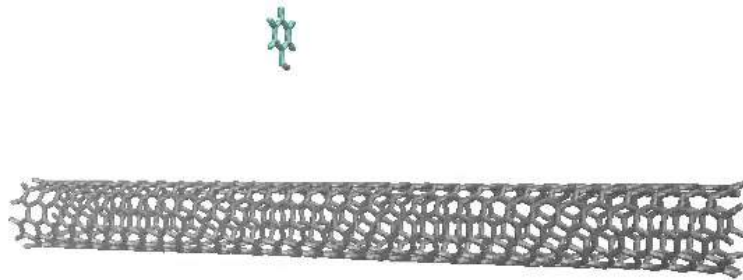
和solvated 系統指令差在 ↓

```
# Constant Pressure Control (variable volume)
useGroupPressure      no    ;#no need in vaccum
useFlexibleCell        no
useConstantArea        no
```

Solvated simulation (CNT)

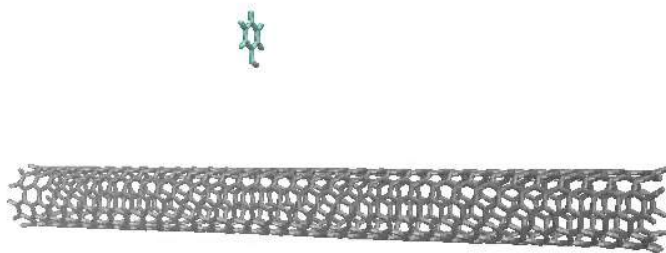


Vacuum simulation (CNT)



計算小分子與CNT的距離

- 在Tk Console輸入 (注意放置路徑):
source distance.tcl



Frame數

0	20.78642318055843
1	20.786781819098746
2	21.43881348889818
3	21.925169780145072
4	22.49873621900503
5	23.05626825633667
6	23.734683241321928
7	23.568969645829565
8	22.944020405255017
9	22.72318032861033
10	22.5102304295546
11	22.045578774801303
12	22.225801333196134
13	22.467258795174175
14	23.599905198295826
15	24.175286507557995
16	24.650381070596282
17	25.717776770317602
18	27.31156748112046
19	28.05961128886521
20	28.292052402519854

phenol相對CNT端點距離

產出distance.dat ↑

phenol相對CNT端點距離 統計次數

7.088063155391156	5
7.449060043707225	3
7.8100569320232935	4
8.171053820339361	1
8.532050708655431	5
8.893047596971499	2
9.254044485287569	1
9.615041373603637	4
9.976038261919706	2
10.337035150235774	0
10.698032038551844	4
11.059028926867912	2
11.42002581518398	7
11.78102270350005	9
12.14201959181612	7
12.503016480132187	9
12.864013368448255	9
13.225010256764325	4
13.586007145080394	2
13.947004033396462	3
14.30800092171253	1
14.6689978100286	0
15.02999469834467	0
15.390991586660737	0
15.751988474976805	1

產出distribution.dat ↑